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ADVANCED STATISTICAL METHODS FOR STUDIES OF LYMPHOMA PROGNOSIS AND TREATMENT OUTCOMES IN A REAL-WORLD SETTING

BY
RASMUS RASK KRAGH JØRGENSEN

PhD Thesis 2024



AALBORG UNIVERSITY
DENMARK

Advanced statistical methods for studies of lymphoma prognosis and treatment outcomes in a real-world setting

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Preface

This PhD thesis comprises three scientific papers that were written during my employment at the Department of Hematology, Aalborg University Hospital from September 2021 until November 2024. During my time as a PhD student, I have had the opportunity and pleasure to meet many competent people whom I would like to acknowledge for their various contributions to this thesis.

First and foremost, I would like to thank my two supervisors, Lasse Hjort Jakobsen and Tarec El-Galaly, for the opportunity to work as a PhD student. Thank you, Tarec, for your always positive attitude, professionalism, and mentorship, and for opening up your network, which has encouraged many collaboration opportunities. Lasse, thank you for always pushing me to do better. I have learned so much with you as my supervisor, and have continued to even after you left for Novo. I am very lucky to have had such competent supervisors, and am very grateful for the opportunities that followed.

Many thanks to all my colleagues at the Department of Hematology, especially Marianne Tang Severinsen for always supporting me and my ideas. I would also like to thank those with whom I have shared an office during my time as a PhD student: Eva, Joachim, Jonas, Mikkel, Lasse, Daniel, Trine, Rasmus B, Nanna, and Lise. Thank you all for contributing to a great and diverse office atmosphere that facilitated excellence through interdisciplinary work and friendship. This is how exemplary great research is created. I would also like to thank my two junior statisticians, Mikkel and Jonas, for all the endless discussion about mathematical properties and life, and helping me finish my third study. Thank you!

Thank you, Sandra Eloranta, for welcoming me during my short stay in Stockholm, and for always providing me with competent and professional feedback. My appreciation also goes out to Mr. Dr. Joshua Entrop, for your excellent help and expertise during my stay at Karoliska and for being an overall decent person. Thank you to Peter Martin and Zhengming Chen for welcoming me in New York.

Finally, I would like to thank my family. Birgitte, thank you for enduring my endless talks about numbers, and for your love and support.

Rasmus Rask Kragh Jørgensen, November 2024

List of papers

The thesis is based on the following papers:

Study I Machine Learning–Based Survival Prediction Models for Progression-Free and Overall Survival in Advanced-Stage Hodgkin Lymphoma
Rask Kragh Jørgensen, R., Bergström, F., Eloranta, S., Tang Severinsen, M., Bjøro Smeland, K., Fosså, A., Haaber Christensen, J., Hutchings, M., Bo Dahl-Sørensen, R., Kamper, P., Glimelius, I., E Smedby, K., K Parsons, S., Mae Rodday, A., J Maurer, M., M Evens, A., C El-Galaly, T. & Hjort Jakobsen, L.
JCO Clinical Cancer Informatics. doi:10.1200/CCI.23.00255

Study II Effectiveness of R-CHOP versus R-CHOEP for treatment of young patients with high-risk diffuse large B-cell lymphoma: A Danish observational population-based study
Rask Kragh Jørgensen, R., Jakobsen, L. H., Eloranta, S., Smedby, K. E., Pedersen, R. S., Jørgensen, J. M., Clausen, M. R., Brown, P., Gang, A. O., Gade, I. L., Larsen, T. S., Jerkeman, M. & El-Galaly, T. C.
European Journal of Haematology **113.** doi:10.1111/ejh.14275

Study III Development of survival prediction models using federated learning.
Rasmus Rask Kragh Jørgensen, Jonas Faartoft Jensen, Tarec El-Galaly, Martin Bøgsted, Rasmus Froberg Brøndum, Mikkel Runason Simonsen and Lasse Hjort Jakobsen. *In preparation*

The following studies were published during my time at the department of hematology at Aalborg University Hospital but are not included in the PhD thesis.

(1) Work Disability and Return to Work After Treatment for Acute Lymphoblastic Leukemia: A Danish Nationwide Cohort Study
Maksten, E., Jørgensen, R., Pedersen, M., Fonager, K., Bech, R., Mølle, I., Ørskov, A., Schöllkopf, C., Overgaard, U., Thomsen, G., El-Galaly, T. & Severinsen, M.
Clinical Epidemiology **Volume 16.** doi:10.2147/CLEP.S444270

(2) MRD-driven treatment with venetoclax-R2 in mantle cell lymphoma: the Nordic Lymphoma Group MCL7 VALERIA trial
Jerkeman, M., Kolstad, A., Hutchings, M., Pasanen, A., Meriranta, L., Niemann, C. U., Kragh Jørgensen, R. R., El-Galaly, T. C., Riise, J., Leppä, S., Christensen, J. H., Sonnevi, K., Pedersen, L. B., Wader, K. F. & Glimelius, I.
Blood Advances **8.** doi:10.1182/bloodadvances.2023011920

(3) Overall survival and causes of death in elderly patients with Hodgkin lymphoma: a Norwegian population-based case-control study

Lia, K., Jørgensen, R. R., L.Wold, B., Fluge, Ø., Fagerli, U.-M., Bersvendsen, H., Bø, I. B., Bhargava, S. & Fosså, A.
Haematologica **109.** doi:10.3324/haematol.2023.283721

English summary

Predicting outcomes and estimating treatment effects among patients with lymphoma are subjects of interest to a broad range of healthcare professionals. New real-world evidence can be obtained by combining the increasing amount of real-world data available from the Nordic countries with advanced statistical methods. The overall aims of this thesis were to develop new prognostic tools that can provide patients with a more accurate prognosis at the time of diagnosis, and to estimate treatment effectiveness among patients in the real-world setting. Currently available prognostic models are often simplistic and were built based on individuals receiving outdated treatment; therefore, there is a need for continuous effort to update these models. Moreover, estimating treatment effectiveness is important, especially when a randomized trial is unlikely.

In the first paper in this thesis, two predictive machine learning models were developed to predict overall survival and progression-free survival in patients with advanced stage classical Hodgkin lymphoma. Notably, the machine learning models outperformed existing prognostic indices upon both internal and external validation, but yielded only slightly higher performance measures compared to the newly developed Advanced Stage Hodgkin Lymphoma International Prognostic Index (A-HIPI) model.

In the second study, young patients with diffuse large B-cell lymphoma (DLBCL) who received R-CHOEP were matched 1:1 to patients receiving R-CHOP who were similar in a large range of clinical characteristics. This study found no evidence that of R-CHOEP had a superior effect over R-CHOP in terms of overall survival or progression-free survival. Analysis of the late effects among patients achieving complete remission revealed, no evidence of an increased risk of cardio vascular disease or secondary primary malignancies in patients treated with R-CHOEP versus R-CHOP.

Protecting patients' right to privacy is important, as there is a risk of re-identifying individuals from predictive models. To address this subject, in the third study, a new method is introduced for training survival models across sites using federated learning. The methods utilized gradients and Hessians to estimate covariate effects, and

included a kernel smoothing process for a Cox proportional hazard model. The novel proposed methods achieved similar performance measures compared to the case where data are pooled at a single site, using both simulated data and real-world data.

Dansk resume

Forudsigelse af prognose og estimering af behandlings effektivitet i patienter med lymfekræft er af særlig interesse for en bred vifte af sundhedspersonale. Ved at kombinere den stigende mænge af real-world data tilgængeligt i de nordiske lande, med avancerede statistiske metoder er det muligt at opnå nye evidens fra den virkelige verden. Formålet med denne afhandling var at udvikle nye prognostiske værktøjer, som kan lave mere præcise prognoser på diagnose tidspunktet, og estimerer effekten af behandling i real-world data. Da nuværende prognostiske modeller ofte er forenklede og trænet på patienter som har modtaget forældet behandling, er der et behov for at opdatere disse eller udvikle nye. Det er også vigtigt at udregne effekten af en behandling når et lodtræknings forsøg ikke kan lade sig gøre.

I det første studie, udviklede vi to machine-learning (ML) modeller til at kunne forudsige overlevelse og progressions fri overlevelse i patienter med avanceret stadie klassisk Hodgkin lymfom. Machine learning modellerne var i stand til at prædiktere mere præcist sammenlignet med allerede eksisterende prognostiske modeller i både intern og extern validerings data, dog var modellerne kun i stand til at få et lidt højere performance mål sammenlignet med den nye udviklede A-HIPI model.

I det andet studie, blev unge patienter med Diffust stortcellet B-celle lymfom (DLBCL), behandlet med R-CHOEP, matchede 1:1 på en lang række kliniske karakteristika til patienter behandlet med R-CHOP. Studiet fandt ikke nogen evidens for at R-CHOEP havde en overlegen effekt sammenlignet med R-CHOP på hverken overlevelse eller progression fri overlevelse.
I analysen for senfølger for patienter der opnåede komplet remission (CR), blev der ikke fundet nogen evidens for en forøget risiko for kardivalkulær sygdom eller sekundær primær malogniteter i patienter behandlet med R-CHOEP sammenlignet med R-CHOP.

At sikre patienters ret til privatliv er vigtigt da risikoen for at identificere individer igennem prædiktive modeller er et problem. I det tredje studie, blev der udviklet en ny metode til at bygge overlevelses modeller på tværs af lande grænser ved at gøre brug

af federated learning. Metoden gjorde brug af gradienter og Hessianer til at estimere covariat effekterne og inkluderede en kerne udglatnings proces for Cox proportionale hazard model. Den nye metode var i stand til at opnår sammenlignelige performance mål sammenlignet med scenariet hvor alt dataene var samlet på et enkelt site, i både simulerede forsøg og i real-world data.

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Abbreviations

95% CI	confidence interval
A-HIPI	Advance-Stage Hodgkin Lymphoma International Prognostic Index
aaIPI	age-adjusted International Prognostic Index
ABVD	adriamycin, bleomycin, vinblastine, and dacarbazine
aHL	advanced-stage Hodgkin lymphoma
ATE	average treatment effect
ATT	average treatment effect on the treated
AUC	area under the curve
BEACOPP	bleomycin, etoposide, doxorubicin, cyclophosphamide, vincristine, procarbazine, and prednisolone
BrECADD	brentuximab vedotin, etoposide, cyclophosphamide, doxorubicin, dacarbazine, and dexamethasone
C-index	concordance index
CAR-T	chimeric antigen receptor T-cell
CCI	Charlson comorbidity index
cHL	classical Hodgkin lymphoma
CNS	Central Nervous System
CPR	Civil Personal Register
CR	complete remission
CRS	Danish Civil Registration System
CV	cross-validation
CVD	cardiovascular disease
DA-EPOCH-R	dose-adjusted etoposide, prednisone, vincristine, cyclophosphamide, and doxorubicin together with standard rituximab
DCDR	Danish Cause of Death Register
DCR	Danish Cancer Register
DHDA	Danish health data authority
DLBCL	Diffuse large B-cell lymphoma

DNPR	Danish National Patient Register
DPR	Danish Pathology Register
E-IPI	Elderly International Prognostic Index
ECOG	Eastern Cooperative Oncology Group
EMA	European Medicines Agency
EPE	expected prediction error
ESMO	European Society For Medical Oncology
FDA	Food and Drug Administration
FL	federated learning
FPS	flexible parametric survival
GDPR	General Data Protection Regulation
HL	Hodgkin lymphoma
HR	hazard ratio
IBS	integrated brier score
ICD-10	International Classification of Disease, 10th Edition
IPI	International Prognostic index
IPS-3	simplified International Prognostic Score
IPS-7	International Prognostic Score
IPTW	inverse probability of treatment weighting
LDH	lactate dehydrogenase level
LYFO	National Danish Lymphoma register
MER	University of Iowa/Mayo Clinic Specialized Program of Research Excellence Molecular Epidemiology Resource
ML	machine learning
MOPP	nitrogen mustard, vincristine, procarbazine, and prednisone
NCCN-IPI	National Comprehensive Cancer Network International Prognostic Index
NHL	non-Hodgkin lymphoma
OS	overall survival
PFS	progression-free survival
PSM	propensity score matching
R-CHOEP	rituximab, cyclophosphamide, doxorubicin, vincristine, etoposide and prednisolone
R-CHOP	rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisolone

R-IPI	Revised International Prognostic Index
RCT	randomized control trial
RKKP	Danish Clinical Registries
RMS	restricted mean survival
RMSE	root mean square error
RWD	real-world data
RWE	real-world evidence
SAP	statistical analysis plan
SAT	single-arm trial
SLR	Swedish Lymphoma Register
SMR	standardized mortality ratio
SNOMED	Systematized Nomenclature of Medicine

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Background

The topic of this thesis is the application of advanced statistical and machine learning (ML) methods within the field of lymphoma research, to enhance our knowledge of prognosis and treatment effects. This is accomplished using real-world data (RWD) derived from national registries, which serves as the data-backbone for Studies I–III. Study III also includes simulated data resembling RWD from registries. The papers included in this thesis were developed through collaboration with national and international partners having diverse sets of skills and areas of expertise. On a personal level, I have had the fortune to develop both my statistical and clinical knowledge through these interdisciplinary collaborations.I have also acquired a more comprehensive understanding of disease- related problems, and how advanced statistical methods can help achieve better outcomes for patients. I hope, that this thesis can help motivate more interdisciplinary collaborations, especially between clinicians and statisticians, in future research.

The first part of this thesis provides an introduction to the various diseases, data-types, clinical prognostic models, epidemiological and statistical concepts, and data sources used throughout this thesis. The second part outlines the overall aims of the thesis. The third part summarizes Papers I –III presenting the key findings and additional methodological considerations or results not presented in the original article, and including additional discussion of problems and findings.

1 Introduction

Lymphomas are a heterogeneous group of hematological malignancies developed in the lymph nodes, and are classically divided in Hodgkin lymphoma (HL) or non-Hodgkin lymphoma (NHL) [6–9]. Lymphoma subtypes generally exhibit vast differences in biology, patient age distribution, and geography. Due to these differences, treatment can vary from intensive chemotherapy followed by stem-cell transplantation, for example, in peripheral T-cell lymphoma and mantle cell lymphoma, to a wait-and-watch strategy in indolent lymphomas, such as follicular lymphoma. Pa-

tients' general health and fitness also have major influences on treatment decisions, as it must be considered whether patients can tolerate intensive treatment. Additional details regarding the lymphomas considered in this thesis are provided below.

1.1 Hodgkin Lymphoma

Historically, HL has been divided into classical Hodgkin lymphoma (cHL) ($\sim$95% of patients) and nodular lymphocyte-predominant HL ($\sim$5% of patients) [7, 10, 11]. Moreover, cHL is sub-divided into four histological subtypes: nodal sclerosis, lymphocyte-rich, mixed cellularity, and lymphocyte-depleted [12, 13]. In Europe, the annual HL incidence is 2.3 per 100,000 [10]. In Denmark, between 2017—2021, there were 162 new cases per year, with a greater proportion of males (95 patients) than females (67 patients) [12, 14, 15]. The age distribution of HL exhibits by a bimodal shape with the first peak around 20–30 years of age, making it a common cancer among young individuals, followed by a second peak around 60–80 years of age. Additionally, patients with a history of Epstein-Barr virus-related infections, have an increased risk of developing cHL ($\sim$2–3 fold) [15].

HL treatment has changed remarkably throughout the years. In 1960, treatment was based on an alkylating agent nitrogen mustard, vincristine, procarbazine, and prednisone (MOPP), along with radiation therapy,; however, newer treatments were sought to reduce adverse events and late toxicities [16, 17]. In the 1980s, a randomized control trial (RCT) compared MOPP against adriamycin, bleomycin, vinblastine, and dacarbazine (ABVD). The long-term results showed that ABVD was associated with increased 7-year freedom from progression (80.8% with ABVD vs. 62.8% with MOPP) [18]. In Denmark, the first-line treatment for advanced-stage Hodgkin lymphoma (aHL) is ABVD or a more intensive regimen including bleomycin, etoposide, doxorubicin, cyclophosphamide, vincristine, procarbazine, and prednisolone (BEACOPP). Treatment selection depends on patient characteristics, such as age at diagnosis, disease stage, and physical condition [19, 20]. Across disease stages, cHL generally shows favorable outcomes with a 5-year relative survival of 93.8% (confidence interval (95% CI): 91.2–96.4%) among males, and 95.3% (95% CI: 92.3 –98.4%) among females [14, 21]. Due to these high cure rates, research has focused on developing less toxic but equally effective treatments. In the recent HD21 trial, BEACOPP was compared to brentuximab vedotin, etoposide, cyclophosphamide, doxorubicin, dacarbazine, and dexamethasone (BrECADD). The results showed that, BrECADD was

better tolerated and more effective than BEACOPP, in terms of progression-free survival (PFS) among adults with cHL [22]. The HD21 trial revealed that BrECADD was associated with an increased 4-year PFS of 94.3%(95% CI: 92.6–96.1%) compared to 90.9% (95% CI: 88.7–93.1%) with BEACOPP, with a hazard ratio (HR) of 0.66 (95% CI: 0.45–0.97, p=0.035). No differences in overall survival (OS) were reported between the two treatment regimens.

HL is staged using the Ann Arbor system, where disease stage ranges from I–IV. Disease stage I is defined as involvement of a single lymph node or the surrounding area on one side of the diaphragm; stage II as the involvement of two or more lymph nodes on the same side of the diaphragm; stage III as the involvement of both side of the diaphragm; and stage IV as the involvement of specific extranodal site(s)[23]. Additional details are often included after the staging, to describe certain disease characteristics, such as B for B-symptoms (fever, weight loss, etc.), X for extranodal disease, and A for no symptoms. Accurate disease staging is critical since stage is the main determinant of treatment. Limited-stage disease (Ann Arbor stage I–II or without risk factors) is generally treated with abbreviated chemotherapy and radiotherapy. Advanced-stage disease (Ann Arbor III–IV or IIB with present risk-factors) is managed using chemotherapy with an extended duration and higher treatment intensity [10, 19]. Important risk factors include presence of large mediastinal mass or extranodal disease, as defined by the German Hodgkin study group [10, 24].

Among young Danish patients (age 18–49 years old), the 5-year OS was 95% (95% CI: 94–96%) with a relapse risk of 13.4% (95% CI: 12.1–14.8%) [25]. Among patients who were alive and relapse-free at 24 months, the risk of relapse decreased to 4.2% (95% CI: 3.8–4.6%) [25]. These findings support the need for less-intensive follow-up after a 24-month period.

1.2 Diffuse Large B-Cell Lymphoma

Diffuse large B-cell lymphoma (DLBCL) is the most common NHL subtype in adults, and accounts for approximately 37% of all NHL cases. In Europe, the DLBCL incidence is about 3.8 per 100,000 per year, and the median age at diagnosis ranges between 65–70 years [26–28]. Most patients with DLBCL (70%) present with advanced-stage disease at diagnosis [29]. The globally accepted standard treatment for DLBCL is multi-agent chemotherapy and immunotherapy with a regimen including rituximab,

cyclophosphamide, doxorubicin, vincristine, and prednisolone (R-CHOP) or similar agents [29].

The standard treatment of DLBCL has not substantially changed over the last 20 years, except for the introduction of more intensive regimens for high-risk disease, and shorter treatment duration for low-risk disease. In the Nordics, the more intensive regimen rituximab, cyclophosphamide, doxorubicin, vincristine, etoposide and prednisolone (R-CHOEP) is widely used. Internationally, dose-adjusted etoposide, prednisone, vincristine, cyclophosphamide, and doxorubicin together with standard rituximab (DA-EPOCH-R) is standard treatment for high-risk patients in many places [28, 29]. However, the use of more dose-dense therapies is generally decreasing worldwide, since no RCT has proven superior efficacy over standard R-CHOP, in terms of OS [30]. A Danish retrospective study from 2012 used RWD to compare 98 young high-risk patients who received R-CHOP, with 61 patients who received R-CHOEP [31]. The results suggested better outcomes with R-CHOEP compared to R-CHOP, but the median follow-up was merely 27 months. Moreover, the groups substantially differed in baseline characteristics, particularly the median age, which was 56 years in the R-CHOP group, and 49 years in the R-CHOEP group.

After initial treatment, the 3-year OS ranges from 59–91% depending on the risk-factors present at diagnosis (Table 1) [32]. Most patients achieve complete remission (CR) after initial treatment, but approximately 30–50% of patients relapse or have refractory disease. The prognosis after relapse remains poor, with a 2-year OS ranging between 20–27% [33–35]. In Denmark, chimeric antigen receptor T-cell (CAR-T) therapy was recently approved as second-line therapy: however, clinical studies show that second-line CAR-T therapy yields a 2-year PFS of 46%, indicating that there is still need for improvement [36, 37]. A Danish study from 2017, investigated 1,621 patients who had been diagnosed with DLBCL from 2003–2011 and were in CR after R-CHOP-like therapy [38]. The study found reduced survival among patients with DLBCL compared to an aged- and sex- matched background population (standardized mortality ratio (SMR) 1.75: 95% CI: 1.60–1.91%, $p<0.001$). However, patients surviving and relapse free at 24 months exhibited decreased excess mortality compared to the matched population (SMR 1.27: 95% CI: 1.12–1.44, $p<0.001$) [38]. Additionally, younger patients (<50 years of age) had favorable outcomes, with only modestly lower survival compared to the matched population. These findings were consistent with an American study from 2014, that included all patients with newly diagnosed

DLBCL, who recieved treatment with rituximab and anthracycline-base chemotherapy, and who were enrolled in either the University of Iowa/Mayo Clinic Specialized Program of Research Excellence Molecular Epidemiology Resource (MER) (2002-2009) cohort or the North Central Cancer Treatment Group NCCTG-N0489 clinical trial (2006-2007) [39]. The analysis revealed that patients deemed event-free after 24 months had a survival rate equivalent to than that of an age- and sex- matched population (SMR: 1.18; 95% CI: 0.89–1.57; p=0.25) [39]. These results Suggesting that there is limited benefit of additional follow-up for patients who are event-free after 24 months.

1.3 Use of Prognostic Models in Clinical Practice

Prognostic Models in Hodgkin Lymphoma

In 1998, the International Prognostic Score (IPS-7) was developed to risk-stratify patients with aHL based on the presence of seven risk factors (Table 1) [40]. The IPS-7 is defined as the number of present risk factors at diagnosis, with a score ranging from 0–7 points. During development of the IPS-7, it was found that all risk factors had approximately equal prognostic effect, enabling the easily usable "1 point per present risk factor" system. Patients with a score of 0 are deemed low-risk, and patients with a score of 5–7 are denoted high-risk.

The IPS-7 model can risk-stratify patients into different risk groups, and thus serve as a tool for assessing patient prognosis. However, rather than providing patients with a personalized prognosis based on their own characteristics, it predicts prognosis based on the average survival of a particular risk group. In other words, it is assumed that, each patient with the same risk-score, has the same prognosis. E.g. a 46-year-old male with an IPS-7 score of 5, and a 95-year-old male with an IPS-7 score of 5, are grouped together in the same survival curve, with similar estimated survival outlook. This binary conversion of continuous covariates consequently results in a huge loss of information, leading to inaccurate prognosis [41–43]. In contrast, using more advanced statistical and ML methods holds great potential for improving model accuracy and enabling personalized predictions [44]. Moreover, risk-stratification is increasingly important as high-risk patients can benefit from new therapeutic approaches while low-risk patients can be treated with reduced therapeutic dosages, thereby decreasing long-term therapy related toxicity [45, 46]. As stated in Study I:

The use of prognostic tools allows clinicians to identify individuals with low predicted survival probabilities who may not respond well to standard treatments and could benefit from novel treatment strategies. Conversely, for those with the best outcomes, trials may aim at treatment reduction to minimize the risk of long-term side effects. With the evolvement of new therapeutic options, precise risk is crucial, particularly for those with the highest risk.[1]

Improved risk stratification can help ensure optimal use of novel therapies, and guide the design of trials that focus on obtaining an optimal balance between efficacy and toxicity for individual patients.

Table 1: Prognostic scores for Hodgkin lymphoma (HL) and diffuse large B-cell lymphoma (DLBCL)

HL	**Risk factors**	**Risk groups (Score)**	**5–year OS** [47–49]
IPS-7	Age $\geq$ 45 years	Low (0)	98%
	Male sex	Low-Intermediate (1)	97%
	Lymphocytes $< 600mm^3$ or $< 8\%$ of Leukocytes	Intermediate Low (2)	91%
	Ann Arbor stage IV	Intermediate High (3)	88%
	Albumin $< 4g/dl$	High-Intermediate (4)	85%
	Hemoglobin level $< 10.5g/dl$	High (5–7)	67%
	Leukocytes $\geq 15.000mm^3$		
IPS-3	Age $\leq$ 45 years	Low (0)	91%
	Hemoglobin level $< 10.5g/dl$	Intermediate (1)	82%
	Ann Arbor stage IV	High-intermediate (2)	73%
		High (3)	53%
DLBCL	**Risk-factors**	**Risk groups (Score)**	**3–year OS** [32, 48, 50, 51]
IPI	Age > 60 years	Low (0-1)	91%
	Elevated LDH	Low-Intermediate (2)	81%
	Ann Arbor stage III-IV	High-Intermediate (3)	65%
	ECOG 2–4	High (4-5)	59%
	Extranodal sites >1		
aaIPI	Elevated LDH	Low (0)	98%
	Ann Arbor stage III–IV	Low-Intermediate (1)	98%
	ECOG 2–4	High-Intermediate (2)	92%
		High (3)	75%
R-IPI	(IPI factors)	Very good (0)	94%(4 years OS)
		Good (1–2)	79%
		Poor (3–5)	55%

Abbreviations: OS, overall survival; IPS, International Prognostic Score; IPI, International Prognostic Index; aaIPI, age-adjusted International Prognostic Index; R-IPI, Revised International Prognostic Index; ECOG, Eastern Cooperative Oncology Group

Since treatments and patient outcomes have changed since the development of the IPS-7, it may no longer accurately reflect patient outcome [47]. Notably, the majority of patients used in the development cohort were treated before 1990. Therefore, in 2015, a new simplified version of the IPS-7 was developed[45]. The simplified International Prognostic Score (IPS-3), includes only three of the previously included

seven risk factors, and shows better discriminatory abilities compared to the original IPS-7 [45]. Another study compared use of the IPS-7 and IPS-3 among 1,191 patients with advance-stage cHL diagnosed between 1980–2018 and treated with ABVD or ABVD-like therapy [52]. The results showed that both models had similar prognostic capabilities for modelling OS, with a concordance index (C-index) of 0.63 for the IPS-7 and 0.66 for the IPS-3 [52]. More recently, the Advance-Stage Hodgkin Lymphoma International Prognostic Index (A-HIPI) was developed, which includes risk factors in their continuous forms, and which outperformed both IPS models for predicting OS and PFS [53]. Notably, the A-HIPI was developed in young patients aged 18–65 years, and therefore it shows sub-optimal ability to provide accurate prognoses for elderly patients [54], particularly since acute treatment-related toxicities are a limiting factor for optimal disease control specifically in elderly patients. The A-HIPI was also validated in Study I of this thesis, and the results aligned with the findings of the original study.

Prognostic Models in Diffuse Large B-cell Lymphoma

The International Prognostic index (IPI) was the first widely used prognostic model in DLBCL, and is still in use today. It was developed in 1993 to serve as a prognostic tool for clinicians to predict OS at the time a patient was diagnosed with large cell lymphoma [50]. The IPI risk-stratified patients using five risk factors that are available at the time of diagnosis (see Table 1). Patients are grouped according to the number of present risk factors, with a score of 0–1 indicating low-risk, and a score of 4–5 indicating high-risk [50]. The IPI was developed using a univariate-multivariable approach [50], which first selects features that are independently significantly associated with the outcome of interest. These significant features are then included in multivariable analysis, followed by a step-down regression analysis. The step-down regression-analysis included a total of 12 features: 7 features that were associated with OS in multivariable analysis, and 5 site-specific features that were deemed clinically relevant by experts [50].

Since the publication of the IPI, other risk scores have been developed to improve upon the IPI, particularly after rituximab was implemented as standard first-line treatment in the mid- 2000s [32, 55]. These include the National Comprehensive Cancer Network International Prognostic Index (NCCN-IPI), Revised International Prognos-

tic Index (R-IPI) and the Elderly International Prognostic Index (E-IPI) for use among elderly patients [51, 56, 57]. ML approaches have also been applied to achieve increased predictive accuracy [44]. The decision algorithm includes these risk scores, and helps decide whether a patient should receive 4 or 6 cycles of chemotherapy, or if a patient should receive aggressive or less-aggressive therapy, in accordance with the Danish guidelines for DLBCL treatment (Table 2) [28]. Additionally, the Central Nervous System (CNS)-IPI, is used as an indicator of CNS involvement, contributes to decision-making regarding behind CNS work-up [28, 58]. Moreover, the European Society For Medical Oncology (ESMO) guidelines recommend different treatment regimens according to the age-adjusted International Prognostic Index (aaIPI) [32, 50].

Table 2: Primary treatment protocol based on the clinical guidelines made by the Danish lymphoma group for diffuse large B-cell lymphoma (DLBCL) [28]

Disease group (DLBCL)	**Treatments**
Ann Arbor stage I–II non-bulk aaIPI = 0–1	R-CHOP × 4
Ann Arbor stage III–IV aaIPI = 1	R-CHOP × 6 Pola-R-CHP
All stages No comorbidity Age $\leq$65 years aaIPI = 2–3	R-CHOEP-14 × 6 DA-EPOCH-R Pola-R-CHP
All stages aaIPI = 2–3 Present comorbidity Age 60–65 years	R-CHOP × 6 Pola-R-CHP

Abbreviation: aaIPI, age-adjusted International Prognostic Index.

As the IPI was developed before rituximab was introduced as standard first-line treatment, later tools that includ molecular features have received increased interest, but the IPI remains the preferred risk score. Studies have evaluated models including the dichotomization of continuous variables, and performance with various prognostic scores (IPI, R-IPI, NCCN-IPI) [41, 43]. However, the dichotomization leads to a loss of information, consequently impacting the overall performance of the model compared to the continuous version.

2 Model Development

In the medical literature, many studies aim at determining the causal effect of a treatment or variable on an outcome of interest. In these studies, a causal theory model is built by applying a statistical model to collected data, to test a predetermined hypothesis [59]. In other studies, a model is created to *predict* the outcome of interest as precisely as possible, based on a number of observed variables in new "unseen" data. The models in these two types of studies are usually referred to as explanatory models and predictive models, respectively [59]. The differences between these model types can be depicted by the bias-variance trade-off, which is quantified by the expected prediction error (EPE), and formally factorized as follows:

$$EPE = Error + Bias^2 + Variance \tag{1}$$

In explanatory studies, the goal, in terms of EPE, is to minimize the *Bias* term and thus ensure the most accurate representation of the underlying relationship between covariates and the outcome. Conversely, in predictive studies, the aim is to minimize the sum of the *bias* and *variance* terms, and to thereby design a model that achieves the highest possible prediction accuracy using unseen data. Although this method, achieves high prediction accuracy, it may sacrifice some theoretical or explanatory qualities, as choosing a more biased model results in a reduction of variance that may lead to a more accurate model [59, 60]. Additionally, explanatory and predictive modelling often differ in the measures that are of interest. Explanatory studies are interested in estimating relative measures, such as hazard ratios or odds ratios, while predictive studies focus on predicting the absolute risk or probability of an event [59]. Throughout this thesis, the terms predictors, variables, features, and covariates are used interchangeably.

2.1 Advanced Model Development Within Lymphoma

When developing a predictive model, one of the key issues is feature selection. There are a vast number of methods for feature selection, ranging from expert domain knowledge to data-driven methods. Notably, the A-HIPI was developed using a backward-elimination technique, which is a more iterative approach [53, 60]. The backward-elimination begins by including all features that are deemed relevant, and then re-

moves features, one by one until some criteria are satisfied. The A-HIPI was developed by removing the feature with the largest p value (>0.05), until all remaining features had reached the significance threshold of <0.05. Then this procedure was repeated 200 times using a bootstrap-method to calculate a sample-95% CI [61]. Extended analyses were performed between features and the survival outcomes, to determine any non-linear relationships. The study found that age and absolute lymphocyte count exhibited a non-linear relationship, which and was modelled using piecewise linear splines. The A-HIPI model was developed in a cohort comprising 4,022 patients who were enrolled in eight different clinical phase III trials from 1996–2014 [53]. By using a more advanced method for model development, and retaining variables in their continuous form, the A-HIPI outperforms both the IPS-3 and IPS-7. External validation was performed in 1,431 patients with aHL derived from four registries, and yielded a C-index of 0.73, compared to 0.66 for the IPS-7 and 0.69 for the IPS-3 [53].

In DLBCL, an ML approach called stacking was used to predict OS [44, 62, 63]. The stacking method aims to combine multiple models into a single "super learner", by assigning a weight to each model. More precisely, it fits a variety of models, and then forms a single combined prediction using a weight average approach. The DLBCL stacking model combined a large range of different survival models including Cox proportional hazard, flexible parametric survival (FPS), IPI, and accelerated failure time models and incorporated different numbers and combinations of covariates. To determine the weights associated with each survival model, a nested cross-validation (CV) scheme was applied, which sought to minimize the integrated brier score (IBS) in a left-out dataset [44, 64]. The stacking model was compiled using 2,759 patients derived from the National Danish Lymphoma register (LYFO), and was externally validated in a separate registry-based cohort comprising 2,414 Swedish patients. In the Swedish cohort, the stacking model attained a C-index of 0.74, compared to 0.66 with the IPI and 0.68 with the NCCN-IPI. Thus, the stacking model outperformed previously mentioned models, yielding substantial improvements in prediction accuracy.

One main reason that the A-HIPI and stacking model outperform the IPS-7 and IPI is that they refrained from dichotomizing continuous covariates. The strength of continuous variables is especially evident in the DLBCL cohorts. Notably, a Cox model including the IPI variables (age, stage, performance score, lactate dehydrogenase level

(LDH), and number of extranodal sites) in their continuous form performed very close to the more sophisticated stacking model, achieving a C-index of 0.738, compared to 0.744 for the stacking model. Similarly, in Study I in this thesis, an FPS model including only age and performance score outperformed all other models [1]. Another great advantage of using covariates in their continuous form is that it also allows for more personalized survival predictions. However, interpretation and clinical decision-making based on personalized outcome predictions can be difficult, as clinical practice is mainly based on models that assign patients to risk groups, which have been, previously used in clinical studies, for treatment decisions. One possible solution is to create risk groups using quantiles of the predicted outcome. With this method, we allow the full potential of the input data to take effect, and thereby achieve higher model accuracy before grouping the predicted outcome into the desired risk percentiles.

2.2 Validation

When developing a prognostic model, it is important to consider the model's validity and ability to predict in a separate population.There are essentially two types of validation: internal validation and external validation [60, 65]. Internal validation refer to the internal *validity* within the setting in which a model was developed, determined by assessing the predictive accuracy [65]. When estimating internal validity, the full cohort is split into two sub-populations, called the training set and test set. One classic method involves randomly sampling 70% of the data into the training set, which is used to fit and train a model, and 30% into the test set, which is used to estimate internal validity [60]. Additionally, the training set can be further split into a training and validation set, which is often done using more comprehensive feature selection techniques when overfitting is an issue. The left-out test set is then used to calculate unbiased performance measures, as the model has not "seen" the test data.

Notably, random splitting of the data carries a risk of not accurately capturing the underlying distribution of the full dataset. The distribution of covariates or outcomes may be skewed by chance creating imbalance between, e.g., the outcome in the training set versus the test set. This is not usually an issue in large datasets: however, it can be a problem with small datasets, or when covariates or outcomes are severely imbalanced. For instance, a binary covariate with a 1:100 occurrence has a high chance of not being evenly distributed in both datasets. As an example, in a dataset with 1,000

patients, by chance, 9 of the 10 occurrences could be allocated to the training set (70% of the data) with only 1 occurrence allocated to the test set (30% of the data). Thus, by chance, the random split could create a skewed distribution, thereby increasing variations in performance estimation [65]. To avoid this issue, another popular method is CV (Figure 1), which enables the whole dataset to be used as an independent test set to create an average prediction error [66].

In CV, the dataset is randomly split into a number of evenly sized sub-datasets, called folds (e.g., k-folds), where the smallest fold can comprise a single subject ("leave-one-out") [60]. The method works by iteratively leaving out one fold, to serve as an "unseen" test set, while the remaining (k-1)-folds are then used for model training. The process is repeated k times, until all folds have been used as a test dataset, such that all subjects are used for assessing performance. Then performance can be estimated as the average performance across all iterations, or all left-out predictions can be pooled and performance calculated in a single iteration [67].

The CV procedure often includes an automatic feature-selection algorithm, such that

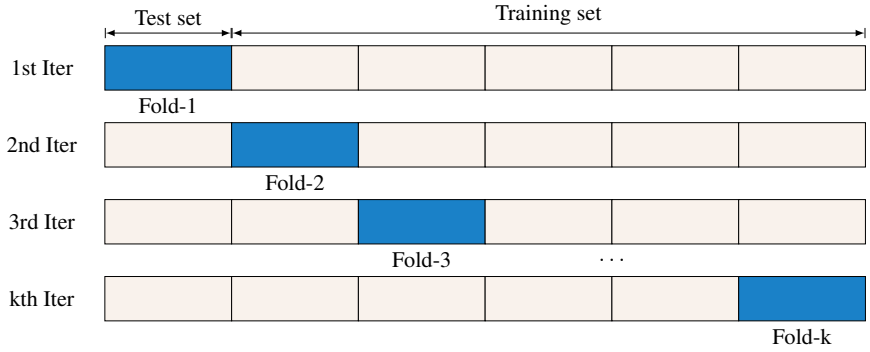


Figure 1: k-fold cross validation schematic

the selected features and effect estimates might not align between each iterations. This means that, the model developed in the first iteration differs from the model developed in the second iteration, as the data included in the training dataset changes between iterations. Therefore, multiple different models including different numbers of covariates and effect estimates are used to predict different left out test sets. This means that no single model is validated, rather it is the method for building the final model (fit on the full dataset) that is validated, meaning that additional external validation is crucial for proper model validation [65, 67].

External validation means testing a model's ability to accurately predict or generalize in a new "related" population of subjects [68]. The external cohort can differ in various manners, but should include some fundamental differences from the development cohort. Including, but not limited to, temporal changes or geographical changes. More diverse external validation provides more information regarding the model's ability to generalize. For example, the A-HIPI was not useful in elderly patients (age > 65), who were not part of the original development cohorts (age 18–65) [54]. In practice, external validation is frequently performed by researchers who have local access to an external dataset, or by directly sharing the predictive model with an external party. Direct sharing of models can allow researchers to potentially circumvent potential legal obstacles, especially when data are shared across borders. However, other issues may arise, as models can indirectly include sensitive patient information [69]. This is usually not an issue for regular parametric models, e.g., logistic regression, since coefficient estimates contain no sensitive information. However, this is not necessarily the case for semi-parametric models, such as the Cox proportional hazard model [70]. The Cox model forms survival predictions using a step-function (cumulative baseline hazard), where each step represents an event time, consequently leaking sensitive information about the outcome for a single individual. One possible solution is to use, extensive anonymization techniques, such as fitting parametric splines to the step function.

2.3 Improving Prognostic Scores

The crude dichotomization of covariates in the IPS-7, IPI, and other prognostic scores, means that the risk-estimates assumes a single fixed-effect increase between each group. For example, with the IPS-7, using a cut-off of 45 years implies that the effect of increasing from 44 to 47 years of age has the same crude effect as increasing from 44 to 95 years, which is highly unlikely. The IPI and IPS-7 were developed using the classic univariate-multivariable method, where features having a p-value < 0.05 in univariate regression analysis, are included in multivariable modelling. Features that retain a significance level of below the 0.05 alpha level in multivariable modelling are further included in the final model fit. This method suffers from the problem of co-linearity between predictors. The issue with co-linearity is that when two correlated predictors which are both significant in univariate analysis, are included in a multivariable regression, this will increase the standard errors on coefficient estimates [71]. The increased uncertainty reduces the accuracy of coefficient estimates [60, 72]. This,

increases the risk of both predictors not reaching statistical significance (especially in smaller datasets), and thus being excluded after multivariable analysis despite containing important information. Moreover, if both predictors were included in the model, accurate interpretation of each predictor would be difficult; therefore careful statistical analysis is warranted [73].

The use of more advanced ML methods holds great potential for improving predictive accuracy, as these methods enable the possibility of modeling non-linearities and combining a multitude of different data types—for example, combining PET/CT images with clinical information—to form predictions [74]. However, ML methods require a large amount of data for training, which can be a limiting factor in clinical research, particularly in rare diseases where data are limited to few subjects. To increase the pool of data, one option is to combine RWD from other registers or countries (e.g., the Nordic countries), which can allow the use of more advanced ML methods. The data used to fit these clinical models comprises of sensitive patient information distributed across a large number of distributors or *sites* (hospitals, countries etc.). Patient records may be vertically distributed, for instance, if one site stores blood samples measurements, and another site stores general patient information, such as age and sex. On the other hand, data can be horizontally distributed, where each site includes full information from patient records, but only for a limited number of patients. With both of these data types a cumbersome legal process involving many institutional bodies is often necessary before a central data merge is possible. This is due to the importance of ensuring patient anonymity and protection against malicious parties, as implemented in the General Data Protection Regulation (GDPR) [75]. Formally, sensitive or micro data are defined as information directly related to an individual, e.g., age, sex, security ID number, debit card information, race, number of children, unemployment benefits received, education, body mass index, outcomes and event-times [76]. This makes data nondisclosure of great importance, as sensitive data could be exploited by malicious users, e.g., to discriminate or blackmail.

As an alternative to central aggregation of these data, studies have proposed the use of methods such as federated learning (FL) for developing predictive models or making inferences [77–79]. FL enables researchers to train models without explicit access to the whole data pool, by iteratively sharing small carefully selected local model updates or likelihood contributions. This, ensures that a subject's data does not leave the

distributor. Whether the data are vertically or horizontally distributed, the idea is to utilize distributed data to fit a model, and to acquire coefficient estimates similar to as if the data were pooled [78]. Formally, FL is a distributed optimization technique that iteratively aggregates local model updates (gradients, Hessians, etc.) to construct a global update, that is applied in some optimization algorithm (Newton-Raphson, gradient decent, etc.) following a new round of iterations, until reaching some convergence criteria [80–82]. The selected convergence criteria often include no larger change in coefficient estimates between iterations $||\hat{\beta}_{p+1} - \hat{\beta}_p|| < \varepsilon$. However, issues may arise because information can be implicitly derived from model updates, e.g., by solving large systems of equations and unknowns, or by using changes in the predicted outcome to make inferences about the input (membership inference attacks) [83]. Malicious parties could exploit these issues raising concerns about patient privacy; However empirical and theoretical evaluations have indicated limited threats to such systems [84–86]. Moreover, these methods often apply neural network models with larger numbers of coefficient estimates than available data points. This is especially an issue when overfitting occurs, as this can lead to large changes between predicted outcomes for two seemingly identical input calls (x_i and $x_i + noise$), indicating the possibility that x_i was used in the training procedure. Additionally, as regular regression models often include a "low" number of covariates, they have limited ability to infer something about the data. Adding a small amount of carefully crafted random noise to local updates has proven effective for preserving privacy [79, 87].

3 Treatment Effect Estimation and Data Design

Since individual lymphoma subtypes are relatively uncommon, using RWD collected from routine clinical setting holds great potential for studies of treatment effectiveness, late toxicities, quality of life, and predictive modelling. However, the use of RWD also increases the risk of bias due to known and unknown confounders. Thus, it is important to consider the use of advanced statistical methods—such as matching, stratification, and multivariable analysis—to minimize the risk of bias in causal inference studies. The following sections introduce, different concepts of data, and their applicability including strengths and weaknesses.

3.1 Randomized-Control Trial Data

An RCT is a highly controlled and selective environment and is often referred to as the golden standard for investigating treatment efficacy [88]. In RCTs, patients are randomly allocated to a specific treatment arm (e.g., Standard therapy vs. experimental therapy) with exact treatment plans, to accurately measure treatment efficacy [89]. The randomization mechanism ensures that confounding variables, both known and unknown, are balanced across treatment arms (Figure 2) and not confounding the treatment efficacy estimates. This protects against bias regarding treatment selection, by removing the path from the confounding variables to the exposure (Figure 4). For example, if an investigator compares one treatment that is primarily given to healthy patients, against another (less toxic) treatment that is administered to less fit patients, the crude comparison induces bias. In this scenario, the apparent treatment effect may be driven only by the patients' health status, which confounds the exposure.

In general, even when adjusting for measurable confounding variables is possible, unmeasured confounding will be present, which cannot be adjusted for. Additionally, combining randomization with blinding can help to avoid additional bias in the selection and allocation of patients in the treatment assignment process [42].

3.2 Real-World Data

There are a number of different definitions for RWD, different data types/sources may be considered RWD depending on the definition used. Classically, RWD is defined

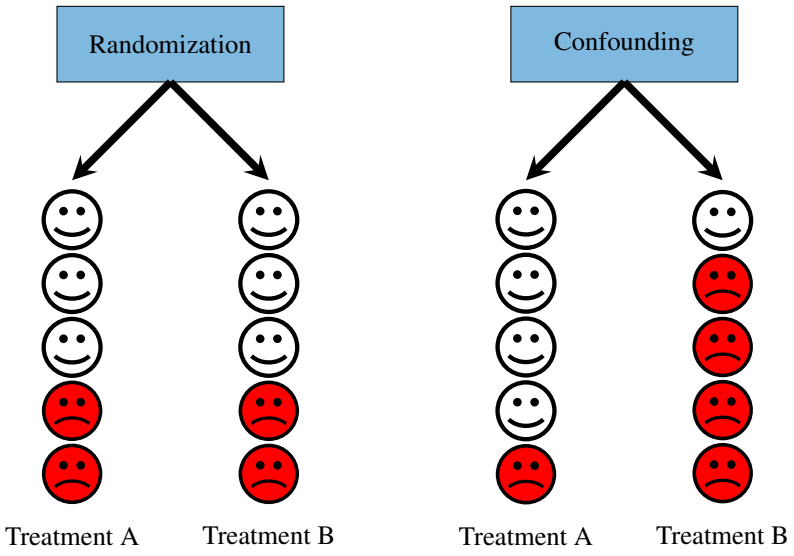


Figure 2: The effect of randomization versus confounding.

as data specifically collected outside of an RCT setting [90]. This definition is quite open, as data from a single-arm trial (SAT) without random treatment allocation would be classified as RWD. A 2017 literature review [91] presented four derived definitions for RWD, as follows:

(Def-1) *Data collected in a non-RCT setting (all health data, except data derived from RCT's)*

(Def-2) *Data collected in a non-interventional/non-controlled setting (data collected without interference with treatment assignment or patient selection)*

(Def-3) *Data collected in a non-experimental setting (a setting with no control of any conditions and no de novo data collection)*

(Def-4) *Other (none of the above)*[91]

The data sources for RWD generation were divided into two categories, as follows:

(A) *RWD derived from data authorities (e.g., registers and administrative data)*

(B) *RWD generated by study design (e.g., observational studies and SAT)*[91]

It is important to consider these varying definitions and sources when discussing the applicability of RWD, since the definition and source can have different implications regarding the bias and limitations (Figure 3).

As in an RCT, patients included in an SAT are also selected using strict inclusion criteria and follows exact treatment plans, but without a randomization process [89, 92]. Therefore, the data derived from an SAT would according to the first RWD definition (Def-1), be classified as RWD. The patient selection poses issues due to the inherent dissimilarities between trial eligible and non-eligible individuals [93], which makes it difficult to translate the efficacy to a broader population outside of trial eligibility criteria. According to the second RWD definition (Def-2), investigators cannot interfere with treatment allocation, but data collection can occur retrospectively or prospective (e.g., health survey before/after treatment initiation). Here, retrospective RWD are data that have already been collected (in medical records, registries, etc.), whereas prospective RWD are newly collected data, such as patient-reported outcome. Applying this definition, all patients (both selective and non-selective) are included but treatment allocation can be confounded, and therefore appropriate statistical adjustments and study design considerations must be considered.

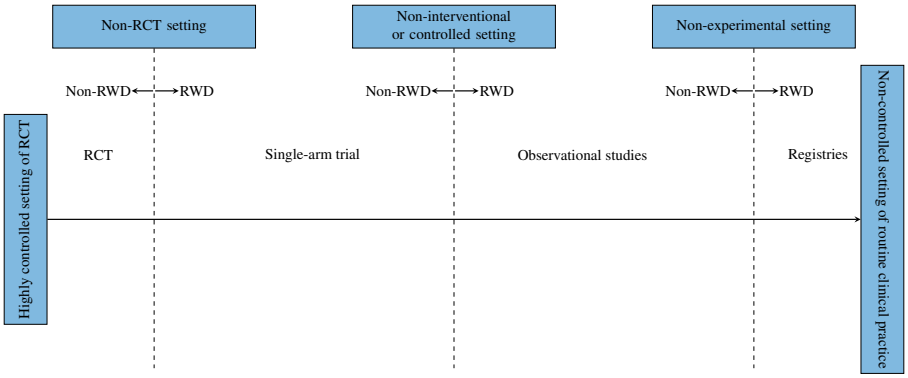


Figure 3: Real-world data (RWD) definition categories, from the controlled setting of an randomized-controlled trial (RCT) to the non-controlled setting of routine clinical practice, adapted from [91].

The Real-World Evidence program from the 2016 Cures Act [94] mandated a new strategic program of the US Food and Drug Administration (FDA) to promote the usage of real-world evidence (RWE) [95]. The US FDA defined RWE (section 21

U.S. Code § 355g(b)) as follows:

"Data regarding the usage, or the potential benefits or risks, of a drug derived from sources other than traditional clinical trials"

Additional guidelines for the use of RWE in decision–making redefined RWE and RWD to better distinguish between data sources (RWD) and evidence (RWE) [96]. Here, RWD is defined as *"data relating to patients' health status and/or the delivery of health-care routinely collected from a variety of sources"* [96]. Consequently, the definition of RWE is changed to *"Clinical evidence about the usage and potential benefits or risks of a medical product derived from analysis of RWD"* [96]. Thus, RWE is used with the objective of evaluating patient populations outside of clinical trials [94, 97]. Notably, RWE derived from RWD has an inherent degree of bias that limits its ability to draw causal inference between exposure and outcome. Therefore, the FDA's recommendation for using RWE in the pool of evidence for assessing the utility of RWE includes the following guidance: carefully made study designs and study protocols (mimicking those of clinical trials) with a pre-made statistical analysis plan (SAP) prior to analyzing RWD, regardless of data collection (retrospective or prospective), to mitigate the potential biases [96, 98]. Similarly, these definitions align with those of the European Medicines Agency (EMA), which formally defined RWD as

"RWD are data that describe patient characteristics (including treatment utilization and outcomes) in routine clinical practice" [99].

This, differs somewhat from the FDA's overall RWD definition, aligning with the following conclusion:

"Persisting variability in stakeholder definitions of RWD may lead to disparities among different stakeholders when discussing RWD use in decision making" [91].

3.3 Real-World Data Versus Clinical Trial Data

As RWE is derived from a broader population, it can potentially answer questions regarding generalizability, expected treatment effect, or overall outcomes among patients [94]. This is especially true in rare diseases, since long survey periods may be required to include a sufficient number of patients and obtain sufficient evidence. Moreover, as RWD has often already been collected (depending on the definition), it

can provide evidence in an inexpensive manner, as it does not require a "real" clinical trial.

Several issues aris when translating the effect of a new drug or intervention outside of trial-eligible patients, since the excluded patients are often those with comorbidities, higher performance score, or older age [92, 100]. Therefore, the observed efficacy during a trial does not always translate to an effect in the a broader patient population [101]. Especially in older and frail patients who are at increased risk of chemotherapy intolerance, postoperative complications and mortality [102]. As stated in Study I:

"From a statistical point of view, it is crucial to include enough patients in an RCT to observe enough events in order to make robust treatment efficacy estimations"[1].

As patients with aHL have more favorable outcomes compared to those with DLBCL, there is a larger numerical difference between the numbers of patients included in trials. For instance, the Alliance/CalGB 50303 trial included 491 patients with DLBCL, who were randomly assigned to receive either R-CHOP or DA-EPOCH-R. At the median follow-up of 5.2 year, 109 deaths were observed across treatment arms [30]. Conversely, the ECHELON-1 trial included 1,334 patients with aHL, and 103 deaths were observed over the entire follow-up period ($\sim$ 8 years) [103]. Due to the more favorable outcome of aHL, nearly three times as many patients were needed to show efficacy, compared to the study of DLBCL. Knowing which patients are more likely to respond poorly to standard treatment could help plan future trials with increased numbers of events and consequently increased study power [1]. Since comparisons between treatment regimens in RCTs are expensive and require long follow-up, it is a valuable option to use already collected RWD.

When conducting any type of study, it is important to consider the type of available data, as data quality can vary. In RCTs, data are specifically collected to answer specific research questions, and are thus of high quality. On the other hand, RWD are collected for different purposes (administration, billing, etc.), which can yield lower quality data. Additionally, systematic errors can occur because data may not be properly registered, neglected, or incomplete [92]. For example, in the LYFO, blood tests (hemoglobin, albumin, etc.) were not consistently registered before 2006; therefore, a substantial number of measurements were missing from that time period compared

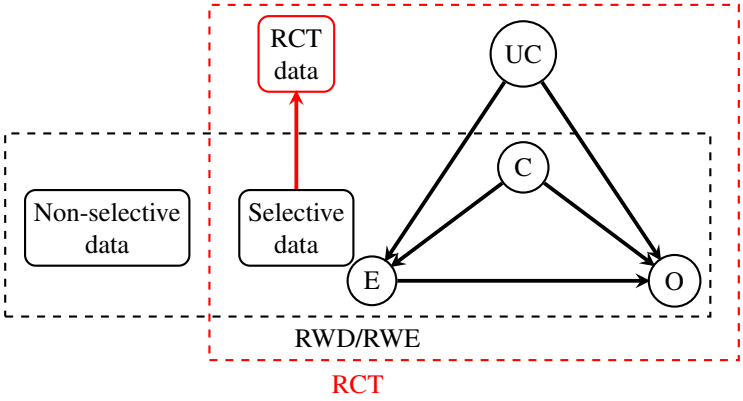


Figure 4: Estimating the effect of the exposure (E) from real-world data (RWD) and randomized-controlled trials (RCT) with measurable confounding (C) variables and unmeasurable confounding (UC) variables. RWE, real-world evidence.

to later periods. Due to this systematic inconsistency, data from before 2006 were not considered in Study I–III.

Epidemiological studies are often designed with the goal of quantifying the effect of an exposure on an outcome of interest in the presence of both measurable and unmeasurable confounders. With RWD, it is possible to adjust for measurable confounders; however, unmeasurable confounders can still bias the results in unknown directions (Figure 4). Moreover, although RWD are often collected retrospectively, prospective analysis is possible, for example, by building a comprehensive SAP before analyzing the data, as recommended by the FDA [98]. When applying a pre-specified SAP, the hypothesis is developed before looking at the data; therefore, the data does not drive the hypothesis, and the chance of, e.g., p-hacking is minimized [104, 105]. Additionally, if the study is properly designed the evidence can help serve as a hypothesis driver for future trials [106].

An SAT can be a valuable option—e.g., for studying rare or advanced-stage diseases, as it still includes scientific rigor, a highly controlled setting, and the possibility of replication [107]. In an SAT, all participants receive the experimental treatment and are prospectively followed over a time period [108]. However, without an official comparison arm, SATs have limited utility compared to RCTs. One option is to use RWD to create a synthetic/historical comparison cohort [109]. Additionally, it is a

possible that RWD may contain a "hidden trial", and that utilizing techniques like, matching may enable emulating of a clinical trial.

3.4 Causality Versus Association

Crudely comparing two treatment regimens may produce misleading results, as there can be underlying systematic differences between the applied treatment regimens in RWD [110]. Causal effect estimates from crude comparisons are often biased. Techniques such as matching or adjustment can be used to minimize bias, and thus improve the causal inference estimates from RWD [111].

In the following example, Z_i is defined as an indicator of the treatment assignment for a subject i ($Z_i = 1$ for treatment, and $Z_i = 0$ otherwise). Within a sample of subjects, the potential outcome is defined as $Y_i(Z_i)$ for each subject i. The difference between the potential outcomes of individual i is then the treatment effect, given by the following:

$$\tau_i = Y_i(1) - Y_i(0). \tag{2}$$

under the *Consistency* assumption, the potential outcome is linked to the observed outcome by $Y_i = (Z_i Y_i(1) + (1 - Z_i)Y_i(0))$ [112]. When estimating the individual treatment effect τ_i, the fundamental issue is that only one of the potential outcomes, $Y_i(1)$ or $Y_i(0)$, can be observed for each subject, and thus it is not feasible to estimate the individual treatment effect in a cohort study [113].

As an alternative, it is more useful to estimate the average treatment effect (ATE) [114], which is a measurement of estimated population treatment effect, given by the following equation:

$$\tau_{ATE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(0)] - \mathbb{E}[Y(1)] \tag{3}$$

Here τ_{ATE} is an estimate of the expected effect on the outcome among the whole population, if everyone was treated versus if everyone was untreated [115]. However, this estimate might not be of interest, as the treatment might not be applicable for some subjects in the whole population. Instead, it may be desirable to estimate the effect of treatment among those for whom the treatment is intended, i.e., the average

treatment effect on the treated (ATT), which is defined as follows:

$$\tau_{ATT} = \mathbb{E}[Y(1) - Y(0)|Z = 1] = \mathbb{E}[Y(1)|Z = 1] - \mathbb{E}[Y(0)|Z = 1] \qquad (4)$$

In an RCT, τ_{ATE} and τ_{ATT} are the same measure, due to the randomization of treatment assignment [114]. In observational studies, this is not apparent, since the treated subjects generally differ from the untreated subjects [110].

There are a number of methods to evaluate causal effect—for example, inverse probability of treatment weighting (IPTW), standardization, and matching. These methods seeks to balance known measurable confounders between treatment arms, by reducing heterogeneity and making the two groups as similar as possible, thereby yielding a more comparable and less biased causal effect estimate. In a randomized experiment or trial, the treatment assignment is random, and thus the potential outcomes are independent of the treatment ($(Y(1), Y(0)) \perp\!\!\!\perp Z$) [116]. This means that the potential outcomes are equally distributed within either treatment arm, and the subjects are considered *exchangeable*. Moreover, if a subject's potential outcome under treatment is equal to the observed outcome, they are deemed *consistent*, i.e., if $Y(Z) = Y$. If the assumptions of *exchangeability* and *consistency* holds, then it is possible to estimate the causal effect [116]. These concepts further elaborated upon in section 6.

4 Data Sources

All three studies (Studies I-III) in the PhD thesis, were conducted using a multitude of both National Danish registers and Nordic registers. Patients were included in these studies according to the criteria presented in each study. Additionally, in the third study (Study III), data were simulated using simulation specifications to mimic registries, as described in the study.

4.1 The Danish Civil Registration System

All persons alive and living in Denmark are assigned a unique 10-digit Civil Personal Register (CPR) number at birth or immigration and registered in the Danish Civil Registration System (CRS) [117–119]. The CRS was established in 1968, and includes individual-level information including name, sex, date of birth, place of birth, citizenship, parents' identity, vital status, place of residence, and spouses [120]. The CPR number is used for many administrative purposes, including appointments at health care facilities, educational services, and financial/tax purposes. The CPR number enables the unique possibility of linkage between Danish registers that capture different data points [118, 121, 122]. Therefore, it allows researchers the opportunity to acquire accurate individual patient-level information with long or complete follow-up.

4.2 Danish Health Authority Registers

The Danish health data authority (DHDA) is an institutional body within the ministry of health established in 2015 and responsible for managing, developing and updating several registers [123, 124]. The following registers managed by DHDA were used for this thesis.

The Danish National Patient Register

The Danish National Patient Register (DNPR) was first established in 1976, and records data regarding all patient admissions. Starting in 1995, data regarding hospital discharges has also been included. The information in the DNPR includes CPR number, sex, age, hospital identification and time of discharge [125, 126]. Disease codes in DNPR are registered according to the International Classification of Disease, 10th Edition (ICD-10) codes. To ensure better and more logical data organization, the DNPR recently underwent modernization from DNPR2 to DNPR3 (LPR2 and LPR3

in Danish), to better accommodate research and statistics [127]. LPR3 was not available before Study II was conducted; therefore, LPR2 was used.

The Danish Pathology Register

In 1997, the Danish Pathology Register (DPR) was established, and pathologists in Denmark were then legally obligated to report pathology data to the Danish National Board of Health and the DPR. The data contained in this register includes CPR number, topography, morphology, type of investigation, description, and diagnosis. Diagnosis is registered using a Danish version of the Systematized Nomenclature of Medicine (SNOMED) [128].

The Danish Cancer Register

The Danish Cancer Register (DCR) was first established in 1943, but reporting to the register was not mandatory until 1987. Since 2004, registration has been performed electronically using the DNPR and DPR. the data in the DCR includes CPR number, cancer type, date of diagnosis, disease stage and histological characteristics [128, 129].

The Danish Cause of Death Register

The Danish Cause of Death Register (DCDR) contains data regarding cause of death, manner of death, date of death, place of death, sex, age at time of death and place of living (municipality level) [130]. In Denmark, it is mandatory for the physician who verified the cause of death to complete and register a death certificate. However, there can be discrepancies that invalidate the cause of death, due to uncertainty, ill-defined conditions and changes in codingm etc. [130].

4.3 Danish Clinical Registries

The Danish Clinical Registries (RKKP) is a Danish organization that administrates about 80 Danish clinical quality databases. RKKP ensures that data is continuously updated, and that registration forms are completed by clinicians within a reasonable time-frame [131].

The Danish Lymphoma Register

The main register used in Studies I-III, is the population-based LYFO [132]. The LYFO contains baseline information regarding Danish patients with lymphoma (18 years of age and older), including clinical features at diagnosis, treatment details, histology, and information about response and relapses. The LYFO was established in 1982 to oversee the quality of lymphoma treatment, but was not implemented nationwide until 2000 [129]. Data in the LYFO are prospectively collected using prespecified registration templates that are completed by hematologists from the various hematological departments in Denmark [129]. Clinicians are required to fill in specific patient information and to ensure high completeness. When data have not been reported DNPR makes requests to the local departments to complete the missing data. A previous Danish study investigated the coverage and data quality of LYFO, and found high completeness across lymphoma types (92–100%), including high predictive values (87–100%) [129]. Moreover, the LYFO is validated against other Danish national registries (i.e., the DCR and DPR) to align the date of diagnosis, histology, and disease stage [129]. Since the LYFO does not contain information regarding other cancers or cardiovascular disease (CVD), such data were retrieved from the DNPR and DCR. Additionally, data regarding comorbidiies were retrieved from the DNPR to estimate the Charlson comorbidity index (CCI) [133–136]. To ensure complete follow-up regarding outcomes, the LYFO was cross-referenced with the Danish Cause of Death Register. The study population for each study were identified from LYFO.

4.4 External registers

Since model validation is a crucial part of developing predictive models, data from Norway and Sweden were used as an external validation cohort. Information regarding Norwegian lymphoma patients with cHL was retrieved from the Cancer Registry of Norway. Patients' clinical information was extracted from the clinical lymphoma database at Oslo University Hospital, which includes about 55% of the lymphoma population in Norway. Information regarding Swedish lymphoma patients was retrieved from the Swedish Lymphoma Register (SLR), and linked with the Swedish Cancer Register to obtain patients' baseline clinical information [137]. Data were extracted from the LymphomaBaSe linkage, which was made accessible by the Clinical Cancer Epidemiology group at the Division of Clinical Epidemiology at Karolinska Institutet. LymphomaBaSe is a matched cohort study comprising of all Swedish pa-

tients with lymphoma diagnosed between 2000–2020. Each lymphoma patient was concurrently matched, based on birth year and sex, to 10 controls from the general population. The Swedish cohort includes information about routinely collected baseline patient characteristics.

4.5 Statistics Denmark

Statistics Denmark is a central authority in charge of collecting, analyzing and publishing statistics in Denmark [138]. All Danish registries, were collected from their respective sources, and transferred to a secure server with remote access hosted by Statistics Denmark. To ensure patient anonymity, data were pseudonymized by changing the CPR-number to a unique PNR-number, such that linkage between registries was still possible. The pseudonymization key is held by Statistics Denmark; therefore, no single researcher can identify an individual using the PNR-number. Only aggregated data can be extracted from Statistics Denmark. Study II was performed at the remote access server made accessible by the Clinical Cancer Research Centre, Aalborg University Hospital.

Aims

The overall aim of the studies included in this thesis was to apply advanced statistical and ML methods to RWD regarding different types of lymphoma, to estimate treatment effects and prognosis. The aims of each individual study were as follows:

Study I: To use advanced ML techniques to increase the accuracy og survival predictions for patients with aHL, compared to existing prognostic models.

Study II: To estimate the treatment effect of R-CHOEP compared to R-CHOP among young patients with high-risk DLBCL.

Study III: To use FL to train survival prediction models, without exposing sensitive patient data, and with particular focus on event times.

Overview of the Thesis

This chapter summarizes, Studies I–III of this thesis, including the key findings, and additional methodological considerations and results that were not presented in the published studies. For Study I, an anonymization technique for a Cox proportional hazard model is described presented. For Study II, the use of IPTW for estimation of ATE and ATT is outlined, with an additional new sensitivity analysis. For Study III, a sensitivity analysis of bandwidth parameters is presented, including a visual of the shared smoothed baseline hazard estimates. Before reading sections 5–7, it is recommended to have read the original manuscripts, as the following section build upon an understanding of Studies I–III.

5 Study I

Machine Learning–Based Survival Prediction Models for Progression-Free and Overall Survival in Advanced-Stage Hodgkin Lymphoma [1]

5.1 Key Findings

In Study I, two ML stacking models were built to predict OS and PFS, using adult patients (aged 18 and above) with aHL, who received curative treatment with ABVD or BEACOPP. The stacking models included a range of different semi-parametric and fully parametric models, and performance was assessed using the C-index, time-varying area under the curve (AUC), and IBS, with additional calibration measures and plots. The models were externally validated in a Nordic cohort comprising Swedish and Norwegian patients, and compared against two prognostic indexes (IPS-7 and IPS-3) and the newly developed A-HIPI. The models outperformed both IPS models, but obtained performance similar to that of the American A-HIPI. Thus, there were limited gains from using an extensive list of clinical variables combined with advanced ML methods. The models were implemented as an online calculator, which is available at `https://lymphomapredictor.rn.dk/`.

5.2 Methodological Considerations and Results

One strength of Study I, was the external validation of the ML stacking model in a separate Scandinavian cohort. The external data (both Swedish and Norwegian) were located in Sweden. Rather than going through the lengthy legal process of acquiring permission to obtain the external data in Denmark, we instead sent the models to Sweden. As the model objects in R contained personally identifiable data—both implicit (e.g., cumulative baseline hazard) and explicit (e.g., a design matrix)—we derived a thorough anonymization process for all models. Notably, this process led to the development and conceptualization of Study III.

Model Anonymity

Formally, the observed event time is $T = \min(C, E)$, where C is the censoring time, E is the event time, and the status indicator $\Delta = \mathbb{1}[E \leq C]$. For each individual $n \in \mathbb{N}$, we have observed covariates and survival outcome (t_i, δ_i, x_i). The Cox proportional hazard model estimated the survival probability of subject i with covariate structure x_i as follows:

$$\hat{S}(t|x_i, \hat{\beta}) = \exp\left(-\int_0^t \hat{h}_0(u)\exp(x_i^\intercal \hat{\beta})du\right) = \hat{S}_0(t)^{\exp(x_i^\intercal \hat{\beta})}, \tag{5}$$

where $\hat{\beta}$ is the coefficient estimates, and $\hat{S}_0(t) = \exp(-\hat{H}_0(t))$ is an estimate of the baseline survival function at time t [139]. A common estimator for the cumulative baseline hazard function is the Breslow estimator [140], calculated as follows:

$$\hat{H}_{0,BR}(t|\hat{\beta}) = \sum_{\substack{i\in\mathcal{D},\\ t_i\leq t}} \left(\frac{1}{\sum_{j\in\mathcal{R}_i}\exp(x_j^\intercal \hat{\beta})}\right). \tag{6}$$

where $\mathcal{R}_i = \{j \in \{1, 2, \ldots, n\} | t_j > t_i\}$ is the risk-set for individuals still at risk at time t_i, and $\mathcal{D} = \{i \in \{1, 2, \ldots, n\} | \delta_i = 1\}$ is the indices for individuals experiencing the event. The Breslow estimator produces a step-function that increases at each event time, thereby exposing individual event times (Figure 5). To ensure that event times were not exposed, a spline model was fitted to the cumulative baseline hazard. Spline-based models require selection of knot placements; therefore, random noise was added to the original event times, and the knot positions were then chosen based

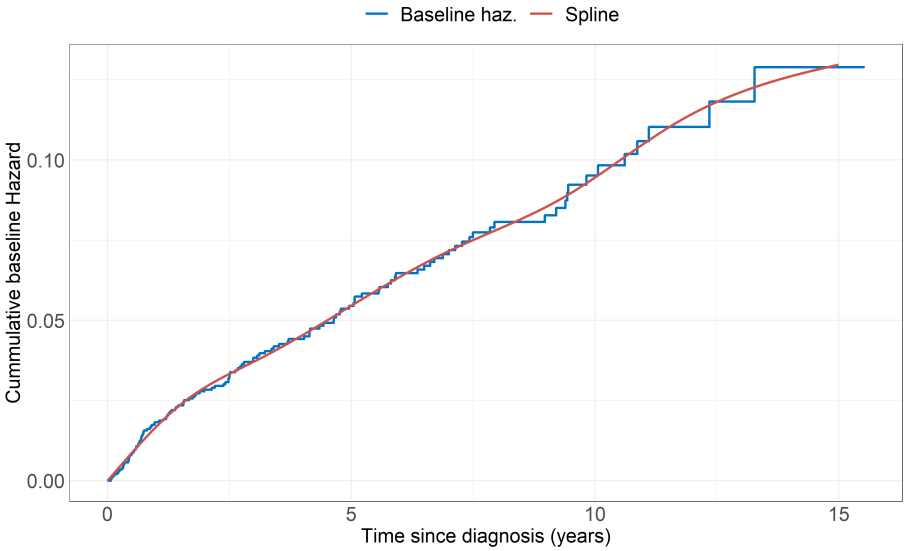


Figure 5: Example of the cumulative baseline hazard with fitted spline

on evenly distributed quantiles. The number of knots was selected using an iterative forward-selection algorithm, wherein the number of knots was increased at each iteration. This was set up using a CV procedure that sought to minimize the root mean square error (RMSE) on a left-out data set, to avoid overfitting [141]. Thereby, the model object comprised the minimum necessary amount of information, i.e., $\hat{\beta}$ coefficient estimates, and a parametric spline function described by knots and coefficients.

5.3 Discussion

Study I utilized the stacking methodology, comprising many different survival models, to provide individual survival and progression prognoses for patients with aHL. The stacking models were built using both parametric and semi-parametric statistical models, although other ML models were initially tested, including neural networks and random survival forest models. Notably, these other models were found to be extremely difficult to anonymize, as it is possible to make inferences about the training dataset by observing the output changes from repeated input queries with slight modifications [83]. Additionally, the random survival forest model bases its survival predictions on Kaplan-Meier estimates for subjects within specific leaf nodes, thus also exposing event times. These models were initially included in the stacking pro-

cedure, but yielded no marginal increases of performance in the internal validation procedure.
Importantly, the the model that achieved the highest performance measures (both internal and external), was a FPS model that included only age and Eastern Cooperative Oncology Group (ECOG) performance status, raising questions regarding the added benefit of more complex models. Rather, a small portion of the utilized clinical features (22 features in total), seemed to constitute the most valuable prognostic information, emphasizing the phrase *garbage in, garbage out* [142]. Notably, it ispossible the the inclusion of genetic information, molecular features, cell of origin, genomics, circulating tumor DNA, and PET-CT data could provide better and more refined input data points, which may improve predictive accuracy beyond what was achieved in the study. Since these types of data can contain vast numbers of data points for single individuals, advanced ML models are required for handling complex non-linearities in data. Such data are rarely available at the time of diagnosis (or in the registries used), but could be applied for in dynamic modeling for continuous updating prognosis during treatment. Moreover, the stacking procedure offer a possibility for combining both the classical statistical models and more advanced ML models to form predictions.

Study I describes the construction of a new ML model that can provide patients with a personalized prognosis based on a range of characteristics. The study demonstrated how big data and ML methods can be combined within the field of hematology, and can help shape the future of clinical decision-making. In particular, these models could be applied in clinical trials to tailor the treatment trajectory, and provide clinicians with useful information for planning patient follow-up.

6 Study II

Effectiveness of R-CHOP versus R-CHOEP for treatment of young patients with high-risk diffuse large B-cell lymphoma: A Danish observational population-based study [2]

6.1 Key Findings

Study II, assessed the effect of adding etoposide to a standard R-CHOP regimen, among young adult patients (aged between 18–60) with high-risk DLBCL (aaIPI ≥ 2) and of relatively good health (ECOG 0–2). To assess the treatment effect among the treated (ATT), a matching scheme were followed, where R-CHOEP-treated patients were matched 1:1 with similar R-CHOP-treated patient without replacement. The treatment effect was quantified based on restricted mean survival (RMS) differences, estimated at 1, 2, and 5 years after diagnosis. The matched cohorts were balanced for most patient characteristics, with the exceptions of B-symptoms and extranodal disease. Unadjusted analysis revealed a difference, in favor of patients receiving R-CHOEP compared to R-CHOP; however, this difference was non-significant, and no difference between treatment arms was observed after matching. Sensitivity analyses were performed with respect to treatment definition and inclusion criteria, and the results were consistent with the primary findings. Secondary analysis of late effects revealed that the addition of etoposide was not associated with increased risk of later CVD or secondary primary malignancies. These findings indicate that the use of more dose-dense therapies should be carefully evaluated at the individual level, given the lack of superior effect and potential for increased toxicity. Before this study was conducted, a through SAP was written and made publicly available at `https://osf.io/dw54j/`.

6.2 Methodological Considerations and Results

In both RCTs and observational studies, there is a balancing score, $b(x)$, which is a function of the baseline covariates x, such that the conditional distribution of x given $b(x)$ is the same for both treated $(Z = 1)$ and untreated $(Z = 0)$, that is $x \perp\!\!\!\perp Z|b(x)$ [110]. If the assumption holds that, *treatment assignment is strongly ignorable* [110] ($Y(Z) \perp\!\!\!\perp Z|x$ and $0 < \mathbb{P}(Z = 1|x) < 1$), then the differences between treated and untreated subjects with the equal balancing score will provide and

unbiased estimate of treatment effect. Consequently, using matching or adjustment to condition the balancing score can produce unbiased estimates of ATT [110, 143], as follows:

$$\tau_{ATT} = \mathbb{E}_{b(x)|Z=1}[\mathbb{E}[Y|b(x), Z=1] - \mathbb{E}[Y|b(x), Z=0]|Z=1] \tag{7}$$

The strong ignorability assumption is also referred to as *unconfoundedness*, which implies that all confounding covariates are observed [113, 144]. Additionally, the assumption of *positivity* ($0 < \mathbb{P}(Z=1|x) < 1$) means that there must be a non-zero probability of being assigned to either treatment arm [116]. In cohort studies, these assumptions do often not hold, due to the inherent presence of unobserved confounding variables. Moreover, the positivity assumption might not hold if the study population is not carefully selected, as the treatment assigning mechanism might be zero for some patients. For example, in Study II, the population was restricted to patients aged 60 years and below, because assigning R-CHOEP to older patients was deemed controversial by co-authors, even though the Danish clinical treatment guidelines suggest its use up to 65 years of age [28].

Propensity score

The "coarsest" balancing score is the propensity score, which is defined as the probability of treatment assignment given observed covariates, x, are give as follows:

$$e = \mathbb{P}(z=1|x).$$

The propensity score can be estimated using a range of modelling techniques, but the most popular is logistic regression [145]. Under the assumption of *conditional exchangeability* ($Y(Z) \perp\!\!\!\perp Z|x$), propensity scores can then be used to determine unbiased estimates of the causal effect [110, 116].

Matching

Matching is a technique that seeks to balance covariate distributions between two groups by "pairing" the treated to the untreated based on a set of shared or similar covariates. For example, a treated male may be matched with an untreated male (exact matching). There are many types of matching techniques, including exact, propensity score matching (PSM), nearest neighbor, etc. In PSM, pairs are formed by matching

treated subjects with untreated subjects having similar propensity scores [110]. Each treated subject is matched to an untreated subject with an equal or similar propensity score, by minimizing all individual distances, $d_{i,j} = |e_i - e_j|$, across all matched subjects. Using propensity scores to match subjects minimizes the imbalance in observed covariate distributions between treated and controls subjects [114]. One issue with PSM is that a matched pair with similar propensity scores ($e_i = e_j$) does not imply equal covariate structure between matched pairs ($x_i \neq x_j$) [111]. On the other hand,, exact matching of the covariates ($x_i = x_j$) does mean that the propensity scores are exactly matched ($e_i = e_j$).

Genetic matching

Genetic matching is a technique in which the generalized Mahalanobis distance is used to determine a weight matrix W, by optimizing a balancing measure using a genetic algorithm [146]. The generalized Mahalanobis distance is defined as follows:

$$d_{i,j} = \Big((x_i - x_j)^T (\Sigma^{-1/2})^T W \Sigma^{-1/2} (x_i - x_j)\Big)^{1/2}, \tag{8}$$

where x_i and x_j are baseline covariates for subjects i and j, respectively; W is a diagonal $p \times p$ positive definite weight matrix; p is the number of covariates; $\Sigma^{1/2}$ is the Cholesky decomposition of the variance-covariance matrix $\Sigma = Cov(x)$; and W represents the weight associated with each covariate [143, 147]. The genetic matching algorithm next seeks to select an appropriate W to maximize the balance of the observed covariates between treated and untreated subjects, through examination of a cumulative probability distribution function, assessed using a non-parametric test statistics [148].

One limitation of Study II was the exclusion of 86 R-CHOEP-treated patients after the matching procedure. These exclusions were caused by the *exact*-matching procedure and the sampling without replacement, as it was not possible to identify matches among the remaining leftover R-CHOP-treated patients. This issue emphasizes the differences between treatment groups, and the complexity of proper comparison. The observational nature of Study II contributed to a likely bias in the crude comparison of the treatment effect between the two treatment groups [149]. Considering the crude comparison as a proper estimation of ATE would provide inaccurate effect estimates, due to the inherent confounding nature present in Study II. To account for

this confounding nature, the use of IPTW is a feasible method for obtaining less biased estimates of ATE, compared to the crude comparison. The key interest is often in the direction of the effect estimates, before and after accounting for the present confounding, as this is a potential indicator of the strength of the confounding effect. Importantly, several sensitivity analysis were performed with respect to treatment definition and inclusion criteria, which are provided in the supplementary material. A natural extension was to include additional sensitivity analysis to test the robustness of the applied matching method.

Inverse Probability of Treatment Weighting

IPTW is another method for assessing treatment effect, which can be used to estimate ATE and ATT without excluding 86 patients [149]. Formally, the weights used for estimating ATE and ATT are given by the following:

$$w_{ATE} = \frac{Z}{e} + \frac{1-Z}{1-e} \tag{9}$$

$$w_{ATT} = Z + \frac{e(1-Z)}{1-e}, \tag{10}$$

where the Z is the treatment assignment indicator, and e is the propensity score [150, 151]. These weights can then be used to "adjust" estimates of OS and PFS for the individual treatment groups [150, 152, 153]. The IPTW Kaplan-Meier estimator for a treatment group is defined as follows:

$$\hat{S}_k(t) = \prod_{t_j \leq t} \left(1 - \frac{d^w_{j,k}}{n_{j,k}}\right) \tag{11}$$

$$d^w_{j,k} = \sum_{i:T_i=t_j} w_i \delta_i \mathbb{1}\left(Z_i = f\right) \tag{12}$$

$$n^w_{j,k} = \sum_{i:T_i \geq t_j} w_i \mathbb{1}\left(Z_i = f\right) \tag{13}$$

where δ_i is an indicator function for a subject i experiencing an event or not, Z_i is the treatment that subject i receives with treatment group indicator f, and w_i is the associated weight for subject i. Depending on the desired treatment effect (ATE or ATT), the weights from Equation (9) and Equation (10) can be substituted in Equation (12) and Equation (13) to produce IPTW Kaplan-Meier survival curves (Figure 6). Additionally, the desired effect estimates can then be calculated by either survival

differences or RMS differences at the desired time-points. Additionally, the 95% CI for the survival difference can be calculated by applying a bootstrap method [61].

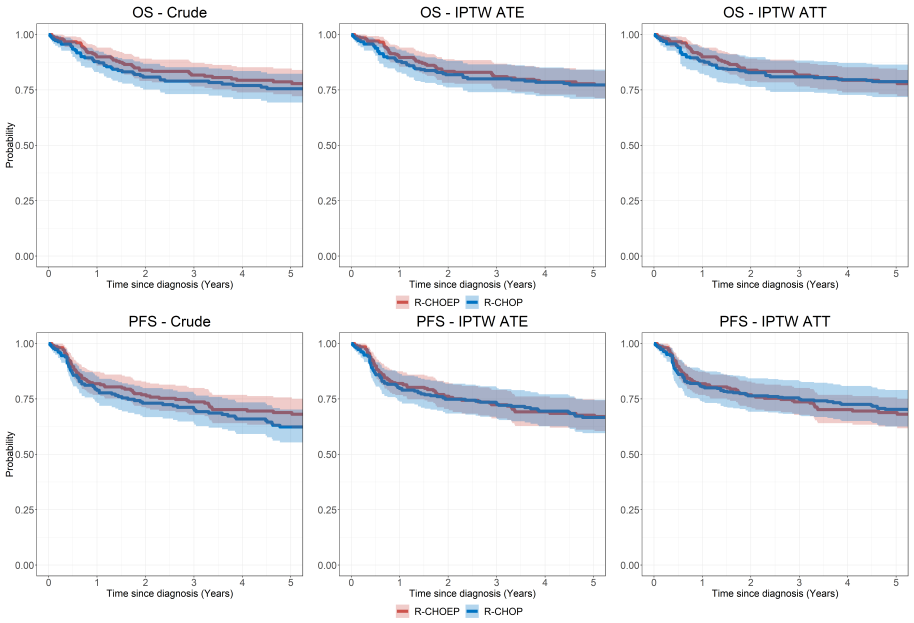


Figure 6: Crude and weighted Kaplan-Meier curves modelling overall survival (OS) and progression-free survival (PFS). IPTW, inverse probability of treatment weighting; ATE, average treatment effect; ATT, average treatment effect on the treated.

6.3 Discussion

In Study II, an inherent concern is the confounding nature between patients receiving the additional etoposide (R-CHOEP), and those who receive standard R-CHOP. To minimize the bias regarding treatment selection, strict inclusion criteria were applied, combined with matching for accurate comparison when addressing treatment effectiveness. Despite the application of a matching scheme, patients receiving R-CHOEP still presented with additional extranodal involvement and B-symptoms compared to patients receiving R-CHOP. This imbalance in covariate distribution after matching introduces some residual confounding, highlighting the difficulty of acquiring sufficiently tight matches. Additionally, it was not possible to obtain information regarding important biological features (MYC, TP53 mutation BCL-2 FISH, or cell of origin), and imbalances of these features could lead to additional residual confounding which

imposed a limitation on the presented study.

Due to the differences between treatment groups, the main aim of the study was to assess ATT rather than ATE, since ATE measures the average effect. Formally, ATE answers the question *"What is the average difference in outcome between those treated with standard R-CHOP and those treated with the additional etoposide?"*. This average effect can be quite misleading when the study includes patients who cannot tolerate the additional etoposide. On the other hand, the ATT estimate answers the question *"Of those treated with etoposide, what would have happened if etoposide were left out?"*. Therefore, estimating ATT was more useful for addressing the effect of etoposide. Coincidentally, the use of IPTW for measuring both ATE and ATT resulted in almost identical survival curves (Figure 6) for both treatment arms in both estimates. Moreover, as the study included young patients ($\leq$60 years), it is likely that a large portion of the patients would have been able to tolerate the addition of etoposide. Thus, the effect estimates for ATE and ATT, seems to align with only minor differences, although the overall conclusion remains.

7 Study III

Development of survival prediction models using federated learning

7.1 Key Findings

In Study III, new methods were developed for utilizing FL to train survival prediction models in a setting where data are distributed across sites, specifically without sharing event times. The method comprised iterative aggregation of site-specific gradients and Hessians, using a Newton-Raphson step for coefficient estimation in a Cox model and a parametric Poisson model. To enable survival prediction for the Cox model, an additional kernel-smoothing algorithm was derived, which aggregated the local baseline hazard across sites. To test the proposed methods, four experiments were conducted using simulated data, as well as an RWD experiment using the aHL dataset from Study I. The FL models were compared to similar non-distributed models fitted on a pooled version of the data used for training, to simulate the case in which data were pooled at a single site. In the four simulated experiments, each model was compared to the true underlying model. For the RWD experiment, the FL models were compared to the pooled version. The results showed that the FL models achieved performance similar to if the data were pooled at a single site, and the kernel-smoothing algorithm was successful in modeling the baseline hazard functions, without exposing event times. These results indicate that it is possible to train survival prediction models in a distributed set up, without explicitly sharing event times, and with minimal loss of predictive accuracy, compared to the pooled model versions.

7.2 Methodological Considerations and Results

In study III, a kernel smoother was applied to the estimates for the baseline hazard function. This was done to ensure that exact event times were not directly exposed when shared across sites. However, the selection of an appropriate bandwidth parameter in the kernel estimator could potentially indirectly expose event times if not carefully selected.

Bandwidth Selection

In the Ramlau-Hansen estimator Equation (14), the bandwidth parameter b is non-negative and determines the degree to which event times further away from t con-

tribute to the new estimates, i.e.., the kernel smoother is given as follows:

$$\hat{h}_{0,RH}(t|\hat{\beta}) = \frac{1}{b}\sum_{i\in\mathcal{D}} \mathcal{K}\left(\frac{t-t_i}{b}\right)\frac{1}{\sum_{j\in\mathcal{R}_i}\exp(x_j^\intercal\hat{\beta})}, \tag{14}$$

where $\mathcal{K}(u) = \frac{1}{\sqrt{2\pi}}\exp(-\frac{1}{2}u^2)$, t_i is the event time for subject i, and t is the desired time-point for prediction, e.g., 1 year or a grid of time-points [154]. A small value of b, indicates that further away event times contribute less, and vice versa for large values of b. However, as the distance between event times increases, the contribution from other event times diminishes, thereby increasing the likelihood of exposing isolated event times. Moreover, if the contribution from other event times is $\sim$0, then the kernel-smoother would create a local maxima at the exact event time. In figure 7, to illustrate the effect of the bandwidth in the kernel smoother, two different bandwidths are chosen ($b = 10$ and $b = 25$), and the estimates for the baseline hazards function are illustrated.

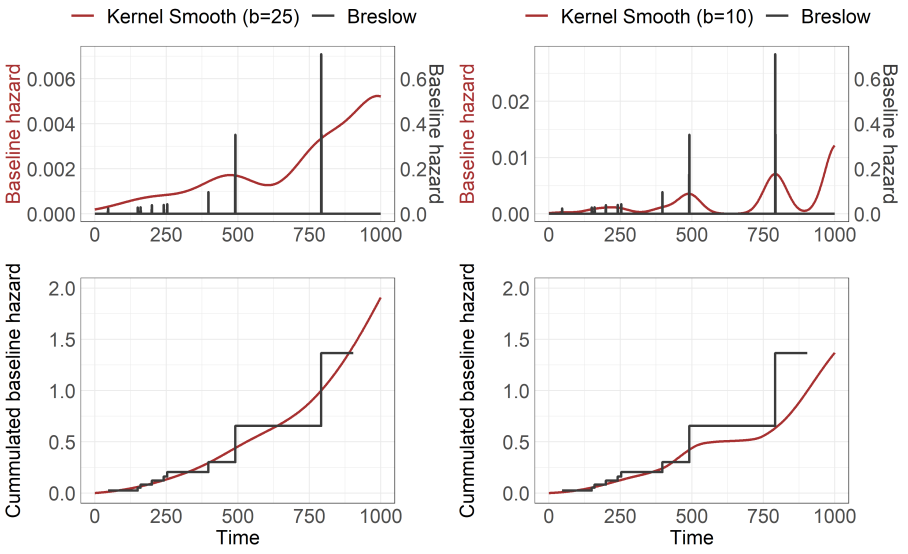


Figure 7: Baseline hazard and cumulated baseline hazard before and after kernel smoothing

When observing the baseline hazard function estimates at $b = 10$, it is apparent that local maxima peaks occur at the exact event time if no other event times are within a certain proximity. This becomes less apparent as b increases, which also seems to align on the cumulative scale. Therefore, careful selection of the bandwidth parameter is

necessary, as it is possible to infer event times based on local maxima in the smoothed version.

7.3 Discussion

Study III presented FL methods for developing survival prediction models without direct exposure of event times, as data non-disclosure is important to protect an individual's right to privacy. Formally, sensitive or micro data are defined as information or records that directly relate to an individual [76]. Consequently, two types of data disclosures are possible, *identity* and *attribute* [155]. An attribute disclosure occurs when it is possible to make an inference about an individual through some combination of variables. On the other hand, an identity disclosure is when an individual is directly identifiable from the data (e.g., names or very old age). It is necessary to balance the utility of health data against the risk of disclosing individuals' data [155]. For example, in an RCT, the utility of two Kaplan-Meier curves (arm A vs. arm B) significantly outweighs the risk of exposing the individual event-times. Conversely, exposing all participants' names (e.g., in an acknowledgments statement) had little utility and poses a high risk for the individual.

In the present study, we weighted the individuals' risk from disclosing event times as higher than the utility of obtaining slightly more accurate coefficient estimates. However, for addressing causal inference questions, precise coefficients estimates are crucial, since explanatory studies primarily focus on minimizing bias terms in the *Bias-variance* Equation (1) to obtain accurate representation of the causal effect estimates [59]. Extending the proposed methods for estimating a causal effect would be a natural continuation, e.g., by building a FL-propensity score model and applying IPTW or matching with the estimated propensity scores, without centralizing data. As the likelihood for a logistic regression is separable, the coefficient estimates would equal a centralized version [156]. Although, the present study focused on not exposing event times, the associated risks are limited, and the utility of such data for malicious users is questionable, as event times in studies of lymphoma would sometimes describe a diseased individual. However, other end-points, such as progression or late effects, may pose a larger treat to the individual, as this information may reveals the state of mind of some patients. In conclusion, the third study of this thesis derived FL methods for training survival prediction models, without the need to directly expose event times, and with a minimal decrease of predictive accuracy.

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Machine Learning–Based Survival Prediction Models for Progression-Free and Overall Survival in Advanced-Stage Hodgkin Lymphoma

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ABSTRACT

PURPOSE Patients diagnosed with advanced-stage Hodgkin lymphoma (aHL) have historically been risk-stratified using the International Prognostic Score (IPS). This study investigated if a machine learning (ML) approach could outperform existing models when it comes to predicting overall survival (OS) and progression-free survival (PFS).

PATIENTS AND METHODS This study used patient data from the Danish National Lymphoma Register for model development (development cohort). The ML model was developed using stacking, which combines several predictive survival models (Cox proportional hazard, flexible parametric model, IPS, principal component, penalized regression) into a single model, and was compared with two versions of IPS (IPS-3 and IPS-7) and the newly developed aHL international prognostic index (A-HIPI). Internal model validation was performed using nested cross-validation, and external validation was performed using patient data from the Swedish Lymphoma Register and Cancer Registry of Norway (validation cohort).

RESULTS In total, 707 and 760 patients with aHL were included in the development and validation cohorts, respectively. Examining model performance for OS in the development cohort, the concordance index (C-index) for the ML model, IPS-7, IPS-3, and A-HIPI was found to be 0.789, 0.608, 0.650, and 0.768, respectively. The corresponding estimates in the validation cohort were 0.749, 0.700, 0.663, and 0.741. For PFS, the ML model achieved the highest C-index in both cohorts (0.665 in the development cohort and 0.691 in the validation cohort). The time-varying AUCs for both the ML model and the A-HIPI were consistently higher in both cohorts compared with the IPS models within the first 5 years after diagnosis.

CONCLUSION The new prognostic model for aHL on the basis of ML techniques demonstrated a substantial improvement compared with the IPS models, but yielded a limited improvement in predictive performance compared with the A-HIPI.

ACCOMPANYING CONTENT

Data Supplement

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INTRODUCTION

Hodgkin lymphoma (HL) is among the most common malignancies in younger individuals and is historically divided into classical HL (cHL; approximately 95%) and nodular lymphocyte–predominant HL (approximately 5%).[1,2] In cHL, cure rates and outcomes across disease stages are generally excellent because of combined modality treatment for limited-stage disease and effective multiagent chemotherapy regimens for advanced-stage disease.[3] Despite these advancements, outcomes still remain variable and 10%-20% of patients will relapse or have refractory disease.[1] A risk-adapted treatment strategy requires accurate prognostic scores on the basis of factors that are available at diagnosis to select the best treatment for an individual patient with HL. With accurate prognostic scores, clinicians can reserve the most intensive and toxic treatment for high-risk patients who need it and spare low-risk patients from unnecessary toxicity. Hence, precise and updated prognostic scores are important for optimal treatment selection for an individual

CONTEXT

Key Objective

To build machine learning (ML) models that predict overall survival and progression-free survival in patients with advanced-stage Hodgkin lymphoma (aHL) and validate it against the International Prognostic Score (IPS) and the newly developed aHL international prognostic index (A-HIPI).

Knowledge Generated

The ML models were able to outperform the IPS in both internal and external validation but yielded limited improvement in predictive performance compared with the A-HIPI. The ML model and the A-HIPI were able to identify high-risk patients with worse survival than those defined as high-risk by the IPS.

Relevance *(J.L. Warner)*

ML-based prognostic algorithms could be implemented at the point-of-care. While in this case the ML model only slightly outperformed the newly developed A-HIPI, it is nevertheless a proof-of-principle for the approach.*

*Relevance section written by *JCO* Editor-in-Chief Jeremy L. Warner, MD, MS, FAMIA, FASCO.

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patient. The International Prognostic Score (IPS-7) is a commonly used risk-stratification score for advanced-stage HL (aHL).[4,5] The IPS-7 consists of seven clinical risk factors and identifies six risk groups. The IPS-3 model (including only age, Ann Arbor stage, and hemoglobin level) was introduced to simplify the IPS-7 while preserving predictive accuracy.[5,6] To simplify the use of IPS-7 and IPS-3, these have been constructed by dichotomizing important prognostic variables and thus provide a risk classification for individual patients. However, the dichotomization of the underlying variables used in both IPS-7 and IPS-3 leads to substantial loss of information, which can result in reduced accuracy of the provided prognostic information.[7] The aHL international prognostic index (A-HIPI)[8] was recently developed using Cox proportional hazards (CPH) models without dichotomizing continuous or categorical clinical factors and indeed demonstrated improvements in predictive capability compared with the IPS models. In the current study, new prognostic models for aHL were developed using machine learning (ML) and an extended list of baseline clinical factors with the aim of further improving predictive performance, which has shown great utility in previous studies for other diseases.[9,10]

PATIENTS AND METHODS

Patients

The ML models were developed using clinical data from the nationwide Danish National Lymphoma Register (LYFO). LYFO contains information on >25,000 patients with lymphoma, including subtype, diagnostic characteristics, laboratory values, treatment, and outcomes.[11,12] Adult patients (18 years and older at diagnosis) who fulfilled the following criteria were included in the study: (1) newly diagnosed cHL, (2) advanced-stage disease defined by Ann Arbor stage III-IV or IIB with extranodal disease,[1] and (3) receipt of curative-intent treatment with BEACOPP (bleomycin, etoposide, doxorubicin hydrochloride, cyclophosphamide, vincristine, procarbazine, and prednisone) or ABVD (doxorubicin, bleomycin, vinblastine, and dacarbazine).

Patients diagnosed between 2006 and 2020 were included from LYFO (referred to as the development cohort). The model was validated using data from the Cancer Registry of Norway and the Swedish Lymphoma Register[13] (SLR). Clinical information for Norwegian patients was extracted from the clinical lymphoma database at Oslo University Hospital (CLDOUH). Validation studies have shown high coverage of the Swedish SLR, the cancer registry of Norway, and Danish LYFO registers.[11,14,15] Patients recorded in the Swedish and Norwegian registers from 2007 until 2020 and 2017, respectively, who met the abovementioned inclusion criteria were pooled to form the validation cohort. Patients with missing information on >2 of the 22 clinical factors included in the models were excluded from the study.

Model Development

Overall survival (OS) was defined as the time from HL diagnosis until death or censoring, and progression-free survival (PFS) was defined as the time from HL diagnosis until progression, relapse, death, or censoring. Censoring occurred on the last date of follow-up or December 31, 2020, in the Danish cohort, December 31, 2020, in the Swedish cohort, and December 31, 2017, in the Norwegian cohort. For the validation cohort, exact relapse and progression dates were not complete for all patients. For these, progression dates were imputed using the last date of first-line treatment and relapse dates were imputed using the start date of second-line treatment.

The ML models for predicting OS and PFS were built using an ensemble method called stacking.[16,17] The stacking method combines a number of survival models into a single survival model through a weighted average approach:

$$S_{\text{Stack}}(t) = \sum_{j=1}^{m} \alpha_j S_j(t), \quad \text{where } \sum_{j=1}^{m} \alpha_j = 1 \text{ and } \alpha_j \in [0,1]$$

where $S_j(t)$ represents the survival probability for the jth model and m is the number of models included. The stacking weights, α_j, were determined by minimizing the integrated brier score (IBS)[18] obtained through cross-validation (CV; Data Supplement, Fig S1).

The following survival models were included in the stacking procedure: a CPH model, penalized CPH model,[19] principal component[20] CPH model, IPS-7, IPS-3, and flexible parametric survival (FPS) models.[21] The models included an extended list of clinical factors, depending on the model type (Data Supplement, Table S1). The survival models incorporated in the ML models, their input, and model abbreviation are described in Table 1. The models were selected to include a broad range of different survival models, including models with time-varying covariate effects.[22]

The initial covariate selection was done by aligning data availability in all cohorts, and only covariates present in all cohorts were included (Data Supplement, Section A4). Missing data on predictors were handled using single value imputation before model development. Further details about the imputation process and methods used for this purpose can be found in the Data Supplement (Section A5).

Model Performance and Validation

Performance measures used were the IBS, time-varying AUC,[23,24] and the concordance index (C-index).[25,26] These measures were evaluated for the first 5 years after diagnosis. Performance measures for the development cohort were calculated by pooling predicted probability estimates in a 10-fold CV setup (Data Supplement, Fig S1). A nested CV setup was applied to avoid overly optimistic performance measures.[27]

To evaluate the ability of the ML models and A-HIPI to identify patients with poor outcomes, patients were ranked on the basis of their predicted 5-year OS and PFS. The IPS high-risk group (IPS score of 5-7) was then compared with the patients with the lowest predicted probability at 5 years (ML high-risk, and A-HIPI high-risk). The probability threshold for the new high-risk group was defined such that the ML high-risk group and the IPS high-risk group included the same number of patients. Outcomes of the two high-risk groups were compared visually using the Kaplan-Meier method. A similar analysis was performed for patients at low risk.

Model calibration was assessed using calibration plots, calibration slope (CS), and calibration-in-the-large (CITL)[28] and was performed in the validation cohort at 5 years postdiagnosis. To investigate the validation of the A-HIPI further, a separate calibration analysis was performed for young patients (age 18-65 years). All analyses were performed using R version 4.1.1 (R Foundation for Statistical Computing, Vienna, Austria; Data Supplement, Table S2).

The study was approved by the North Denmark Region (ID: 2021-002), the Swedish Regional ethics committee in Stockholm (Dnr 2019-00242), and the Regional Committee for Medical Research Ethics South East Norway (No. 2018/2209).

RESULTS

Among a total of 4,582 patients with cHL, 3,115 were excluded (Fig 1) and 1,467 patients were included (707 in the

TABLE 1. The Prediction Models Incorporated in the ML Model

Model	Type	Variable Subset	Notes
CPH-Enet	CPH model with elastic net regularization	1	CV was applied to estimate regularization parameters
CPH-LASSO	CPH model with LASSO regularization	1	Same as CPH-Enet
CPH-LASSO-Int	CPH-LASSO including all possible first-order interaction terms	1	Same as CPH-Enet
PC-CPH	CPH model with PC vectors as predictors	1	A PC rotation was applied to all covariates
FPS	FPS models with time-varying covariates	2	Coefficient estimates for the baseline and time-varying coefficients were modeled by natural cubic splines with 2 degrees of freedom
CPH-IPS	CPH model with IPS covariates	3	IPS covariates were used in their continuous form
IPS-7	IPS risk group Kaplan-Meier estimates	3	
IPS-3	IPS risk group Kaplan-Meier estimates with 3 covariates	4	
FPS-age-PS	FPS model with age and PS as time-varying covariates	5	Same as for the FPS model

Abbreviations: CPH, Cox proportional hazard; CV, cross-validation; Enet, elastic net; FPS, flexible parametric survival; Int, interactions; IPS, International Prognostic Score; LASSO, least absolute shrinkage and selection operator; ML, machine learning; PC, principal component; PS, performance status.

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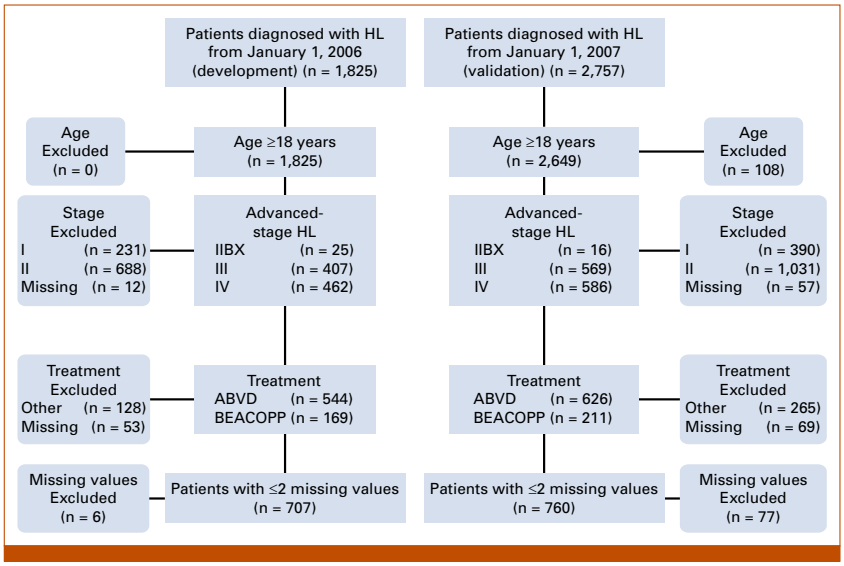


FIG 1. CONSORT diagram of patient flow in the development and validation cohorts. ABVD, doxorubicin, bleomycin, vinblastine, and dacarbazine; BEACOPP, bleomycin, etoposide, doxorubicin hydrochloride, cyclophosphamide, vincristine, procarbazine, and prednisone; HL, Hodgkin lymphoma; IIBX, Ann Arbor stage II with B-symptoms and extranodal involvement.

development cohort and 760 in the validation cohort). The median follow-up (reverse Kaplan-Meier method[29]) was 7.2 years in the development cohort. The 5-year OS was 83% (95% CI, 81 to 87), and the 5-year PFS was 73% (95% CI, 70 to 77; Data Supplement, Fig S2A). For the validation cohort, the median follow-up was 8.6 years. The 5-year OS was 91% (95% CI, 88 to 93), and the PFS was 87% (95% CI, 85 to 90; Data Supplement, Fig S2B). Clinical characteristics in the cohorts were similar except for age (Table 2 and Data Supplement, Table S3). The weights in the ML models showed that a small number of models obtained non-negligible weights ($\alpha_j > 0.001$, Data Supplement, Table S4).

Validation

IPS Versus ML

For OS in the development cohort, the ML model, IPS-7, and IPS-3 achieved a C-index of 0.789, 0.608, and 0.650, respectively. In the validation cohort, the corresponding C-index was 0.749, 0.700, and 0.663, respectively. For PFS, the C-index in the development cohort was 0.665 for the ML model, 0.549 for the IPS-7, and 0.576 for the IPS-3. The corresponding estimates in the validation cohort were 0.691 for the ML model, 0.672 for the IPS-7, and 0.640 for the IPS-3. The ML model attained a lower IBS for OS in both the development and validation cohorts compared with the IPS-7 and IPS-3 except for PFS in the validation cohort (Table 3). For both OS and PFS, the time-varying AUC in the development cohort was consistently higher for the ML model compared with the IPS-7 and IPS-3 (Fig 2A). Similar results were seen in the validation cohort (Fig 2B).

Patients classified as high risk (13.9% of patients) in the development cohort, according to the ML model, had a 5-year OS of 51% (95% CI, 41 to 63), whereas the 5-year OS was 70% (95% CI, 61 to 80) in patients defined as high risk by the IPS-7. The 5-year PFS was 50% (95% CI, 40 to 62) for the ML high-risk group and 58% (95% CI, 49 to 69) for the IPS-7 high-risk group (Fig 3A). In the validation cohort, patients in the ML high-risk group (9.2% of patients) had a 5-year OS of 70% (95% CI, 59 to 81), whereas patients defined as high risk by the IPS-7 had a 5-year OS of 74% (95% CI, 64 to 86). The corresponding estimates for PFS were 65% (95% CI, 55 to 78) and 69% (95% CI, 60 to 82), respectively (Fig 3B). Patients classified as low risk in both cohorts, according to the ML model, had similar OS and PFS as the IPS-7 (Data Supplement, Figs S3A and S3B).

A-HIPI Versus ML

In the development cohort, the A-HIPI achieved a C-index of 0.768 for OS and 0.649 for PFS. The corresponding estimates for the validation cohort were 0.741 for OS and 0.677 for PFS. The time-varying AUC was similar to the ML model for OS and PFS in the development cohort (Fig 2A). For the validation cohort, the time-varying AUC estimates for A-HIPI were similar for OS but attained similar estimates as the IPS-7 for PFS (Fig 2B). The CITL and CS for the A-HIPI in the

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TABLE 2. Characteristics of Patients With Advanced Hodgkin Lymphoma in the Development and Validation Cohort

Clinical Factor	Development Cohort (n = 707)	Validation Cohort (n = 760)
Age, median (range)	44 (18-87)	37 (18-74)
Sex, No. (%)		
Female	292 (41.3)	295 (38.8)
Male	415 (58.7)	465 (61.2)
Ann Arbor, No. (%)		
IIB	20 (2.8)	10 (1.3)
III	324 (45.8)	360 (47.4)
IV	363 (51.3)	390 (51.3)
PS, No. (%)		
0 and missing[a]	466 (65.9)	484 (63.7)
1	187 (26.4)	217 (28.6)
2	27 (3.8)	42 (5.5)
3	18 (2.5)	14 (1.8)
4	9 (1.3)	3 (0.4)
B-symptoms, No. (%)		
No	234 (33.1)	291 (38.3)
Yes	462 (65.3)	460 (60.5)
Missing	11 (1.6)	9 (1.2)
Nodal, median (range)	4 (0-11)	4 (0-15)
Extranodal, median (range)	1 (0-5)	0 (0-4)
Tumor diameter, No. (%)		
<10 cm	505 (71.4)	564 (74.2)
≥10 cm	87 (12.3)	124 (16.3)
Missing	115 (16.3)	72 (9.5)
Treatment, No. (%)		
ABVD	538 (76.1)	566 (74.5)
BEACOPP	169 (23.9)	194 (25.5)
Radiotherapy, No. (%)		
No	621 (87.8)	676 (88.9)
Yes	86 (12.2)	84 (11.1)

Abbreviations: ABVD, doxorubicin, bleomycin, vinblastine, and dacarbazine; BEACOPP, bleomycin, etoposide, doxorubicin hydrochloride, cyclophosphamide, vincristine, procarbazine, and prednisone; PS, WHO performance status.
[a]Missing values are pooled together with PS 0 because of data sharing regulations. In total, there were 21 missing values for PS.

TABLE 3. Performance Measures for the ML Model, IPS-3, IPS-7, and A-HIPI in the Development and Validation Cohorts

	Development Cohort				Validation Cohort			
	OS		PFS		OS		PFS	
Model	IBS	C-Index	IBS	C-Index	IBS	C-Index	IBS	C-Index
ML	0.073	0.789	0.141	0.665	0.049	0.749	0.085	0.691
IPS-3	0.085	0.650	0.149	0.576	0.054	0.663	0.078	0.64
IPS-7	0.087	0.608	0.151	0.549	0.052	0.700	0.077	0.672
A-HIPI	0.076	0.768	0.142	0.649	0.052	0.741	0.091	0.677

Abbreviations: A-HIPI, advanced-stage Hodgkin lymphoma international prognostic index; C-index, concordance index; IBS, integrated brier score; IPS, International Prognostic Score; ML, machine learning; OS, overall survival; PFS, progression-free survival.

development cohort were −0.14 and 0.99, respectively, for OS and 0.09 and 1.22, respectively, for PFS (Data Supplement, Fig S4). In the validation cohort, the ML model for OS attained CITL and CS values of 0.14 and 1.02, respectively, whereas the A-HIPI for OS attained CITL and CS values of 0.27 and 0.88, respectively. The ML model for PFS attained a CITL of 0.86 and a CS of 1.30, whereas the CITL and CS for the A-HIPI model for PFS were 0.89 and 1.61, respectively (Data Supplement, Figs S5 and S6). Restricting to patients age 18-65 years results in both cohorts remaining largely unchanged (Data Supplement, Figs S7 and S8). No substantial difference was found for patients classified as low-risk compared with the IPS-7 (Data Supplement, Figs S3A and S3B).

DISCUSSION

In the present study, we developed and validated a new prognostic model for patients with aHL using ML models that included more details on clinical factors than the conventional risk scores used for HL. The model outcompeted the conventional IPS-7 and IPS-3 in the development and validation cohorts with respect to both OS and PFS predictions. The ML model identified a high-risk group with a worse OS than the high-risk group identified through IPS-7. The improved performance of the ML model compared with IPS-7 and IPS-3 is not unexpected as the ML model uses a more flexible modeling approach and avoids dichotomization of clinical factors, thereby leading to more accurate prediction of outcomes. However, the ML models developed in this study exhibited only a minor improvement in performance compared with the A-HIPI.

Prognostic tools, such as the proposed model, have the potential to assist clinicians in patient counseling and the development of future trials for improved risk-adapted treatment approaches. The use of prognostic tools allows clinicians to identify individuals with low predicted survival probabilities who may not respond well to standard treatments and could benefit from novel treatment strategies. Conversely, for those with the best outcomes, trials may aim at treatment reduction to minimize the risk of long-term side effects. With the evolvement of new therapeutic options, precise risk is crucial, particularly for those with the highest risk. Thus, the introduction of new treatments that substantially change disease outcomes provides a need for updating and revalidating the existing prognostic models. This will also be the case for the ML model and the A-HIPI model as new treatments for aHL are introduced, particularly, as immune checkpoint inhibitors (anti-PD1) are made available in the first-line setting.

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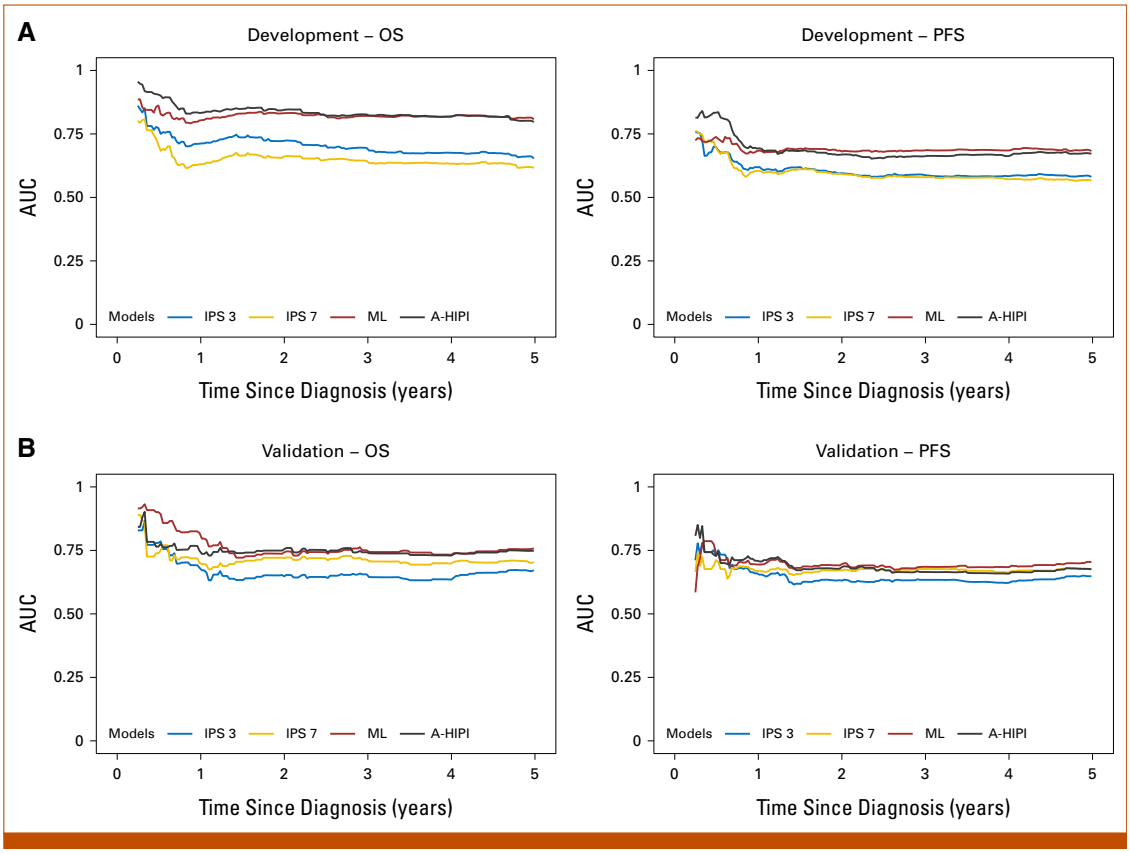


FIG 2. (A) Time-varying AUC for predicting OS and PFS for the ML model, the IPS-7, IPS-3, and A-HIPI. For the development cohort, the calculations for the ML model were performed by pooling predictions from each validation set in the cross-validation scheme. (B) Time-varying AUC for predicting OS and PFS for the ML model, the IPS-7, IPS-3, and A-HIPI in the validation cohort. A-HIPI, advanced-stage Hodgkin lymphoma international prognostic index; IPS, International Prognostic Score; ML, machine learning; OS, overall survival; PFS, progression-free survival.

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Personalized treatment with response-adapted treatment strategies has been introduced to decrease long-term therapy-related toxicity and improve outcomes.[5,30-34] Recent trials in aHL have shown the challenge of identifying patients at sufficient risk and assessing the treatment effect in a group of patients with an overall favorable prognosis. For instance, both the HD15 and HD18 trials have required a high number of patients to formally show noninferiority of strategies with response-adapted reduction of treatment.[32,35] Similarly, the ECHELON-1 trial required over 1,300 patients to show a 6.9% increase in PFS with brentuximab vedotin, doxorubicin, vinblastine, and dacarbazine (A + AVD) over ABVD, a result that has not led to change of practice in all countries in the western world. The recent SWOG-S1826 trial included 994 patients and showed an approximately 8% increase in 1-year PFS in favor of nivolumab-AVD over A + AVD. From a statistical point of view, obtaining a sufficient number of events is crucial for obtaining robust estimates of the treatment effect in clinical trials, which is especially relevant within HL. Improving prognostic tools to more accurately identify patients at the highest risk could thus also lower the number of patients needed in aHL trials. The A-HIPI was recently developed as a new prognostic model for aHL in the modern era. The A-HIPI was developed using data from eight international phase III clinical trials and validated in four international cancer registries, demonstrating high performance compared with the IPS-7 and IPS-3. In our study, the A-HIPI outperformed the IPS models on most performance metrics for both OS and PFS, with the exception of a slightly higher IBS for PFS in the validation cohort (Table 3). This confirms the generalizability of the A-HIPI, which may be replacing the conventional IPS.

Calibration in the development cohort was better than that in the validation cohort for both the ML model and A-HIPI, especially for PFS where predictions from both models were substantially lower than the observed PFS in the validation

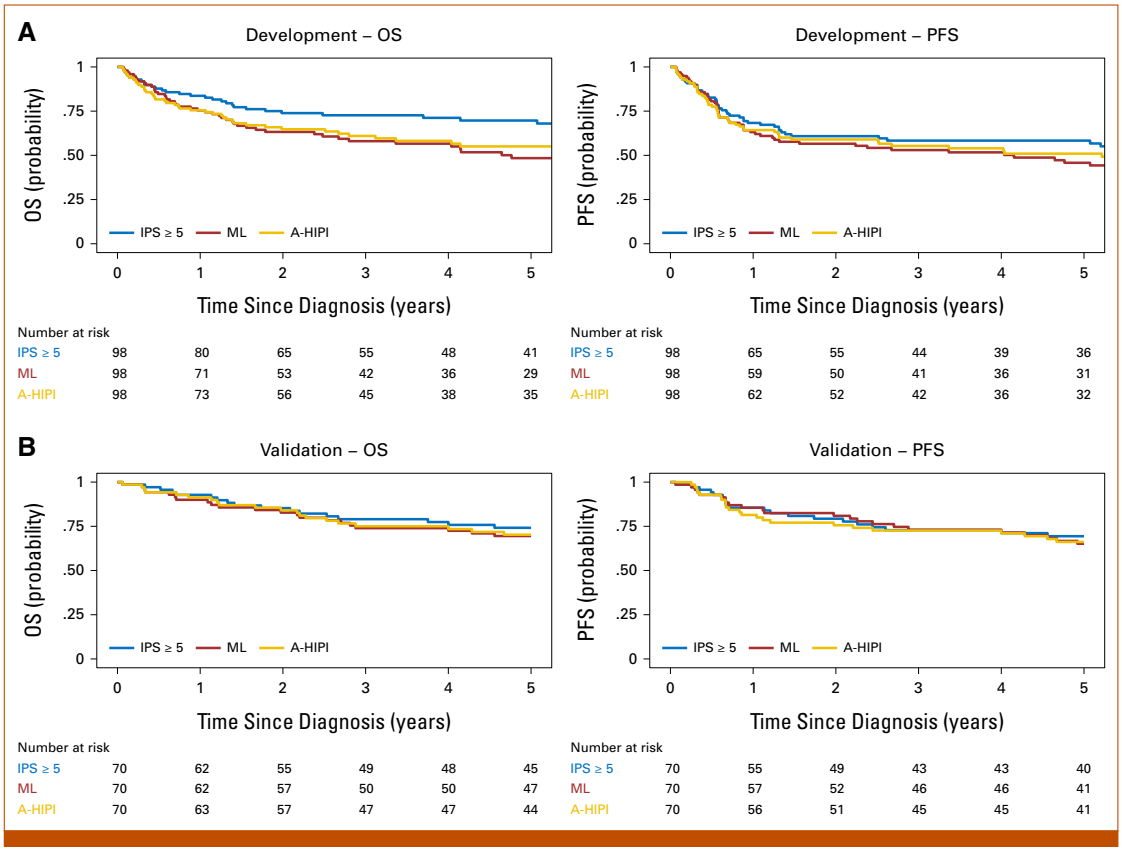


FIG 3. (A) Kaplan-Meier estimates of OS and PFS for high-risk patients (IPS-7, 5-7) and the same number of patients with the lowest predicted probability according to the ML model and the A-HIPI in the development cohort. (B) Kaplan-Meier estimates of OS and PFS for high-risk patients (IPS-7, 5-7) and the same number of patients with the lowest predicted probability according to the ML model in the validation cohort. A-HIPI, advanced-stage Hodgkin lymphoma international prognostic index; IPS, International Prognostic Score; ML, machine learning; OS, overall survival; PFS, progression-free survival.

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cohort (Data Supplement, Figs S5 and S6). This may be explained by patient selection in the validation cohort, which is best exemplified by the higher OS and PFS in the validation cohort compared with the development cohort and the rather large difference of 7 years in median age at diagnosis. A substantial proportion of patients in the validation cohort had incomplete treatment information and were therefore excluded. This might have led to important differences in the development and validation cohorts that directly affect performance metrics. In addition, exact progression dates were incomplete in the validation cohort, and for patients with disease progression without an exact progression date, the progression date was imputed using the end of treatment date. The observed differences between the development and validation cohorts limit the generalizability of the external validation results seen. This data incompleteness limits analyses of external validation and highlights some of the challenges of using nationwide registries. In addition, the A-HIPI was only developed for patients age 18-65 years and is thus not designed for older patients, which might have also affected model performance in the present study. However, when limiting the cohort to 18- to 65-year-olds, results regarding calibration were largely unchanged (Data Supplement, Figs S7 and S8).

In this study, it was investigated whether a more flexible modeling strategy could increase the predictive performance over the IPS-7, IPS-3, and A-HIPI. Consistently, with a previous study within diffuse large B-cell lymphoma (DLBCL), we found that avoiding dichotomization and applying a flexible model strategy led to a substantial improvement in predictive performance.[9,36] However, the DLBCL study also found that compared with simple prognostic models for survival data, using ML to model outcomes only led to small increases in predictive performance. A similar finding was made in the present study where only limited differences between the developed ML model and the A-HIPI model were seen. Given the additional complexity of the ML approach developed in the current study, this suggests that ML models may hold limited

utility for outcome predictions in lymphoma when only a limited number of clinical factors are available to inform predictions. The clinical management of patients with aHL could potentially be guided by having accurate prognostic models for other survivorship end points. For instance, providing accurate prognostic information for patients in complete remission could inform the choice of post-treatment follow-up duration and intensity, whereas models for other important survivorship end points (like secondary malignancies and cardiovascular disease) would be helpful in planning individual patient follow-up programs. With the wide range of data that could be collected for aHL, ML models still have an important role to play in developing tools for these end points.

In the ML models, the most dominant underlying model (stacking weights; $\alpha_j = 0.807$; Data Supplement, Table S4) for OS was an FPS model including only age and WHO performance status. For PFS, the most important submodels (stacking weights; $\alpha_j > 0.1$) were the same FPS model, the CPH-LASSO-Int (least absolute shrinkage and selection operator with all possible interaction terms), and the CPH-IPS model, suggesting that accurate prediction of PFS requires more data points compared with OS. However, no model performed as well for PFS as they do for OS. This indicates that the clinical factors considered in the models can only to a limited degree predict response mechanisms in aHL or that patients with progressive aHL are kept alive for a long time because of effective salvage treatments. In addition, we found that using a CPH model including the continuous version of the IPS variables led to high performance metrics compared with all models in terms of both OS and PFS (Data Supplement, Table S5). This suggests that a limited number of clinical factors constitute the most important prognostic information for disease control in patients with aHL. Although this study focused on using clinical variables available at the time of diagnosis to predict the outcome of patients with aHL, studies in other lymphoma entities have shown that genetic information and molecular features are predictive. Thus, incorporating data such as cell of origin, genomics, radiomics, and histology data may improve performance beyond what can be achieved by the A-HIPI and similar models.[37-40] In addition, including positron emission tomography and computed tomography scan data, circulating tumor DNA would allow for a more refined tumor size staging system in regard to disease burden and has been shown to have great prognostic ability in HL.[41] These data were not available for the current study but may potentially improve PFS predictions in the future.

The strength of this study is the use of nationwide data with cohorts that are less prone to selection bias and is more representative of the entire HL patient population, ultimately improving measures of calibration. In addition, the few missing values in the data set were imputed on the basis of other clinical factors available at the time of diagnosis. Despite registry-based data from three countries, the limited number of patients with aHL and the high OS and PFS are not ideal for ML modeling and may thus affect the overall benefit of using more advanced predictive models. In addition, progression and relapse may not be fully captured in both the development and validation cohorts, but the excellent calibration measures of A-HIPI in the development cohort indicate that the missingness may be negligible for that cohort.

In conclusion, we have developed a new prognostic model using ML and compared this with existing prognostic models for aHL. The model provides personalized predictions on the basis of patient characteristics available at the time of diagnosis and is able to outperform the existing IPS models with respect to OS and PFS. However, the ML model only performs slightly better than the recently developed A-HIPI and a CPH model including the continuous version of the IPS variables. Given the simplicity of the A-HIPI, this model may be preferable to use in clinical practice.

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DATA SHARING STATEMENT

Data are only accessible through the official data owners in Denmark, Sweden, and Norway. Therefore, it is not possible to share the data used in this study.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Data supplement

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A1 - Data cleaning and standardization

The development cohort was restricted to advanced-stage Hodgkin lymphoma patients diagnosed from 2006 in order to ensure nationwide standardization of lactate dehydrogenase (LDH) values and included only patients treated with either bleomycin, etoposide, doxorubicin hydrochloride, cyclophosphamide, vincristine, procarbazine, and prednisone (BEACOPP) or doxorubicin, bleomycin, vinblastine and dacarbazine (ABVD). In the Danish data, the LDH upper limit of normal (ULN) was 255 for patients aged ≥70 and 205 for patients <70[1]. In the Swedish cohort the LDH ULN was 4.3 for patients aged ≥70 and 3.5 for patients aged <70 years. For the Norwegian cohort the LDH ULN was the same as for the Danish cohort. Location variables where fewer than 5 patients in the development cohort had experienced involvement of that location were excluded from the analysis. PS groups 3 and 4 were combined due to the low number of patients with a PS of 4. The continuous version of tumor diameter had a substantial amount of missingness and therefore a categorical variable (≤10 cm and >10cm) was used. Continuous covariates in the development cohort were standardized by subtracting the mean and dividing by the standard deviation.

A2 - Modelling

Overall survival (OS) and progression free survival (PFS) was truncated at five years, meaning that patients were censored at five years after diagnosis if no event had occurred before that time point. This was done in order to ensure that the effect of lab measurements was not neglected in the modelling process, due to the large impact of age on long-term outcomes.[2]

A3 - Model anonymization

To ensure the safety and anonymity of patient sensitive data, an anonymization scheme was implemented that allows distribution of the model, also programmatically. Cumulative baseline hazards and Kaplan-Meier curves were smoothed using cubic-splines. The number of knots for the spline were chosen by a forward-selection setup seeking to minimize the root mean square error. Additionally, a minor random value was added to the survival time variables when determining the knot positions.

A4 – Covariates used in modelling

All available covariates were considered in the penalized-CPH models and PC-CPH model. To give an indication of the performance lost due to dichotomizing the IPS covariates was applied in their continuous form in a CPH model. Additionally, including elevated Lactate dehydrogenase (LDH) and performance score as these have previously been shown to be predictive.[3, 4] Also, age and

performance score was solely considered in a FPS model due to their predictive power found in other lymphomas.[5]

A5 - Missing value imputation

To handle missing data, single-value imputation was performed by constructing a hybrid majority mean vote penalized regression model.[6] This approach was chosen to allow for anonymization of the imputation procedure. The method is a two-step procedure: 1) an initial majority or mean imputation step and 2) a forecasting step. The imputation model was built using least absolute shrinkage and selection operator (LASSO) regression . For discrete covariates, the imputation model was set up using logistic or multinomial regression, while for continuous predictors a linear model was used for imputation. All covariates listed in Table S1 were used to construct the imputation models except the outcome variables (OS and PFS). Imputation in the development cohort was performed within the cross validation loop by applying the imputation model from the training data to the test data. Imputation in the validation cohort was performed by applying the imputation model trained on the entire development cohort.

A6 – performance measures

The IBS summarizes the time-varying Brier score up to five years. The C-index provides a measure of the overall discriminatory capability of a model, whereas the time-varying AUC assesses a model at specific time points. The time-varying AUC and C-index were calculated using the five-year predicted survival probability as the risk ranking. The CS measures the agreement between the observed outcomes and predicted outcomes, while CITL compares the means between the observed risk and the predicted risk, indicating the degree to which predictions are systematically too low or too high.[7]

A7 – R packages

All analyses were conducted using R version 4.1.1 (R foundation for Statistical Computing, Vienna, Austria). The IBS was calculated using the pec[8] package, the C-index was calculated using the SurvC1[9] package, and the time-varying AUC was calculated by the timeROC[10] package. Other specific packages and functions used in this study are listed in the supplementary material (Table S5).

Flow chart of Model setup

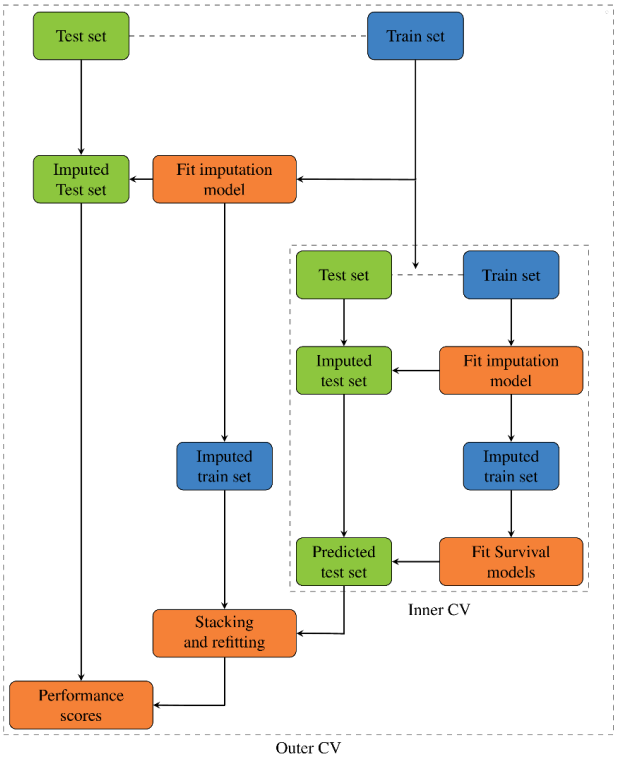


Figure-S1: Flowchart of the nested cross-validation setup for the stacking procedure and internal validation.

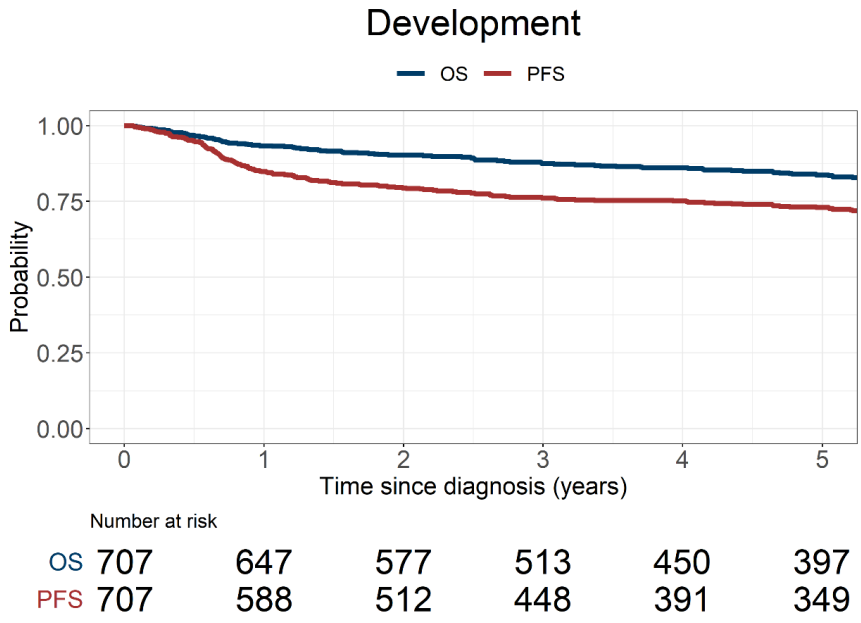


Figure S2A: Kaplan-Meier estimates of OS and PFS in the development cohort. Abbreviations: OS, Overall survival; PFS, Progression free survival.

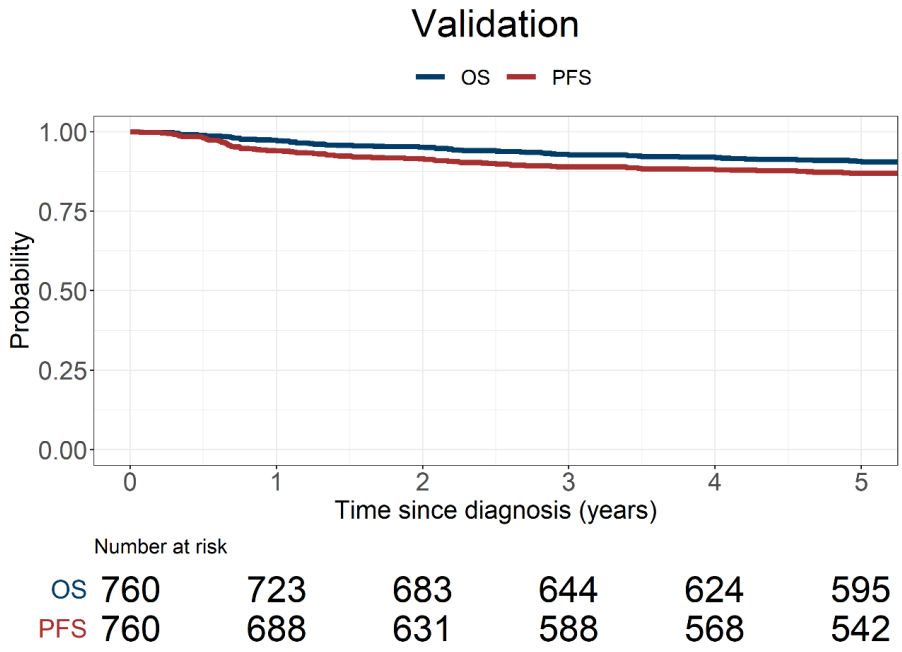


Figure S2B: Kaplan-Meier estimates of OS and PFS for the validation cohort. Abbreviations: OS,

Overall survival; PFS, Progression free survival.

Low risk groups

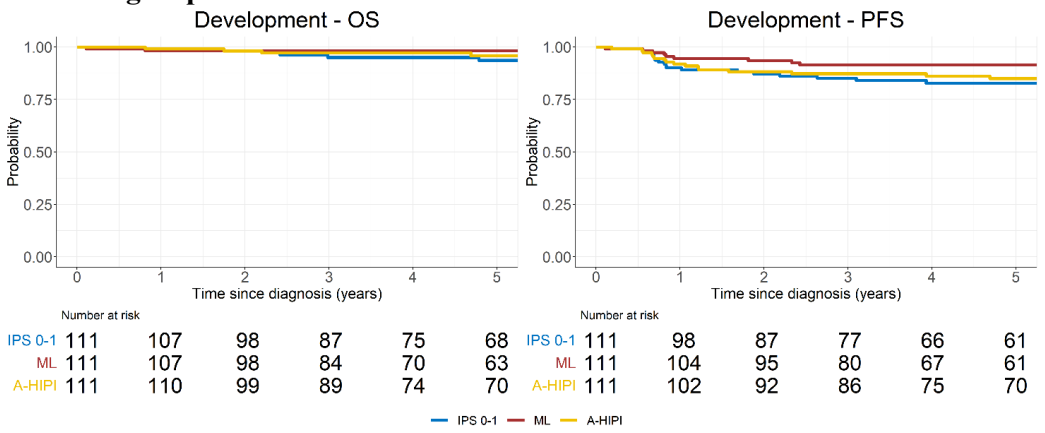


Figure S3A: Kaplan-Meier estimates of OS and PFS for high-risk patients (IPS-7, 0-1) and the same number of patients with the highest predicted probability according to the ML model and the A-HIPI in the development cohort. Abbreviations: OS; Overall survival, IPS; International prognostic score, PFS; Progression free survival; ML, Machine learning; A-HIPI, Advance-stage Hodgkin lymphoma international prognostic index.

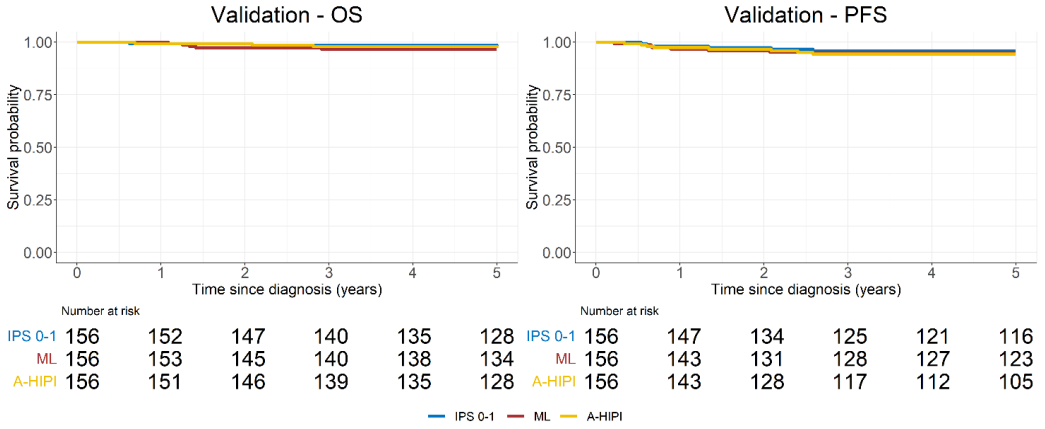


Figure S3B: Kaplan-Meier estimates of OS and PFS for high-risk patients (IPS-7, 0-1) and the same number of patients with the highest predicted probability according to the ML model and the A-HIPI in the validation

cohort. Abbreviations: OS; Overall survival, IPS; International prognostic score, PFS; Progression free survival; ML, Machine learning**;** A-HIPI**,** Advance-stage Hodgkin lymphoma international prognostic index**.**

Calibration

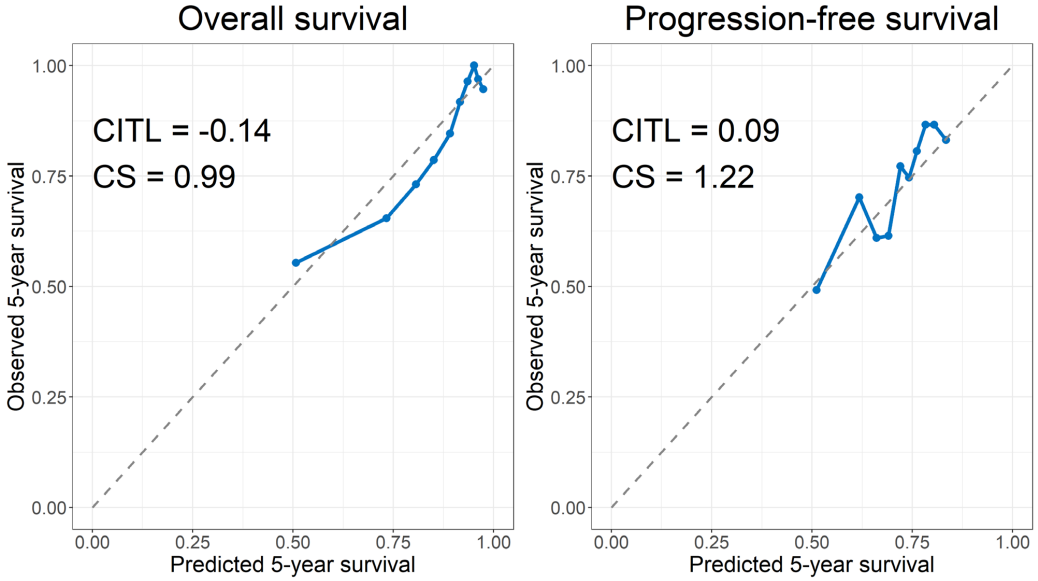


Figure – S4: Calibration plot, including CS and CITL for the A-HIPI model, calculated in the development cohort. CITL, Calibration in the large; CS, Calibration slope.

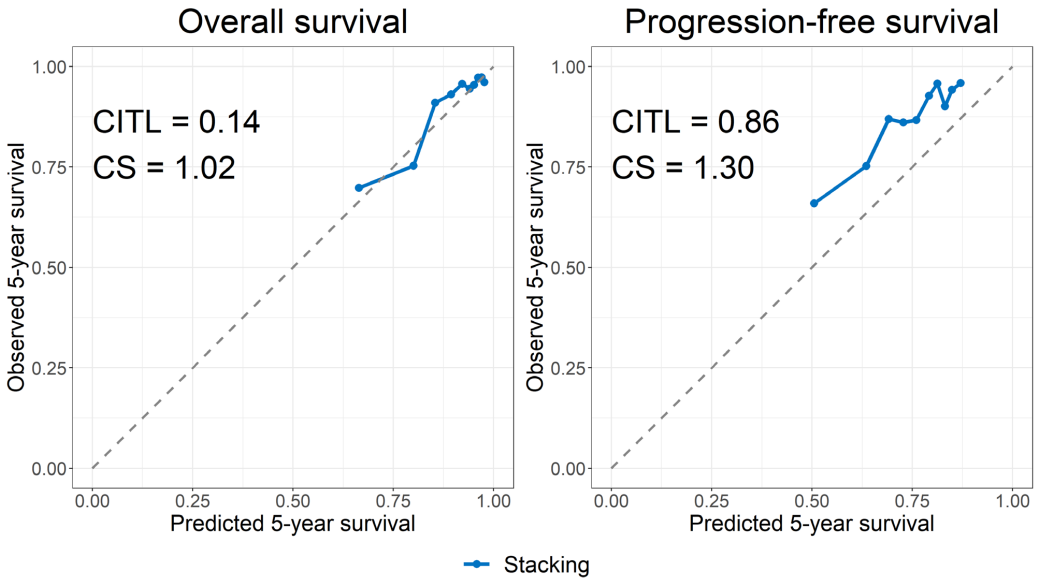


Figure – S5 Calibration plot, including CS and CITL for the ML model, calculated in the validation cohort. CITL, Calibration in the large; CS, Calibration slope.

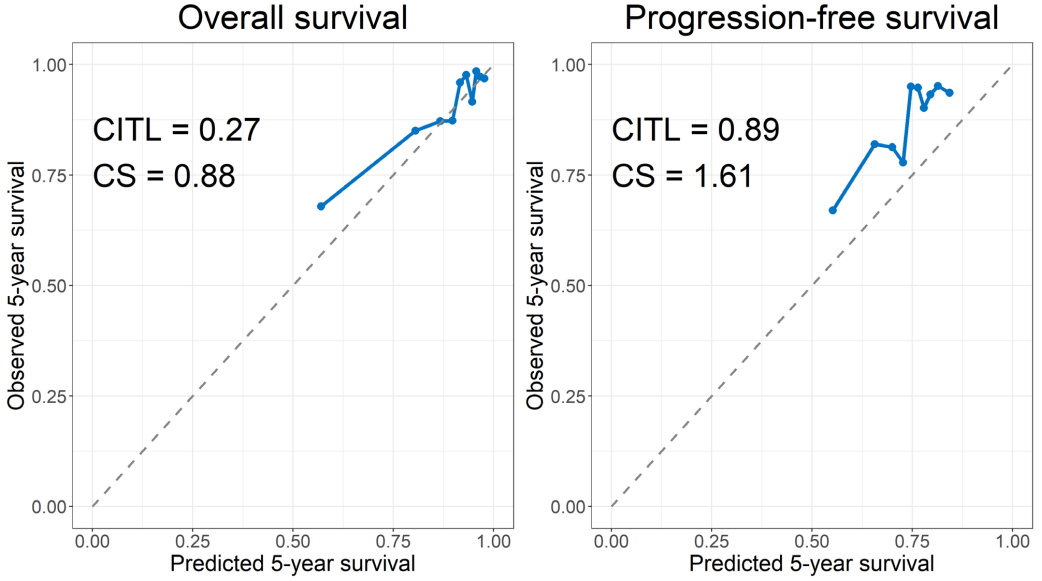


Figure – S6: Calibration plot including CS and CITL for the A-HIPI model, calculated in the validation cohort. CITL, Calibration in the large; CS, Calibration slope.

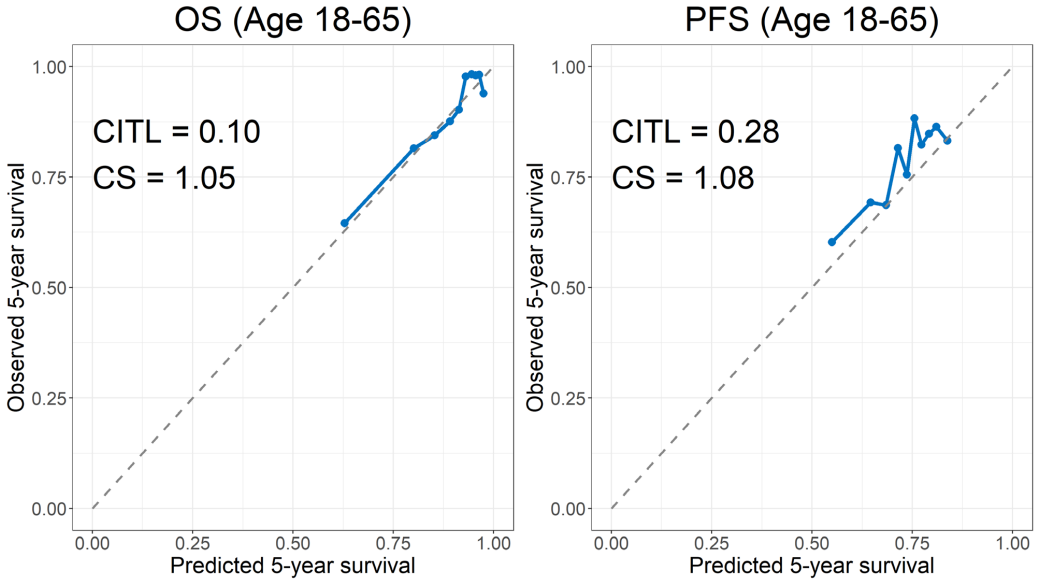


Figure – S7: Calibration plot, including CS and CITL for the A-HIPI model, calculated in the development cohort for patient aged between 18-65 years. CITL, Calibration in the large; CS, Calibration slope.

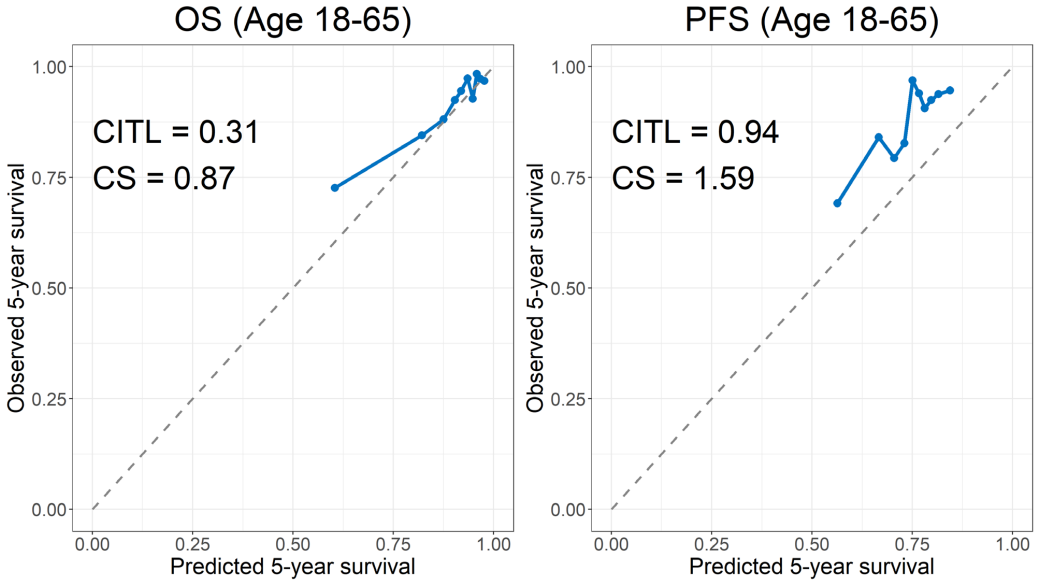


Figure – S8: Calibration plot, including CS and CITL for the A-HIPI model, calculated in the validation cohort for patient aged between 18-65 years. CITL, Calibration in the large; CS, Calibration slope.

Tables

Table S1: Included predictors and their units. Each predictor is associated with one or more variable subsets which is used to define the variables included in each prognostic model.

Covariates	Functional form and unit	Variable subsets
Age	Numeric: Years	1, 2, 3, 4, 5
Sex	Categorical: Male/Female	1, 2, 3
Ann Arbor stage	Categorical: I, II, III, IV	1, 2, 3, 4
Albumin	Numeric: g/L	1, 2, 3
Hemoglobin	Numeric: g/L	1, 2, 3, 4
Leukocytes	Numeric: Count 10^9/L	1, 2, 3
Lymphocytes	Numeric: Count 10^9/L	1, 2, 3
LDH	Numeric: Measured LDH divided by the upper reference level	1, 2
Tumor diameter	Categorical: ≤10cm, >10 cm	1
PS	Categorical: 0, 1, 2, 3, 4	1, 2, 5
B-symptoms	Categorical: Yes/No	1
Number of nodal sites	Integer: 0, 1, 2, 3, 4...	1
Number of extra nodal sites	Integer: 0, 1, 2, 3, 4...	1
Extra nodal location sites: bone marrow, muscular system, lungs, colon, kidney, liver, skin, bones	Categorical for each site: Yes/No	1

Abbreviations: PS, Performance status; LDH, Lactate dehydrogenase.

Table S2: R packages, functions and version used throughout this study.

Package	**Functions**	**Usage**	**Package Versions**
glmnet	cv.glmnet, glmnet,	Penalization models (Cox lasso and Elastic net) and cross-validation	4.1-3
rstpm2	stpm2	FPS models	1.5.2
Pec	calPlot, ibs	Calibration plots and the integrated brier score	2021.10.11
Rsolnp	solnp	Optimizing stacking weights	1.16
survival	coxph	Cox model	3.4-0
survC1[9]	Est.Cval	Estimation of the c-index	1.0-3
TimeROC[21]	timeROC	Estimation of the time-varying area under the curve	0.4

Table S3: Summary of clinical characteristics.

Clinical factors	**Units**	**Development cohort (n = 707)**	**Validation cohort (n = 760)**
Hemoglobin, median (range)	g/L	120.00 [62.40, 169.60]	123.00 [8.00, 175.00]
Ratio of LDH/ULN, median (range)		0.98 [0.05, 2.78]	1.00 [0.37, 12.29]
Albumin, median (range)	g/L	37 [10, 70]	37 [3, 384]
Lymphocyte, median (range)	Count 10^9/L	1.36 [0.00, 110.00]	1.50 [0.00, 36.00]
Leukocyte, median (range)	Count 10^9/L	9.40 [0.30, 117.00]	15.40 [0.30, 982.00]
Bone marrow, n (%)			
	No	583 (82.5)	641 (84.3)
	Yes	124 (17.5)	119 (15.7)
Lung, n (%)			
	No	594 (84.0)	648 (85.3)
	Yes	113 (16.0)	112 (14.7)
Colon, n (%)			
	No	699 (98.9)	760 (100.0)
	Yes	8 (1.1)	0 (0)
Kidney, n (%)			
	No	702 (99.3)	>755* (99.3)
	Yes	5 (0.7)	<5 (0.7)
Liver, n (%)			
	No	628 (88.8)	692 (91.1)
	Yes	79 (11.2)	68 (8.9)
Skin, n (%)			
	No	702 (99.3)	755 (99.3)
	Yes	5 (0.7)	5 (0.7)
Muscular system, n (%)			
	No	678 (95.9)	755 (99.3)
	Yes	29 (4.1)	5 (0.7)
Bone, n (%)			
	No	538 (76.1)	644 (84.7)
	Yes	169 (23.9)	116 (15.3)

Abbreviations: LDH, Lactate dehydrogenase; ULN, Upper limit of normal**.**

*Due to patient confidentiality and the low numbers with Kidney involvement in the validation cohort, Kidney involvement was reported to agree with the legal limits.

Stacking weights

Table S4: The weights of each individual model included in the ML model for both overall survival and progression-free survival.

	Stacking Weights	
Models	**OS**	**PFS**
IPS-3	0.040	<0.001
IPS-7	<0.001	0.049
CPH-IPS	0.135	0.155
CPH-LASSO-Int	<0.001	0.472
CPH-LASSO	<0.001	<0.001
CPH-Enet	<0.001	<0.001
FPS	0.018	<0.001
FPS-Age-PS	0.807	0.292
PC-CPH	<0.001	0.032

Abbreviations: OS, Overall survival; PFS, Progression free survival; IPS, International prognostic score; CPH, Cox proportional hazard; Lasso, Least absolute shrinkage and selection operator; Int, Interactions; Enet, Elastic net; FPS, Flexible parametric survival; PS, Performance status; PC, Principal component; ML, Machine learning.

Performance measures

Table S5: Performance measures for overall survival and progression-free survival in the development and validation cohorts, for all models listed in Table 2.

	Development cohort				Validation cohort			
	OS		PFS		OS		PFS	
Models	IBS	C-index	IBS	C-index	IBS	C-index	IBS	C-index
FPS-Age-PS	0.073	0.793	0.141	0.656	0.050	0.747	0.082	0.701
ML	0.073	0.789	0.141	0.665	0.049	0.749	0.085	0.691
CPH-LASSO	0.075	0.782	0.142	0.661	0.050	0.735	0.086	0.676
PC-CPH	0.075	0.779	0.141	0.67	0.052	0.745	0.093	0.678
CPH-IPS	0.077	0.773	0.142	0.673	0.049	0.745	0.089	0.685
CPH-LASSO-Int	0.077	0.771	0.140	0.666	0.053	0.708	0.086	0.679
CPH-Enet	0.078	0.777	0.145	0.636	0.05	0.734	0.086	0.681
FPS	0.079	0.72	0.159	0.613	0.051	0.737	0.086	0.693

IPS-3	0.085	0.650	0.149	0.576	0.054	0.663	0.078	0.64
IPS-7	0.087	0.608	0.151	0.549	0.052	0.700	0.077	0.672
A-HIPI	0.076	0.768	0.142	0.649	0.052	0.741	0.091	0.677

Abbreviations: OS, Overall survival; PFS, Progression free survival; C-index, Concordance index; IPS, International prognostic score; CPH, Cox proportional hazard; LASSO, Least absolute shrinkage and selection operator; Int, Interactions; Enet, Elastic net; FPS, Flexible parametric survival; PS, Performance status; PC, Principal component; ML, Machine learning.

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Study II

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ORIGINAL ARTICLE

European Journal of Haematology

WILEY

Effectiveness of R-CHOP versus R-CHOEP for treatment of young patients with high-risk diffuse large B-cell lymphoma: A Danish observational population-based study

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Abstract

Purpose: Etoposide to standard R-CHOP is used for high-risk diffuse large B-cell lymphoma (DLBCL) in some countries. Due to the lack of randomized trials, a real-world data study using matching methods was used to test the potential effectiveness of R-CHOEP over R-CHOP.

Patients and Methods: This study included patients from the Danish Lymphoma Register diagnosed between 2006 and 2020 at the age of 18–60 years with de novo DLBCL and age-adjusted IPI ≥2. R-CHOEP treated patients were matched 1:1 without replacement to R-CHOP treated patients using a hybrid exact and genetic matching technique. Primary endpoints were progression-free survival (PFS) and overall survival (OS).

Results: In total, 396 patients were included; 213 received R-CHOEP and 183 received R-CHOP. Unadjusted 5-year PFS and OS for R-CHOEP were 69% (95% Confidence intervals [CI]; 63%–76%) and 79% (CI;73%–85%) versus 62% (CI;55%–70%) and 76% (CI;69%–82%) for R-CHOP (log-rank test, PFS $p = .25$ and OS $p = .31$). A total of 127 patients treated with R-CHOEP were matched to 127 patients treated with R-CHOP. Matching-adjusted 5-year PFS and OS were 65% (CI; 57%–

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74%) and 79% (CI; 72%–84%) for R-CHOEP versus 63% (CI; 55%–73%) and 79% (CI;72%–87%) for R-CHOP (log-rank test, PFS $p = .90$ and OS $p = .63$).

Conclusion: The present study did not confirm superiority of R-CHOEP over R-CHOP for young patients with high-risk DLBCL.

KEYWORDS

DLBCL, lymphoma, optimizing treatment outcome, R-CHOP vs R-CHOEP, real world data, toxicity after treatment, treatment of high-risk patients

Novelty Statements

What is the new aspect of your work?

The study challenges that value of more intensive alternatives to R-CHOP for younger patients with high-risk DLBCL from a population-based setting.

What is the central finding of your work?

When using novel statistical methods to emulate a clinical trial setting and balance confounders, more R-CHOEP is not more effective than R-CHOP.

What is (or could be) the specific clinical relevance of your work?

Use of intensive therapies, even for high-risk DLBCL, should be carefully evaluated for each patient due to limited evidence for clinical benefit.

1 | INTRODUCTION

Diffuse large B-cell lymphoma (DLBCL) is the most common type of non-Hodgkin lymphoma (NHL) with an annual incidence of 3.8/100000 per year in Europe.[1,2] Given a median age at diagnosis of approximately 70 years, most patients will be eligible for standard curative intent R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone) or Pola-(polatuzumab vedotin)-R-CHP.[3–5] While the majority achieve durable remissions after standard therapy and become long-term survivors, up to 30% of patients have primary refractory disease or relapse after initial response.[6] The risk for suboptimal outcomes after R-CHOP is particularly high among patients with high-risk disease according to the international prognostic index (IPI) where complete remission (CR) rates are in the range 60%–75%.[7–9] Biological high-risk features, such as *MYC* and *BCL2* rearrangement, are also associated with poorer outcomes after R-CHOP with 2-year progression-free survival (PFS) and overall survival (OS) of 40% and 49%, respectively.[10,11] Once refractory/relapsed (R/R) DLBCL occurs, outcomes are less favorable with 2-year OS of 20–27% for refractory patients and a 2-year PFS of 19% for patients eligible for high-dose therapy.[12,13] Outcomes are better when chimeric antigen receptor T-cell therapy (CAR-T) are used in second line, but a 2-year PFS of 46% remains unsatisfactory.[14]

Given sub-optimal outcomes of R/R DLBCL, optimization of frontline therapy remains a key priority in DLBCL. Several alternative treatment strategies have been developed and tested to improve outcomes over and above standard R-CHOP in the rituximab era, but without success, despite initial encouraging phase 2 results of DA-EPOCH-R (dose-adjusted etoposide, prednisone, vincristine, cyclophosphamide, and doxorubicin together with standard rituximab) with a higher than expected 5-year PFS of 79%.[15] However, superiority over R-CHOP was not confirmed in the later Phase 3 Alliance/CALGB 50303 study where 2-year PFS was 75.5% for R-CHOP versus 78.9% for DA-EPOCH-R.[16] Intensified rituximab schedule during induction and as maintenance therapy in DLBCL was also not superior to standard dose R-CHOP.[17] However, in the pre-rituximab era, in the NHL-B1 trial, dose-intensified chemotherapy with etoposide added to CHOP (CHOEP) improved outcomes in younger patients with a 5-year EFS of 69.2% in the CHOEP arm versus 57.6% in the CHOP arm ($p = .004$).[18] Subsequently, for more than a decade, R-CHOEP has been a recommended standard therapy for younger patients with high-risk DLBCL in Denmark (since 2007), Sweden (since 2003), and the European Society for Medical Oncology (ESMO).[3,19,20] This practice was also supported by a Danish retrospective study from 2012 confirming that younger, high-risk patients (<60 years) treated with R-CHOEP-14 had better OS compared than those treated with R-CHOP-14 (4-year OS 74% vs 62%), but there was a median age difference of 7 years between the groups.[21] The aim of this study was to use real-world data (RWD) recorded in recent years and modern statistical matching-techniques to emulate a clinical trial of R-CHOEP versus R-CHOP in younger patients with high-risk DLBCL.

2 | PATIENTS AND METHODS

2.1 | Patients

This study used individual level patient data from the population-based National Danish Lymphoma Register (LYFO).[22] LYFO contains

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data on adult patients 18 years and above with lymphomas in Denmark, including histology, clinical features (such as IPI, Ann Arbor stage, performance scores, and extra nodal involvement) and treatment details. LYFO has a high completeness (92%–100% across lymphoma types) and high predictive values (87%–100%).[23] Patients with de novo DLBCL, diagnosed between 2006 and 2020 were eligible for inclusion if the following criteria were met: (1) aged 18–60 years at diagnosis; (2) received at least 1 cycle of R-CHOP or R-CHOEP, (3) Eastern Cooperative Oncology Group [ECOG] 0–2, and 4) age-adjusted-IPI[7] (aaIPI) ≥2. These criteria resemble those used for R-CHOEP eligibility in Denmark and the ESMO guidelines.[3,20] Patients with central nervous system involvement (CNS), Richter's syndrome, discordant lymphoma, and included in a clinical trial or treated outside trials with the Nordic Lymphoma Group (NLG-LBC-05) CHIC-like regimen were excluded (2× R-CHOP followed with 4× R-CHOEP or DA-EPOCH-R).[24]

Eligible patients were grouped according to the treatment given in the first cycle of therapy (R-CHOP or R-CHOEP) to mimic the intention-to-treat principle. R-CHOEP treated patients were matched with R-CHOP patients on a 1:1 basis without replacement using a hybrid between exact- and genetic matching. Exact matching was done using age, sex, Ann Arbor stage, ECOG, and IPI. The exact matching variables were aggregated in pre-defined groups to minimize patient exclusion. Additionally, B-symptoms (Yes, No), lactate dehydrogenase (LDH) divided by upper limit of normal level (LDH-ratio), largest tumor diameter (≤10 cm, >10 cm), number of extranodal sites (count of all sites), number of nodal sites (count of all sites), the Charlson comorbidity index (CCI) measured 180 days before diagnosis, year of diagnosis, and propensity scores were included as variables in the genetic matching scheme.[25] Additional information on the matching scheme can be found in the statistical analysis plan (SAP).

A unique identification number registered in the Danish Civil Registration System (CRS) is assigned to all Danish citizens at birth or immigration. This number enables complete linkage between the Danish health registries.[26] The Danish National Patient Register (DNPR) contains information on all patients treated in the public healthcare system. Disease codes in the DNPR are registered by the International Classification of Disease, 10th Edition (ICD-10) codes.[27] Data on secondary primary malignancy (SPM) and cardiovascular disease (CVD) events were retrieved from the DNPR and the Danish Cancer Register. Last date of follow-up for CVD and SPM was January 1, 2019, but last day of follow-up for survival was December 31, 2020. Comorbidities were identified using DNPR. Data were pseudonymized and analyzed on a secure server hosted by Statistics Denmark.

The primary endpoint was 5-year OS defined as time from diagnosis until death from any cause or censoring. Secondary endpoints were 5-year PFS (defined as time from diagnosis until progression, death from any cause, or censoring), duration of complete remission (CR)/CRu from end of therapy, and incidence of CVD and SPM. Before this study was carried out a SAP was made publicly available https://osf.io/dw54j/. Some analyses were amended as per feedback from co-authors. Disease codes for CVD can be found in the SAP, and SPM includes all cancers registered in the Danish cancer register.

2.2 | Statistical methods

The Kaplan–Meier method was used to estimate OS and PFS and the log-rank test was used to test for survival differences. The absolute effect of treatment on OS and PFS was assessed using restricted mean survival time (RMS), defined as the area under the survival curve.[28] RMS-differences were calculated at 1, 2, and 5 years post-diagnosis. Wald tests were performed to evaluate level of significance at each time point and 95% confidence intervals (CI) were calculated by pooling standard errors from each estimate. Sensitivity analyses were performed to test impact on treatment definitions on the results. Following analyses were performed (1) majority treatment where patients were allocated based on what treatment they received most of, (2) restriction to patients exclusively either receiving R-CHOP or R-CHOEP with exclusion of all receiving both, and (3) allocation of patients to the R-CHOEP group if they had received R-CHOEP at some point during treatment regardless of timing. Moreover, sensitivity analyses were performed to test the impact of study population selection on outcomes. The following tests were performed: (1) increase of inclusion to 18–65 years, (2) inclusion of all ECOG stages 0–4, and (3) inclusion of all aaIPI levels. Sensitivity analysis was done using the aforementioned matching technique and using multivariable Cox proportional hazard (CPH) analysis on all patients where treatment was adjusted for age, sex, aaIPI, extranodal involvement, ECOG, CCI, year of diagnosis, and LDH ratio.

Time to CVD or SPM events were defined as time from end of treatment until the event of interest or censoring. Cumulative incidence was estimated using the Aalen-Johansen estimator with death and progression as competing events. Cumulative incidence was estimated for all patients in CR/CRu and in rematched patients in CR/CRu from end of treatment until CVD or SPM event after diagnosis. Patients with CVD prior to lymphoma diagnosis were excluded in the CVD analysis, and patients with previous malignancy prior to lymphoma diagnosis were excluded in the SPM analysis. Cause-specific hazard ratios (HR) for CVD and SPM were estimated and patients with progression or death prior to events were censored. Finally, a sensitivity analyses of time period was performed, where only patients treated up to start of January 1, 2016 were included in order test R-CHOEP versus R-CHOP prior to recommended routine testing for double hit lymphomas according to the 2016 revision of the WHO classification.[29]

All analyses were performed using R version 4.2.2 (R foundation for Statistical Computing, Vienna, Austria) and SAS (SAS institute Inc, Gary, NC). Matching of patients was performed using the MatchIt package,[30] and RMS was estimated using the SurvRM2 package.[31] OS, PFS and cause-specific HR were estimated using the survival package,[32] and cumulative incidence was calculated using the prodlim package.[33]

3 | RESULTS

3.1 | Unadjusted baseline characteristics

In total, 396 young patients with high-risk DLBCL were identified (Supplementary Figure S1, consort diagram). First treatment cycle was

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R-CHOEP in 213 patients and R-CHOP in 183. In the R-CHOEP group, 162 patients were treated with 14-day cycles versus 132 in the R-CHOP group, whereas 41 in the R-CHOEP group were treated with 21-day cycles versus 46 in the R-CHOP group. Ten had incomplete information in the R-CHOEP group and five in the R-CHOP group. Median age was 50 years in the R-CHOEP group and 55 years in the R-CHOP group. Differences were also observed for presence of B-symptoms, extranodal involvement, IPI, and year of diagnosis (Table 1 and Supplementary Figure S2). Remaining variables were balanced between groups. Overall response (CR/CRu/PR) rates were 88.3% (95%CI; 83.2%–92.0%) for R-CHOEP and 85.8% (95%CI; 79.9%–90.2%) for R-CHOP (Supplementary Table S1). CR/CRu rates were 84.5% (95%CI; 79.0%–88.8%) for R-CHOEP versus 74.9% (95% CI; 68.0%–80.7%) for R-CHOP.

The median follow-up (reverse Kaplan–Meier[34]) was 5.9 years (inter quantile range [IQR]; 5.3–7.0 years) for the R-CHOEP group versus 9.6 years (IQR; 8.3–11.2 years) for the R-CHOP group. Unadjusted 5-year PFS and OS estimates were 69% (95%CI; 63%–76%) and 79% (95%CI; 73%–85%) for patients treated with R-CHOEP versus 62% (95%CI; 55%–70%) and 76% (95%CI; 69%–82%) for patients treated with R-CHOP (Table 1, Supplemental Figures S3 and S4; Table S1). The unadjusted 5-year RMS estimates for PFS were 3.9 years (YR) (95%CI; 3.6–4.1) for R-CHOEP versus 3.7YR (95%CI: 3.4–4.0) for R-CHOP with an RMS difference of 0.2 (95%CI; −0.2–0.5, $p = .39$, Table 2). For OS, 5-year RMS estimates were 4.3YR (95%CI; 4.1–4.5) for R-CHOEP versus 4.1YR (95%CI: 3.9–4.4) for R-CHOP with an RMS difference of 0.2YR (95%CI; −0.2–0.5, $p = .36$).

3.2 | Matched survival analysis

A total of 127 patients treated with R-CHOEP were matched to 127 treated with R-CHOP. Baseline characteristics were balanced between treatment groups with exception of B-symptoms, year of diagnosis and extranodal disease (Table 1). Characteristics of unmatched patients are shown in Supplementary Table S2. Median follow-up for matched patients were 5.5 years (IQR; 4.9–6.6 years) for R-CHOEP and 8.8 years (IQR; 6.8–10.3 years) for R-CHOP. The 5-year PFS and OS estimates were 65% (95%CI; 57%–74%) and 76% (95%CI; 69%–84%) for R-CHOEP versus 63% (95%CI; 55%–73%) and 79% (95%CI; 72%–87%) for R-CHOP (log-rank for PFS $p = .90$ and OS $p = .63$, Figure 1A,B; Table 1). The 5-year RMS estimates for PFS were 3.7 YR (95%CI; 3.3–4.0) for R-CHOEP versus 3.8YR (95% CI; 3.4–4.1) for R-CHOP with an RMS-difference of −0.1YR (95%CI; −0.6—0.4, $p = .89$, Table 2). No significant differences were observed between groups in subgroup analyses (Figure 2). The 5-year RMS estimates for OS were 4.2YR (95%CI; 3.9–4.4) for R-CHOEP versus 4.2YR (95%CI; 3.9–4.5) for R-CHOP with an RMS difference of 0.0YR (95%CI; −0.4—0.4, $p = .89$). Subgroup analyses for RMS-differences in OS showed no consistent differences overall (Supplementary Figure S5).

3.3 | Sensitivity analyses

In the sensitivity analyses of treatment definition (majority, etoposide at some point and exclusive), no differences were observed between treatment groups regardless of treatment group definitions. Survival curves are shown in Supplementary Figures S6–S8. Consistently, there were no treatment effects in multivariable analysis for OS and PFS regardless of treatment group definitions (Supplementary Table S3). When age, ECOG, and aaIPI criteria were amended according to the methods section in unadjusted and matched analyses, there were no consistent differences between treatment groups when analyzed according to the ITT principle (Supplementary Figure S9–S12). The borderline/significant differences in unadjusted analysis disappeared after matching in all cases. Also, restricting to patients diagnosed between 2006 and 2016 did not alter the results (Supplementary Figure S12).

3.4 | Cardiovascular disease and secondary primary malignancies

In unadjusted analysis of CVD (prior to matching), 153 patients treated with R-CHOEP and 112 patients treated with R-CHOP achieved CR/Cru and were without CVD events prior to DLBCL diagnosis. A total of 20 patients experienced CVD events during 10 years post-remission (7 for R-CHOEP vs. 13 for R-CHOP). The 5- and 10-year cumulative incidence of CVD was 4.9% (95%CI: 0.6%–9.2%) and 10.2% (95%CI; 1.5%–19.0%) for R-CHOEP versus 9.3% (95%CI: 3.5%–15.2%) and 12.3% (95%CI; 5.4%–19.3%) for R-CHOP ($p = .055$, Supplementary Figure S13A). Unadjusted and adjusted cause-specific HR for CVD after R-CHOP showed no association between treatment groups and CVD (Supplementary Table S4). Results were consistent in the matched analysis (Supplementary Figure S14A).

In analyses of SPM (prior to matching), 159 patients treated with R-CHOEP, and 106 patients treated with R-CHOP achieved CR/CRu and did not have other cancer diagnoses prior to first DLBCL diagnosis. A total of 25 patients developed a SPM after 10 years (10 for R-CHOEP vs. 15 for R-CHOP). The 5- and 10-year cumulative incidence of SPM was 5.0% (95%CI; 1.4%–8.7%) and 10.8% (95%CI: 3.1%–18.5%) for R-CHOEP versus 6.8% (95%CI: 1.5%–12.0%), and 17.0% (95%CI; 8.4%–25.6%) for R-CHOP ($p = .04$, Supplementary Figure S13B). Again, unadjusted and adjusted cause-specific HR for SPM showed no association between treatment groups and SPM (Supplementary Table S4). Results were consistent in the matched analysis (Supplementary Figure S14B).

In the total patient population, 10 cases out of 396 patients developed myelodysplastic syndrome (MDS)/acute myeloid leukemia (AML) (<5 for R-CHOP and >5 for R-CHOEP). For patients in CR/Cru, six patients out of 317 developed MDS/AML.

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TABLE 1 Patient characteristics for all patients before and after matching.

		All patients			Matched		
	Levels	R-CHOEP	R-CHOP	ASM-Differences	R-CHOEP	R-CHOP	ASM-Differences
Total number of Patients	*n*	213	183	-	127	127	-
Age at diagnosis	Median [range]	50 [18, 60]	55 [19, 60]	0.550[b]	53 [25, 60]	53 [21, 60]	0.044[b]
Sex *n* (%)	Female	79 (37.1)	71 (38.8)	0.023	48 (37.8)	48 (37.8)	0
	Male	134 (62.9)	112 (61.2)		79 (62.2)	79 (62.2)	
Ann Arbor *n* (%)	I-II[a]	<5	<5	0.078	0 (0)	0 (0)	0
	III-IV	213 (100)	183 (100)		127 (100)	127 (100)	0
ECOG PS *n* (%)	0	110 (51.6)	87 (47.5)	0.076	68 (53.5)	68 (53.5)	0
	1	75 (35.2)	70 (38.3)	0.044	49 (38.6)	49 (38.6)	0
	2	28 (13.1)	26 (14.2)	0.052	10 (7.9)	10 (7.9)	0
Tumor diameter *n* (%)	≤10 cm and missing[a]	146 (68.5)	132 (72.1)	0.080	93 (73.2)	99 (78.0)	0.072
	>10 cm	67 (31.5)	51 (27.9)		34 (26.8)	28 (22.0)	
B-symptoms *n* (%)	No and Missing[a]	75 (35.2)	80 (43.7)	0.151	46 (36.2)	59 (46.5)	0.210
	Yes	138 (64.8)	103 (56.3)		81 (63.8)	68 (53.5)	
CCI *n* (%)	0	65 (30.5)	42 (23.0)	0.090[b]	43 (33.9)	29 (22.8)	0.038[b]
	1-2	103 (48.4)	85 (46.4)		55 (43.3)	65 (51.2)	
	≥3	45 (21.1)	56 (30.6)		29 (22.8)	33 (26.0)	
IPI *n* (%)	2–3	202 (94.8)	170 (92.9)	0.096	122 (96.1)	122 (96.1)	0
	4	11 (5.2)	13 (7.1)		5 (3.9)	5 (3.9)	
aaIPI *n* (%)	2	190 (89.2)	163 (89.1)	-	119 (93.7)	118 (92.9)	-
	3	23 (10.8)	20 (10.9)		8 (6.3)	9 (7.1)	
Year of diagnosis *n* (%)	2006–2010	59 (27.7)	94 (51.4)	0.527[b]	36 (28.3)	57 (44.9)	0.464[b]
	2011–2015	73 (34.3)	43 (23.5)		43 (33.9)	33 (26.0)	
	2016–2020	81 (38.0)	46 (25.1)		48 (37.8)	37 (29.1)	
Extra nodal *n* (%)	0	57 (26.8)	56 (30.6)	0.124[b]	25 (19.7)	43 (33.9)	0.372[b]
	1-2	110 (51.6)	95 (51.9)		68 (53.5)	63 (49.6)	
	≥3	46 (21.6)	32 (17.5)		34 (26.8)	21 (16.5)	
LDH ratio	Median [range]	1.68 [0.77, 17.48]	1.60 [0.74, 17.77]	0.087[b]	1.68 [0.77, 17.48]	1.63 [0.86, 17.77]	0.093[b]
Lymphocytes (10^9/L)	Median [range]	1.08 [0.00, 32.70]	1.05 [0.0, 5.7]	-	0.99 [0.0, 21.5]	1.05 [0.0, 5.7]	-
Total WBC (10^9/L)	Median [range]	7.40 [0.55, 39.30]	7.20 [1.24, 41.50]	-	6.90 [1.50, 35.70]	7.10 [1.24, 41.50]	-
Hemoglobin (g/L)	Median [range]	121.60 [56.00, 169.60]	123.2 [76.8, 171.2]	-	120.0 [56.0, 168.0]	124.8 [76.8, 161.6]	-
Nodal *n* (%)	0	14 (6.6)	11 (6.0)	0.009[b]	12 (9.4)	7 (5.5)	0.006[b]
	1-2	55 (25.8)	47 (25.7)		29 (22.8)	32 (25.2)	
	≥3	144 (67.6)	125 (68.3)		86 (67.7)	88 (69.3)	

Abbreviations: ASM, Absolute standardized mean; CCI, Charlson comorbidity index; ECOG, Eastern Cooperative Oncology Group; IPI, International prognostic index; LDH, lactate dehydrogenase; WBC, White blood cell.

[a]Missing values or small sample sizes were aggregated to other subgroups due to the limited number of patients in order to ensure patient anonymity when numbers were <5.

[b]Absolute standardized mean difference calculated in continuous form.

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TABLE 2 Restricted mean survival time and differences at 1, 2, and 5-years for all patients and matched patients.

	All patients				
	Time (Years)	**R-CHOP**	**R-CHOP**	**RMS-Difference**	***p*-Value**
PFS	1-YR	0.91YR (CI:0.88–0.94)	0.89YR (CI:0.85–0.92)	0.02YR (CI:−0.03–0.07)	.40
	2-YR	1.7YR (CI:1.62–1.78)	1.65YR (CI:1.55–1.74)	0.06YR (CI:−0.07–0.18)	.38
	5-YR	3.86YR (CI:3.61–4.11)	3.69YR (CI:3.41–3.98)	0.16YR (CI:−0.21–0.54)	.39
OS	1-YR	0.96YR (CI:0.94–0.98)	0.93YR (CI:0.91–0.96)	0.02YR (CI:−0.01–0.06)	.19
	2-YR	1.83YR (CI:1.77–1.89)	1.77YR (CI:1.7–1.85)	0.06YR (CI:−0.04–0.15)	.24
	5-YR	4.26YR (CI:4.06–4.47)	4.11YR (CI:3.87–4.36)	0.15YR (CI:−0.17–0.47)	.36
	Matched patients				
	Time (Years)	**R-CHOEP**	**R-CHOP**	**RMS-Difference**	***p*-Value**
PFS	1-YR	0.90YR (CI:0.86–0.94)	0.89YR (CI:0.85–0.93)	0.01YR (CI:-0.05–0.07)	.75
	2-YR	1.67YR (CI:1.56–1.77)	1.66YR (CI:1.55–1.77)	0.01YR (CI:−0.14–0.16)	.90
	5-YR	3.68YR (CI:3.34–4.01)	3.75YR (CI:3.41–4.08)	−0.07YR (CI:−0.55–0.4)	.77
OS	1-YR	0.95YR (CI:0.93–0.98)	0.94YR (CI:0.91–0.97)	0.01YR (CI:−0.03–0.05)	.62
	2-YR	1.81YR (CI:1.73–1.89)	1.78YR (CI:1.7–1.87)	0.03YR (CI:−0.09–0.15)	.67
	5-YR	4.16YR (CI:3.87–4.44)	4.19YR (CI:3.9–4.48)	−0.03YR (CI:−0.43–0.38)	.89

Abbreviations: CI, confidence intervals; OS, overall survival; PFS, progression-free survival; RMS; restricted mean survival time; YR, year.

4 | DISCUSSION

Etoposide inhibits the DNA unwinding enzyme, topoisomerase II, leading to impaired re-ligation of double strand breaks and consequently apoptosis.[35] The present study investigated if adding this agent to R-CHOP in younger patients with high-risk DLBCL lead to improved outcomes. The study was based on retrospective data where confounding by indication and selection bias can limit the ability to draw firm conclusions. However, modern matching-techniques were applied to reduce the risk of such bias. Overall, patients treated with R-CHOEP were younger (median age 50 vs. 55) and had lower CCI compared to patients treated with R-CHOP, demonstrating non-random treatment assignment in the real-world setting. The difference in 5-year PFS observed for R-CHOEP, albeit non-significant, (69% vs. 62%, $p = .25$, Table 1 and Supplementary Figure S4) in unmatched analyses was reduced after a matching that accounted for differences in baseline covariates (65% vs. 63%, $p = .9$, Table 1 and Figure 1B). Sensitivity analyses where treatment assignments were defined using different rules lead to similar conclusions.

The present study applied an individual patient-level matching strategy to minimize the impact of clinical factors such as age, comorbidities, and performance score as well as disease-related features. This approach is different from those used in the previous population-based studies.[21,36] The matching strategy was two-fold with an exact and a genetic matching part. The exact part ensured that the available important factors were precisely matched in each treatment group, whereas the genetic part balanced several covariates and covariate combinations.[37,38] The latter separates this methodology from the more commonly used propensity scores matching which ensure average covariate balance but not in combinations.[39] The matching

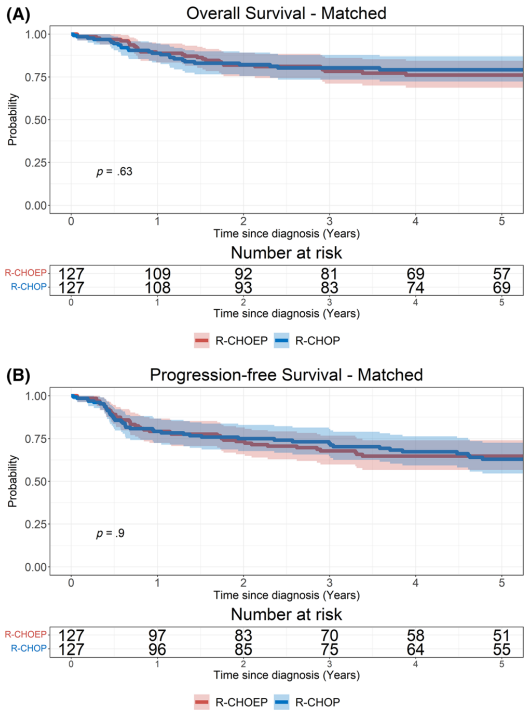


FIGURE 1 (A) Kaplan–Meier estimates of overall survival for R-CHOEP and R-CHOP patients after matching. (B) Kaplan–Meier estimates of progression-free survival for R-CHOEP and R-CHOP patients after matching.

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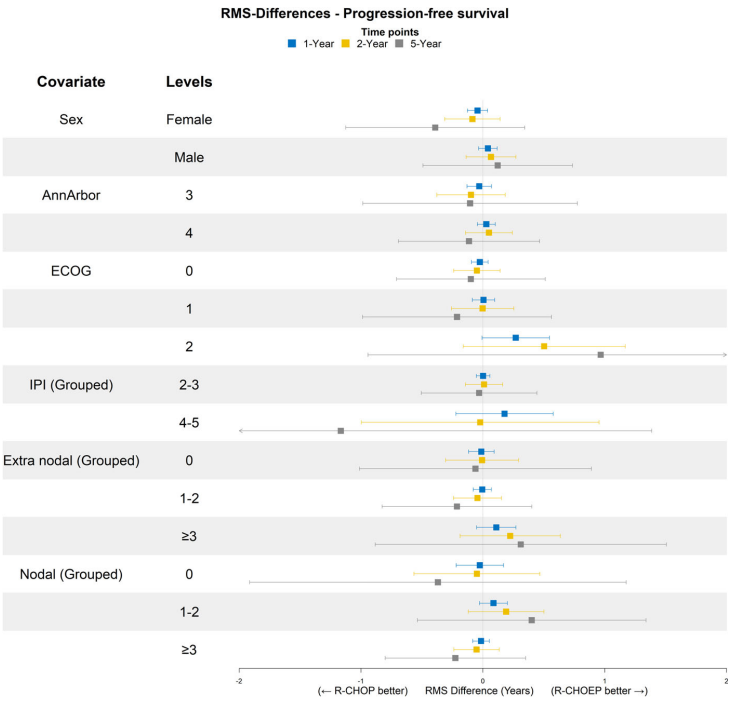


FIGURE 2 Forest plot analysis of pre-specified subgroups for risk differences in RMS modelling progression-free survival. CCI, Charlson comorbidity index; ECOG, Eastern Cooperative Oncology Group; IPI, International prognostic index; RMS, Restricted mean survival time.

strategy used in the present study led to exclusion of 86 R-CHOEP treated patients, due to the use of exact matching and sampling without replacement as some patients treated with R-CHOEP could simply not be matched to the remaining patients treated with R-CHOP. This was documented by the marked difference in median age between unmatched patients in the two groups (45 years for R-CHOEP vs. 58 years for R-CHOP, Supplementary Table S2). Despite the successful balancing of known confounders, unmeasured or unknown confounders cannot be accounted for in observational studies and could still bias the results.

A previous Danish study by Gang et al. of 159 younger (≤60 years) patients with high-risk DLBCL in the period 2005–2008 reported better outcomes after R-CHOEP-14 as compared to R-CHOP-14.[21] The study period was overlapping 3 years with the surveyed period of the present study, but median follow-up was shorter (2 years versus >6 years in the present study). In contrast to the present study, the previous Danish study observed a superior 4-year OS of 75% with R-CHOEP-14 versus 62% with R-CHOP-14 ($p = .04$).[21] PFS was also significantly better after R-CHOEP-14 ($p = .02$). In that study, patients treated with R-CHOEP were also younger, but by a median age of 7 years. As age is directly related to the risk of all-cause mortality, poorer treatment tolerability, and less likelihood of successful curative salvage of R/R DLBCL, a difference in age is an important confounder.[40] When adjustments for age as a dichotomized covariate and other confounders were performed, a differences remained (HR 0.5, 95%CI; 0.2–1.0 for OS, $p = .04$ and HR 0.5, 95%CI; 0.3–0.9 for PFS, $p = .03$).[21] However, with the limited number of patients (98 for R-CHOP and 61 in R-CHOEP) and the shorter follow-up in that study, outcomes are sensitive to random effects. The present study found no survival advantage associated with R-CHOEP in young high-risk DLBCL which is likely explained by difference in inclusion period and different statistical methodology.

A Swedish register-based study by Wästerlid et al. investigated the effect of R-CHOP-21, R-CHOP-14, and R-CHOEP-14 in a cohort of 3443 patients with DLBCL diagnosed between 2007 and 2012.[36] After a median follow-up of 47 months, the 5-year OS estimates for patients ≤65 years of age were 84% for R-CHOEP-14, 78% for R-CHOP-14, and 85% for R-CHOP-21. The difference in median age at time of diagnosis was 9 years between patients treated with R-CHOEP or R-CHOP.[36] Using a flexible multivariate analysis with age modeled by splines and a larger number of covariates such as performance score, upper normal limit of LDH, sex, bulky disease, and Ann Arbor stage, OS was significantly better with R-CHOEP-14 compared to R-CHOP-21 (HR 0.49, 95%CI; 0.3–0.9, $p = .028$) but not against R-CHOP-14 (HR 0.76, 95%CI; 0.5–1.2, $p = .3$). The substantial effect modification in adjusted analyses documented the selective use of R-CHOEP-14 for high-risk patients.

Response rates and duration of responses are often interpreted as evidence for clinical activity in early phase studies.[41,42] In the present study, CR/CRu rates were 84.5% (95%CI; 79.0%–88.8%) for R-CHOEP and 74.9 (95%CI; 68.0%–80.7%) for R-CHOP prior to matching and largely unchanged after matching (82.7% 95%

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CI;75.0%–88.4% for R-CHOEP and 78.0% 95%CI; 69.8%–84.4% for R-CHOP). Since response assessment has changed over time in particular PET-CT, there may have been changes in response rates, as well. However, when limiting to patients diagnosed after 2010 (Supplementary Table S5), the response rates align between the matched patients (CR/CRu, 85.7% for R-CHOEP vs. 84.3% for R-CHOP). Overall, the CR rate increased after 2010, which indicate that patients that would previously have been categorized with PR would be assessed as CR in the PET-CT era.

The Nordic single arm, phase II study CHIC (NLG-LBC-05) reported 5-year PFS and OS of 81% and 83% in younger high-risk patients with DLBCL after dose-dense immunochemotherapy with early systemic CNS prophylaxis.[24] The 5-year OS was comparable to the 79% (95%CI; 73%–85%) observed for the 213 R-CHOEP treated patients (prior matching) in the present study. The 139 patients enrolled in CHIC had comparable characteristics (93% with aaIPI ≥2 and median age of 56 (range 20–64 years)) to patients in the present study. Acknowledging limitations in cross-trial comparisons, the matched R-CHOP cohort in the present study had a 5-year OS of 76% (95%CI; 69%–82%), which is not significantly different from outcomes in CHIC. Finally, while outcomes in CHIC may have been better than expected from unselected historical cohorts of high-risk patients, it is known that trial-eligible are generally more healthy and have better outcomes that patients treated in the real-world setting if biases are not accounted for.[43–45]

In the present study, R-CHOEP was not associated with increased risk of later CVD or SPM despite addition of etoposide. This was consistent with observations for CVD with the NHL-B1 trial,[18] where CVD after CHOEP-21 was 2.8% versus 0.6% after CHOP-14 or CHOP-21 ($p = .232$). Consistently, no significant difference was observed between DA-EPOCH-R and R-CHOP in the Alliance/GALGB 50303 for CVD events (6 for R-CHOP and 2 for DA-EPOCH-R).[16] Alliance/GALGB 50303 also did not show increased risk of AML after etoposide exposure with 1 patient in R-CHOP and in 2 in DA-EPOCH-R diagnosed with AML during follow-up. The present study identified 10 patients with AML/MDS in total, but no significant difference between groups.

The strength of this study was the population-based setting and long follow-up of 7.9 years with loss to follow-up occurring only in a minuscule number of patients (due to emigration). The matching to balance confounders came at the costs of lower power due to loss of patients in the matching process. Hence, the power to detect a 10% difference in 5-year OS at alpha level .05 was 65%. Despite the matching, the study could still be biased by unmeasured confounders which is a limitation inherent to the retrospective design. Furthermore, the matching did not result in perfect alignment of all variables. For example, the number of extranodal sites, a known high-risk feature, was a bit higher in patients treated with R-CHOEP after matching. In addition, LYFO does not contain data on important biological risk factors such as cell of origin or *MYC* and *BCL-2 FISH* rearrangements and TP53 mutation arrangements where imbalances could also lead to biased results. Hence, this study cannot exclude that the numerically higher survival of R-CHOP patients in the last part of the inclusion study period is explained by preferential selection of biological poor risk patients to R-CHOEP. Other clinical features, which could have affected the treatment choice between R-CHOP and R-CHOEP, like metabolic tumor volume, could not be extracted from the LYFO database either, although the matching by extranodal numbers, LDH levels and stage will likely be surrogate for tumor volume to some extent. Another limitation was the missing information on dosing schedules, but the inclusion young patients (≤60 years) makes it likely that the vast majority initiated standard dose treatment. In terms of relevant toxicities, the present study could not measure adverse events such as fatigue, nausea, and other important symptoms experienced by patients, which is more common with R-CHOEP. For example, patients treated with DA-EPOCH-R in the Alliance/GALGB 50303 experienced higher rates of febrile neutropenia, mucositis, neuropathy, and other non-hematological toxicities.[16] In NHL-B1, patients treated with CHOEP also experienced significantly more hematological adverse events.[18] Full treatment plans are not documented in LYFO and only received treatments are registered. Therefore, treatment definitions used in the study could lead to systematic bias. However, the conclusions of the present study were robust to various changes in treatment definitions and eligibility criteria (Supplementary Figures S8–S14 and Table S4).

In conclusion, the present study did not show superior effect of R-CHOEP over R-CHOP on PFS or OS in young DLBCL patients. The non-significant difference in favor of R-CHOEP in unadjusted analysis was biased by differences in baseline characteristics between treatment groups. Therefore, the use of dose-dense therapies such as R-CHOEP should be based on careful evaluations of benefits/risks on individual patient level given the lack of firm evidence of superior effect and added toxicity. However, with current frontline therapies up to 50% of patients with high-risk DLBCL have refractory disease or relapse after initial disease. Therefore, more studies are needed to optimize the frontline treatment of young patients with high-risk DLBCL. Intriguing strategies include better risk stratifications using biological high-risk features and ctDNA guided treatment intensifications.

AUTHOR CONTRIBUTIONS

RRKJ, SE, LHJ, and TC wrote the statistical analysis plan. RRKJ, TC, and LHJ developed the concept and design. RSP, JMJ, MRC, PB, AOG, TSL, IG, and TC provided the patient data. RRKJ wrote all code and statistical analysis. RRKJ and TC wrote the manuscript. All authors have critically revised the manuscript and approved the manuscript for publication.

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CONFLICT OF INTEREST STATEMENT

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DATA AVAILABILITY STATEMENT

Data used cannot be shared as this is prohibited for privacy reasons governed by Statistics Denmark. A statistical analysis plan is publicly available at https://osf.io/dw54j/. Due to privacy rules, numbers including less than five patients cannot be explicitly stated and <5 is used instead.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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Supplementary

Figures

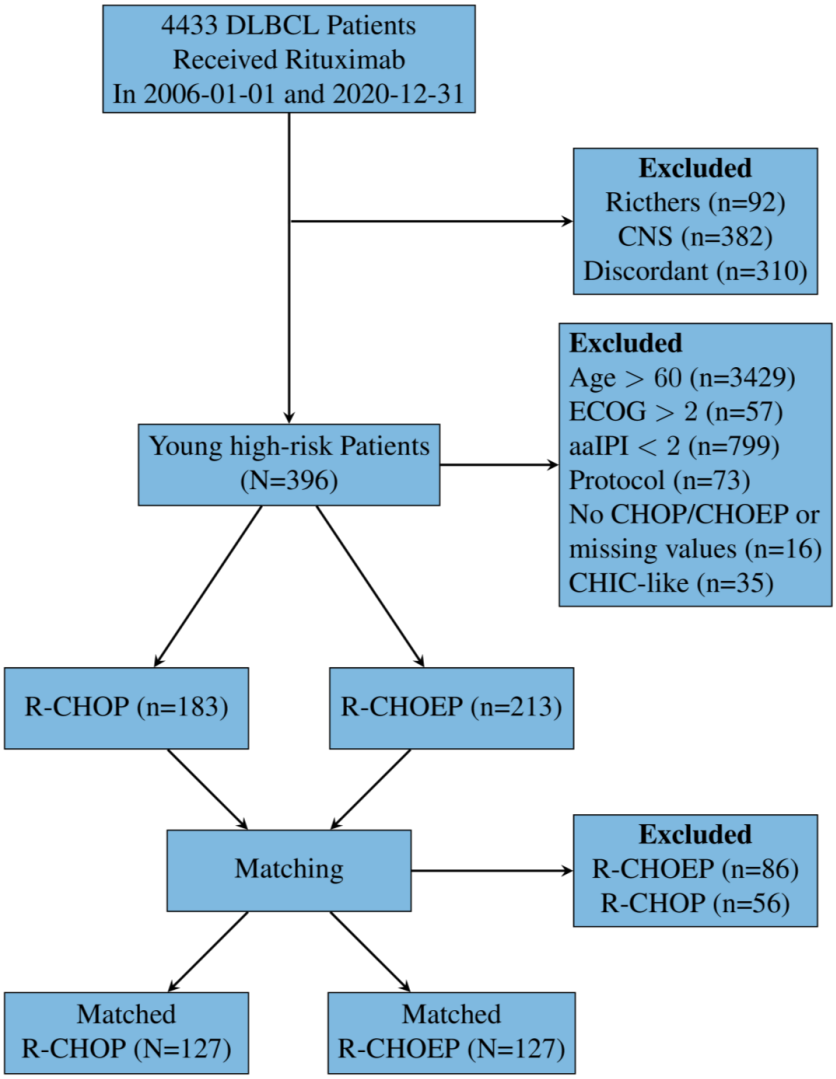


Figure S1: Consort diagram.
Abbreviations: CNS, Central nervous system; ECOG, Eastern Cooperative Oncology Group; aaIPI, Age-adjuster international prognostic index; CHIC-like, 2x R-CHOP and 4x R-CHOEP or DA-EPOCH-R

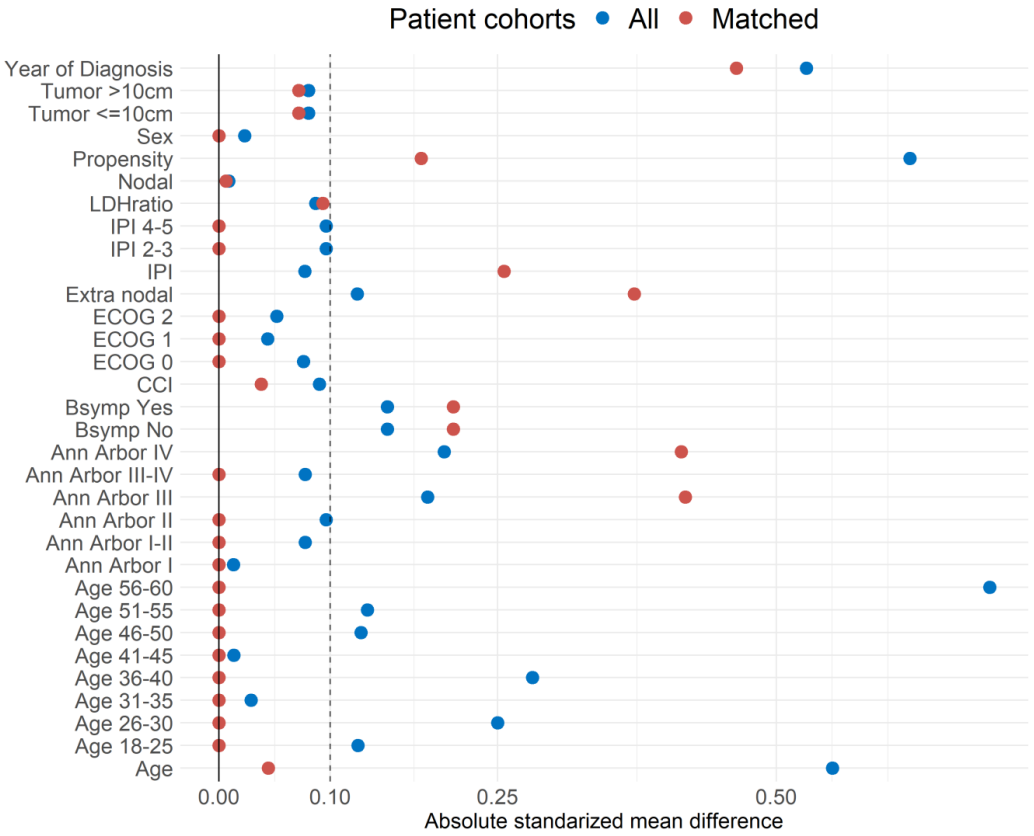


Figure S2: Absolute standardized mean differences between treatment regimens in all patients and after primary matching.

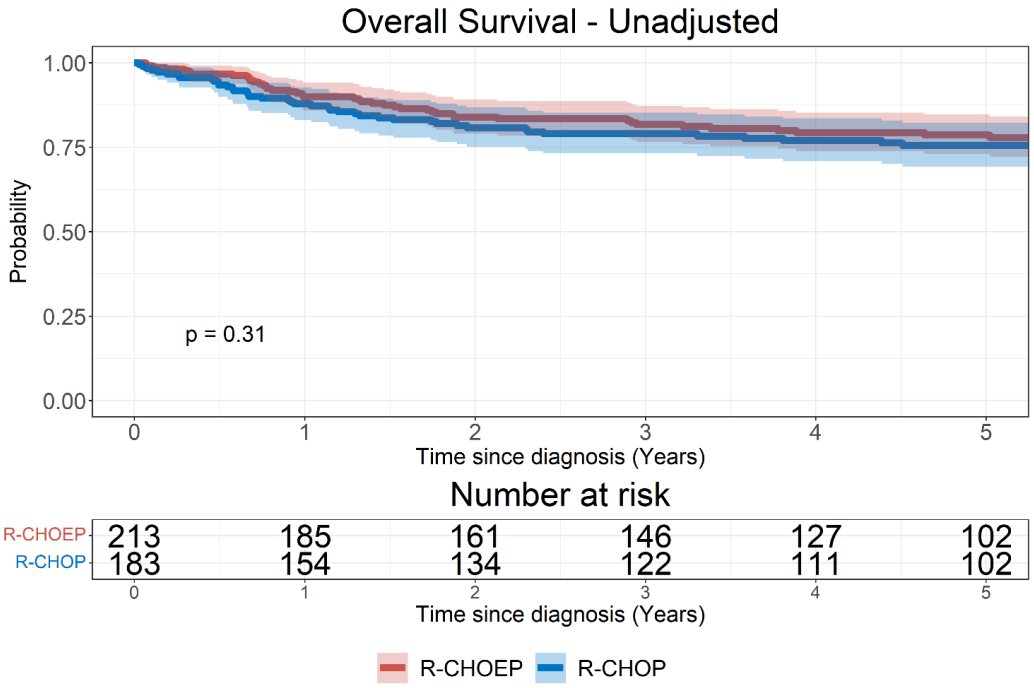


Figure S3: Kaplan-Meier estimates of overall survival for all patients treated with R-CHOEP and R-CHOP.

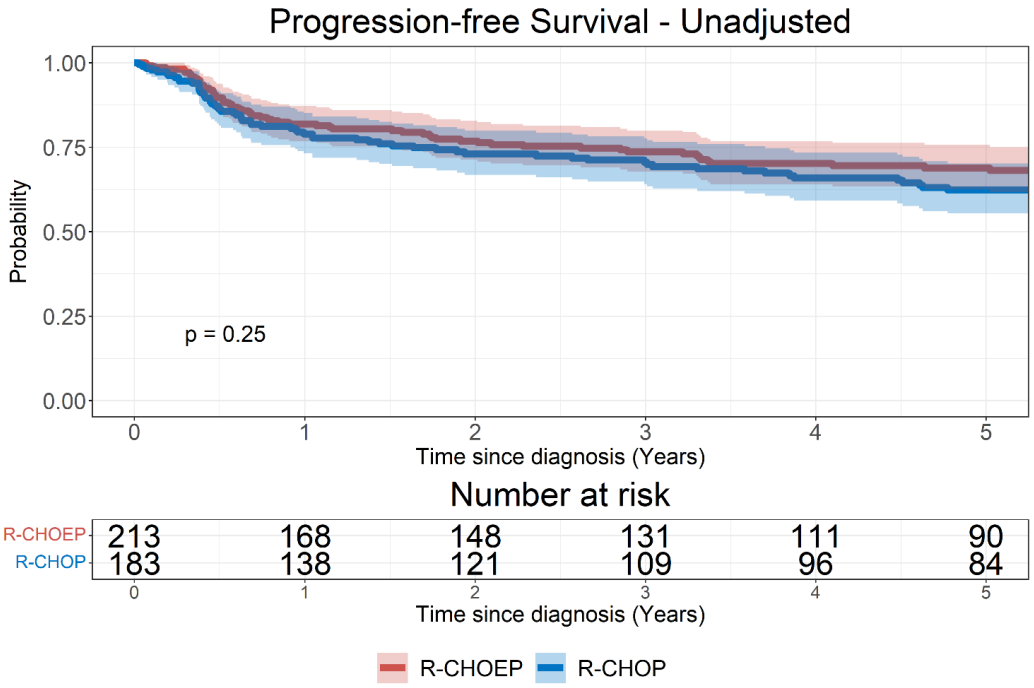


Figure S4: Kaplan-Meier estimates of progression-free survival for all patients treated with R-CHOEP and R-CHOP.

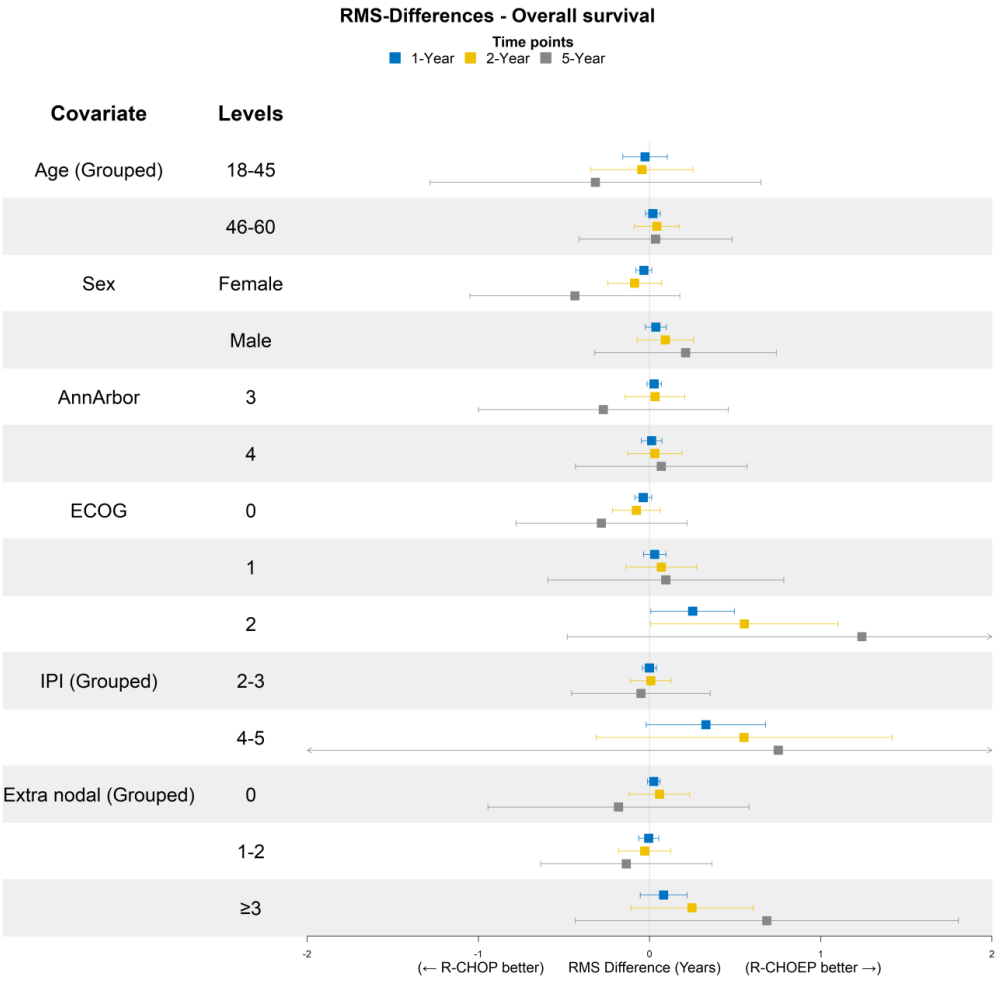


Figure S5: Forest plot analysis of prespecified subgroups for risk differences in RMS modelling overall survival.
Abbreviations: ECOG, Eastern Cooperative Oncology Group; IPI, International prognostic index;

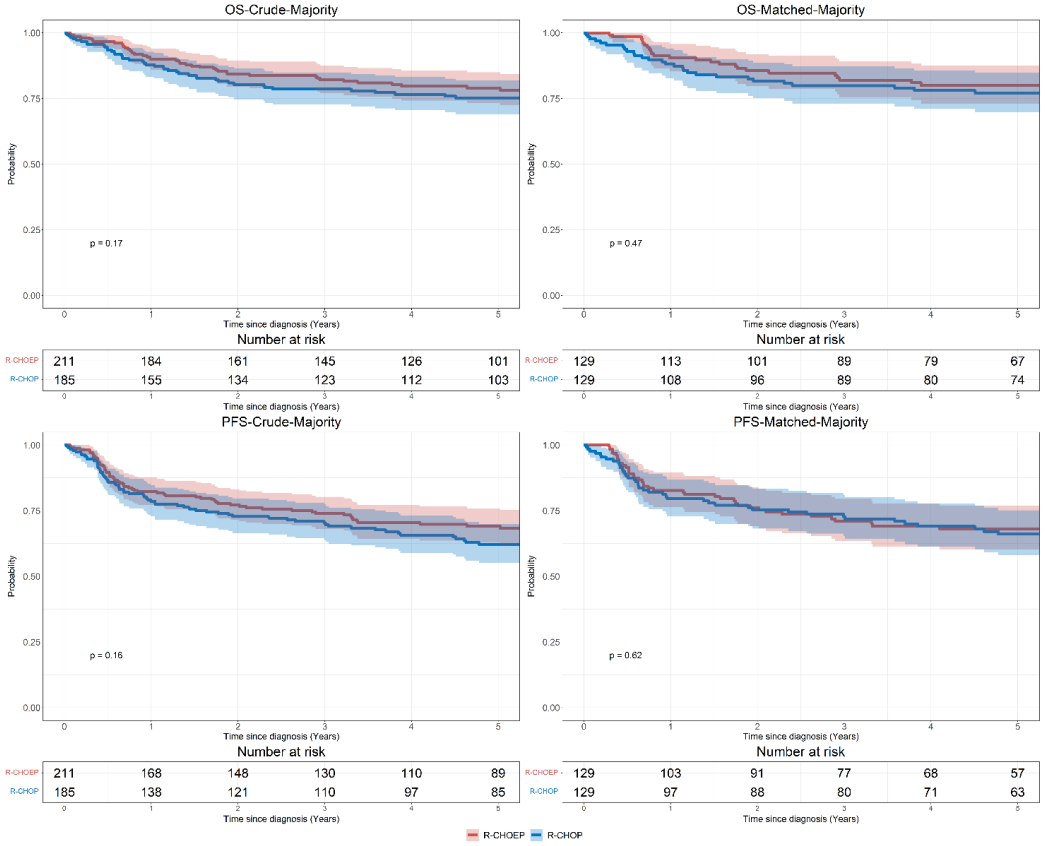


Figure S6: Kaplan-Meier estimates of overall survival and progression-free survival before and after matching for all patients, where patients were allocated based on what treatment they received most of.

Abbreviations: OS, Overall survival; PFS, Progression-free survival.

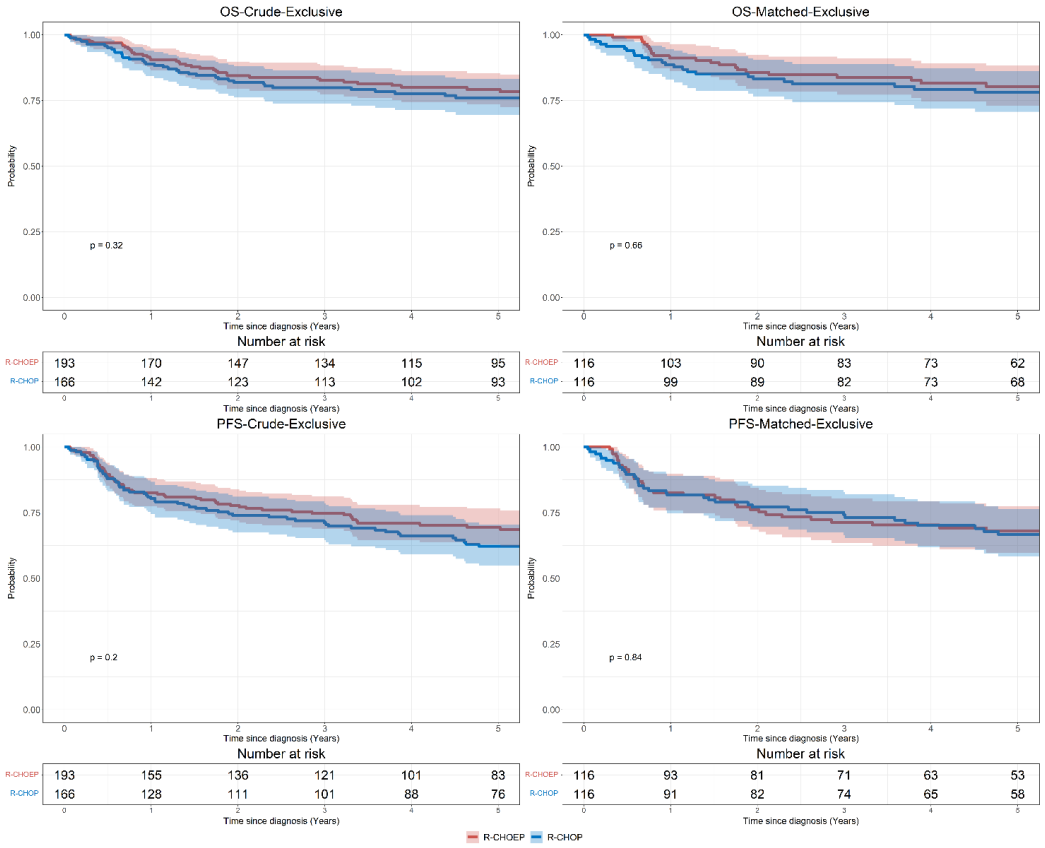


Figure S7: Kaplan-Meier estimates of overall survival and progression-free survival before and after matching for patients restricted exclusively either receiving R-CHOP or R-CHOEP with exclusion of all receiving both.

Abbreviations: OS, Overall survival; PFS, Progression-free survival.

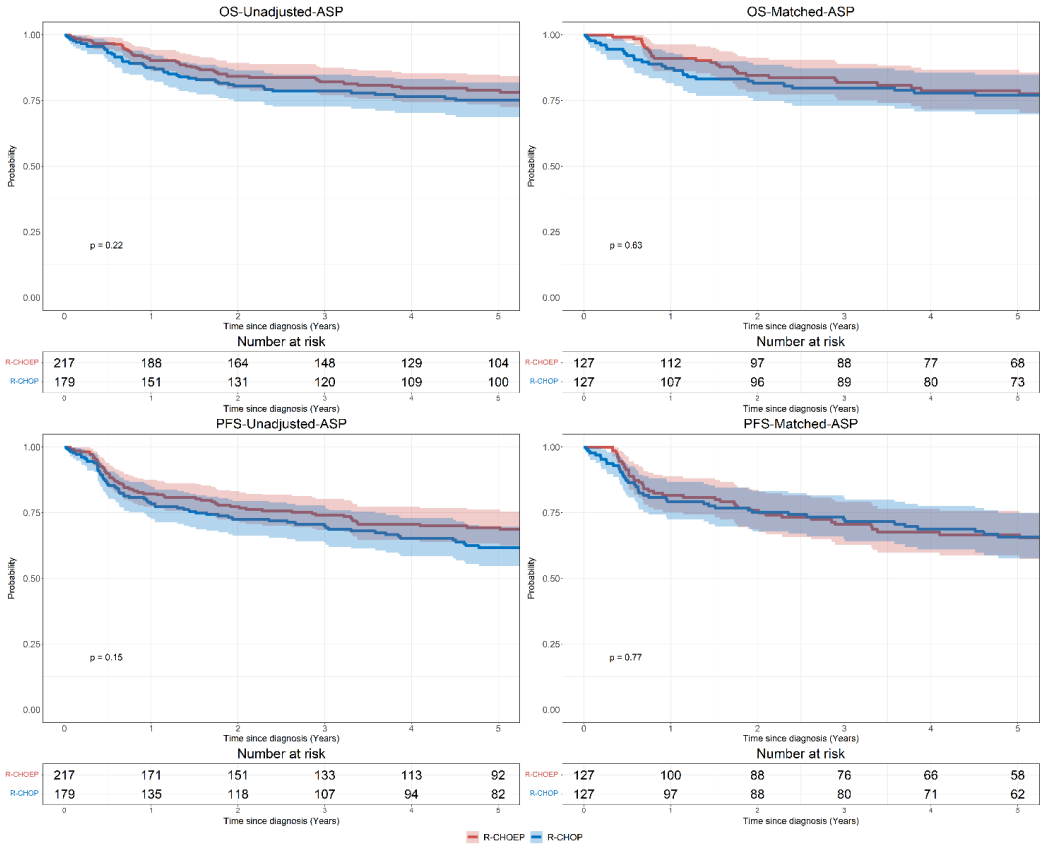


Figure S8: Kaplan-Meier estimates of overall survival and progression-free survival before and after matching were patients are allocated to the R-CHOEP group if they had received R-CHOEP at some point during treatment regardless of timing.

Abbreviations: OS, Overall survival; PFS, Progression-free survival; ASP, R-CHOEP at some point.

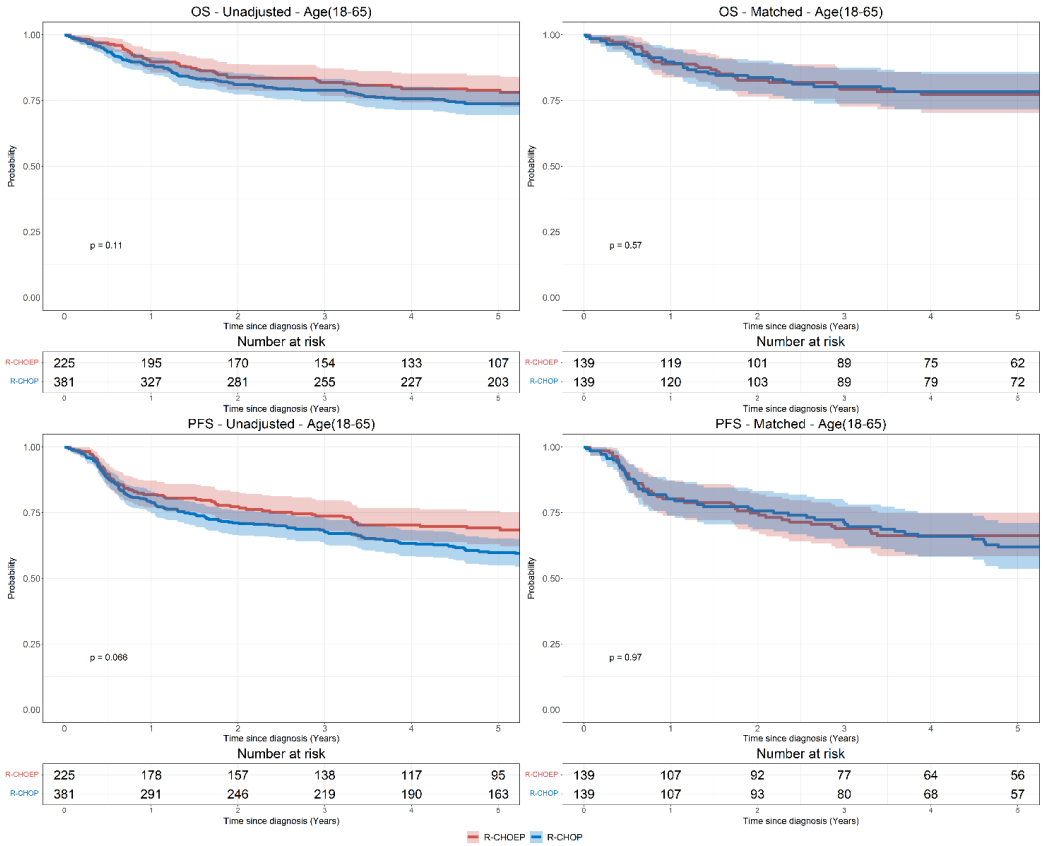


Figure S9: Kaplan-Meier estimates of overall survival and progression-free survival for all patients aged 18-65 and matched patients treated with R-CHOEP or R-CHOP.

Abbreviations: OS, Overall survival; PFS, Progression-free survival.

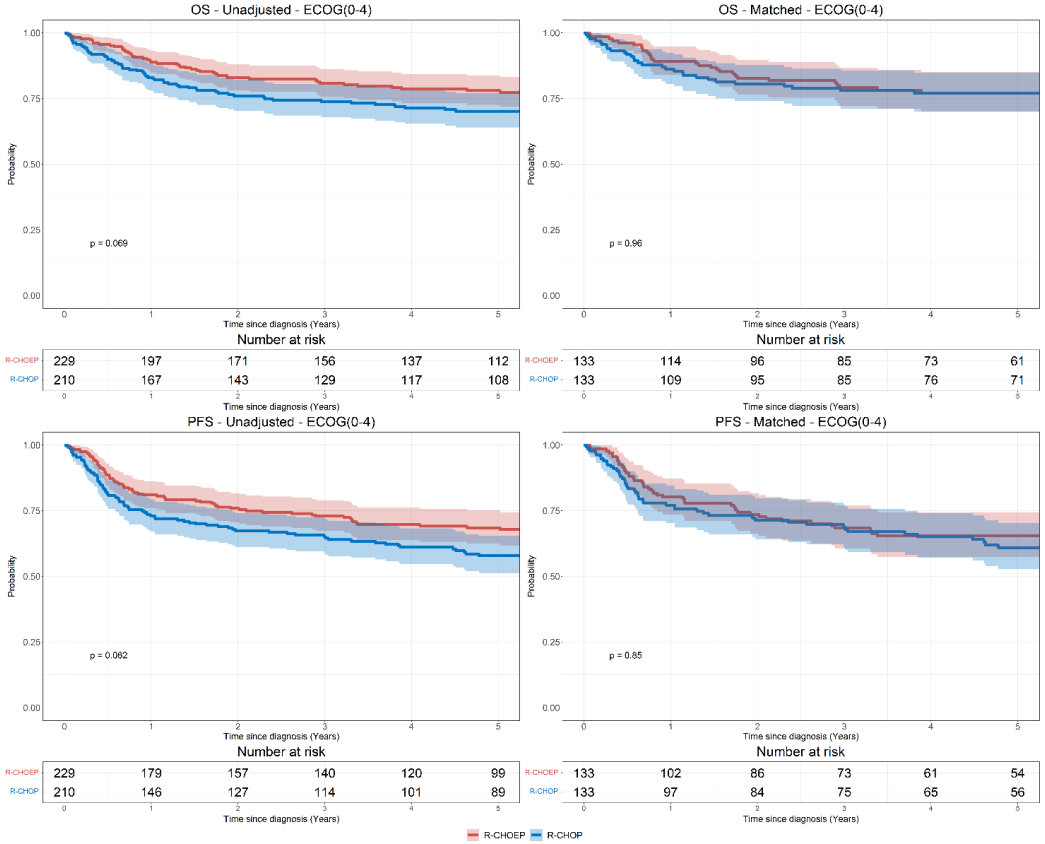


Figure S10: Kaplan-Meier estimates of overall survival and progression-free survival for all patients with ECOG 0-4 and matched patients treated with R-CHOEP or R-CHOP.

Abbreviations: OS, Overall survival; PFS, Progression-free survival; ECOG, Eastern Cooperative Oncology Group.

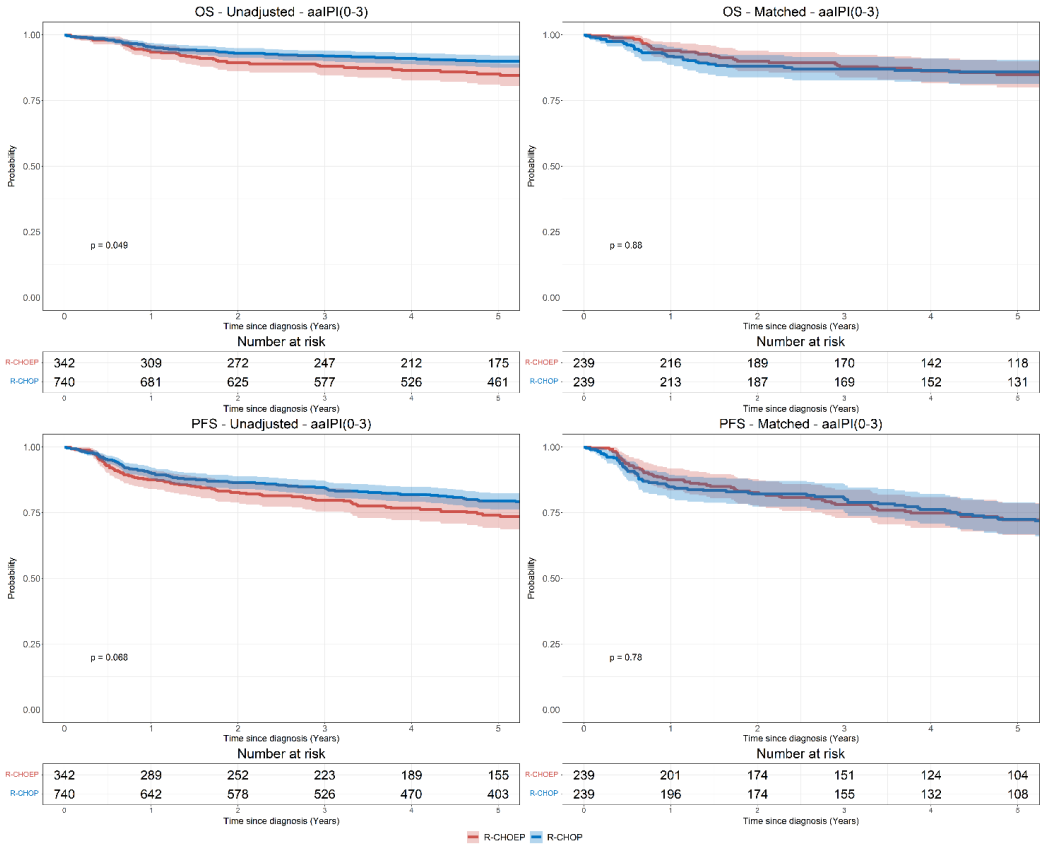


Figure S11: Kaplan-Meier estimates of overall survival and progression-free survival for all patients with aaIPI 0-3 and matched patients treated with R-CHOEP or R-CHOP.

Abbreviations: OS, Overall survival; PFS, Progression-free survival; aaIPI, Age-adjusted international prognostic index.

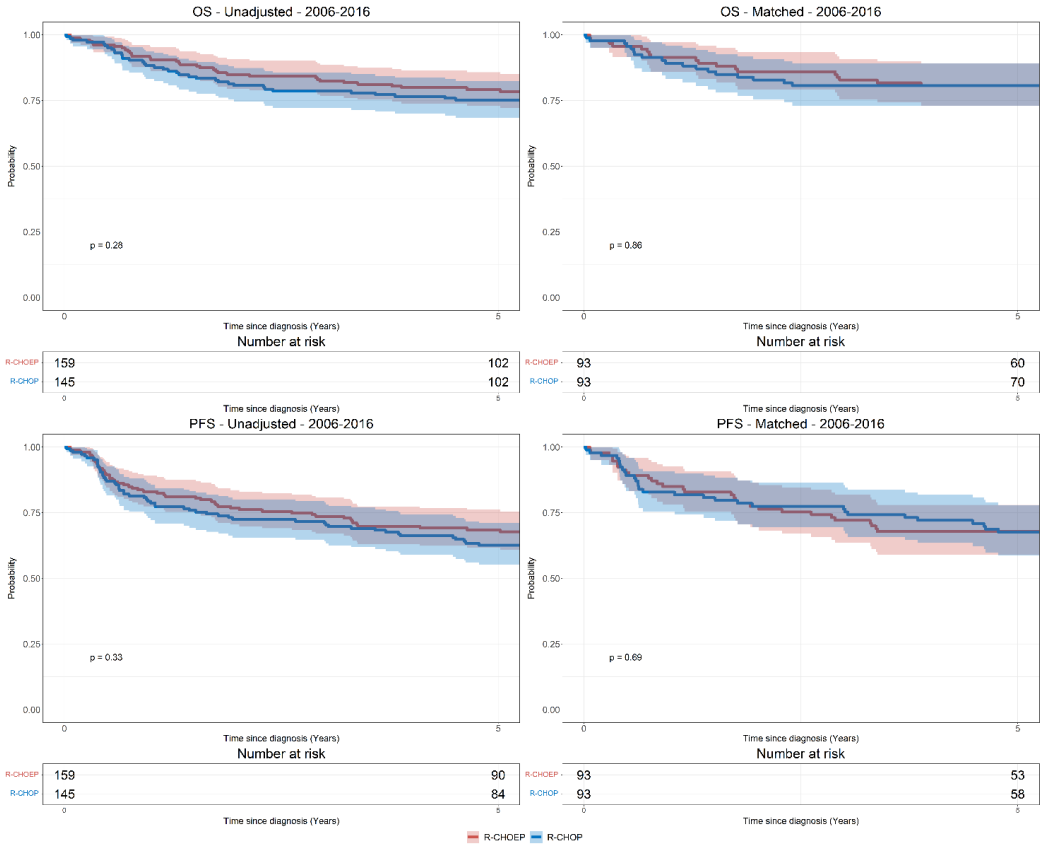


Figure S12: Kaplan-Meier estimates of overall survival and progression-free survival before and after matching for all patients diagnosed between 2006-2016.

Abbreviations: OS, Overall survival; PFS, Progression-free survival;

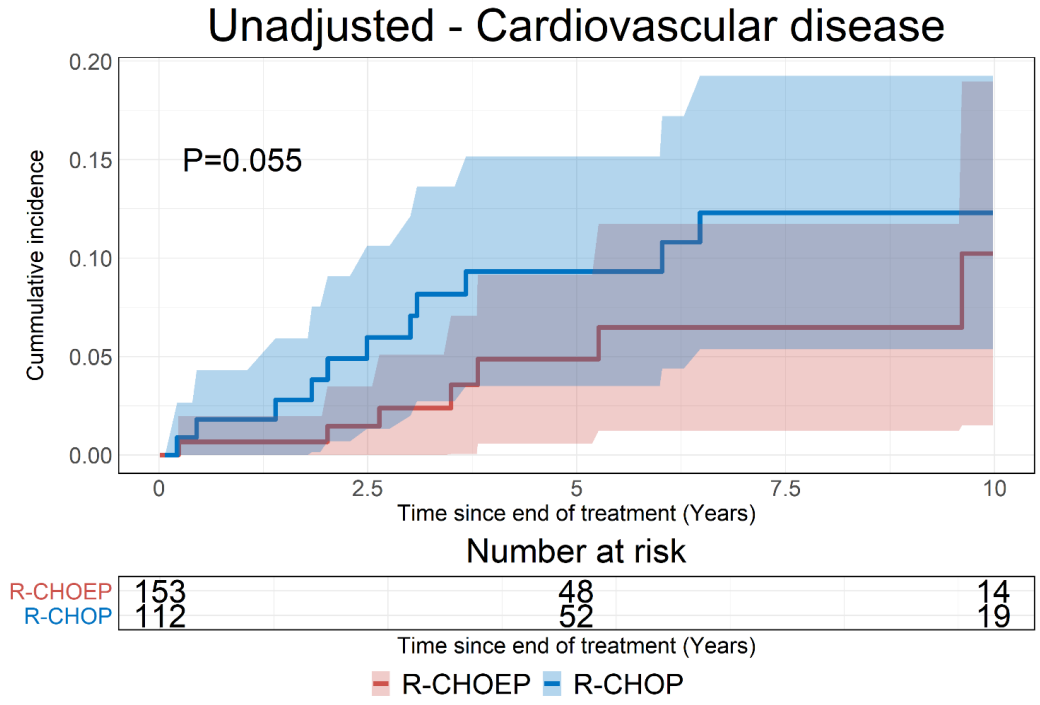


Figure S13A: Cumulative incidence for cardiovascular disease in patients without prior

cardiovascular disease and in complete remission.

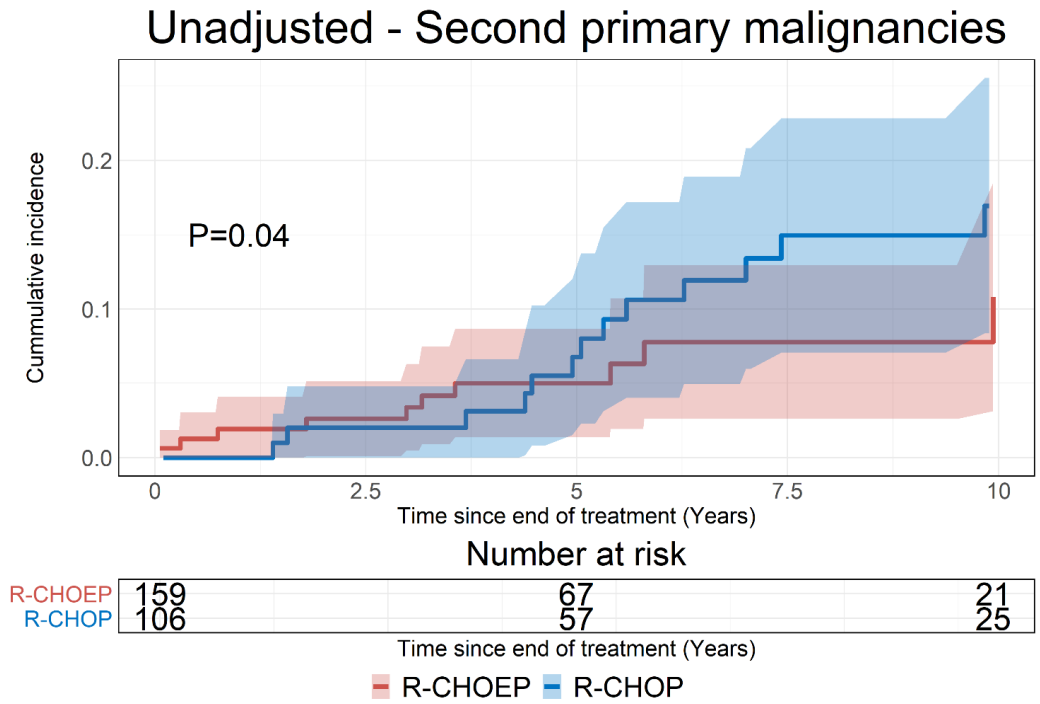


Figure S13B: Cumulative incidence for second primary malignancies in patients without prior cancer and in complete remission.

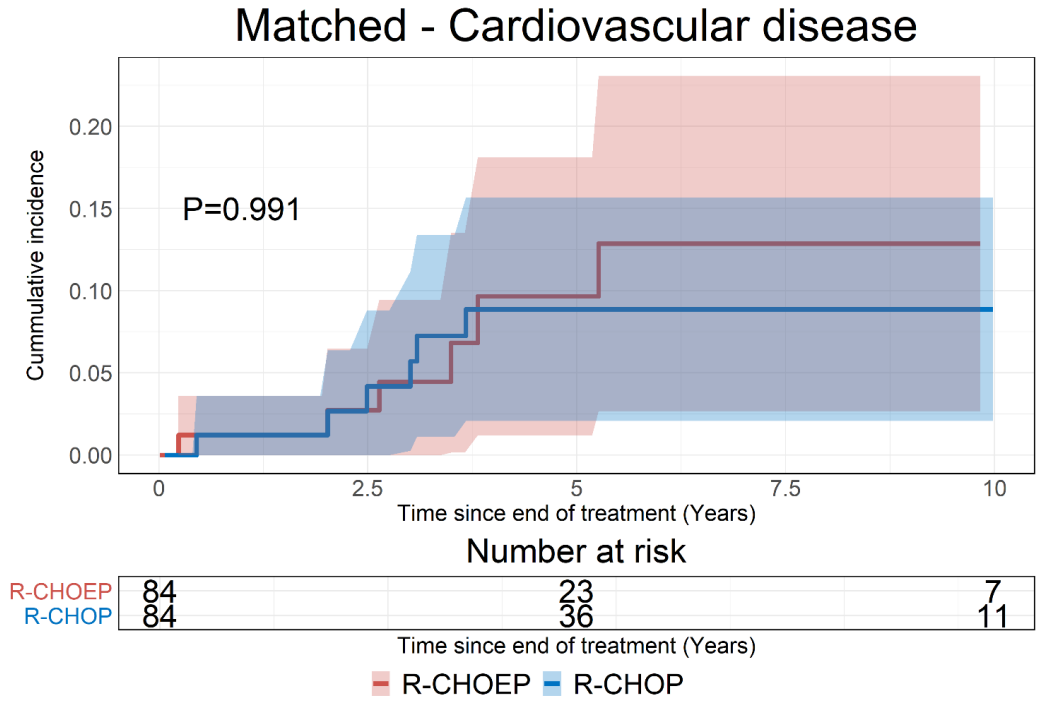


Figure S14A: Cumulative incidence for cardiovascular disease in matched patients without prior cardiovascular disease and in complete remission.

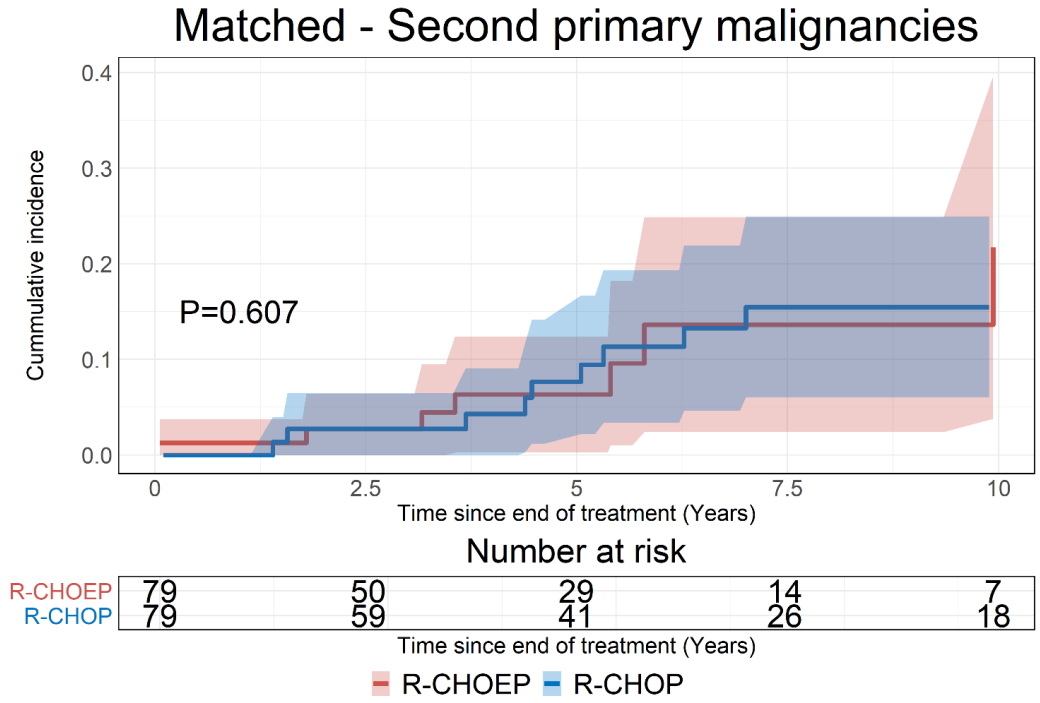


Figure S14B: Cumulative incidence for second primary malignancies for matched patients without prior cancer in complete remission.

Tables

Table S1: Outcomes for all patients before and after matching.

		All Patients		Matched	
	levels	**R-CHOEP**	**R-CHOP**	**R-CHOEP**	**R-CHOP**
5-year OS	% [95% CI]	79% (73-85)	76% (69-82)	76%(69-84)	79% (72-87)
5-year PFS	% [95% CI]	69% (63-76)	62% (55-70)	65% (57-74)	63%(55-73)
Median follow-up	Years [IQR]	5.9 [5.3-7.0]	9.6 [8.3-11.2]	5.5 [4.9-6.6]	8.8 [6.8-10.3]
Response n(%)	CR/CRu	180 (84.5)	137 (74.9)	105 (82.7)	99 (78.0)
	NC/missing†	12 (5.6)	12 (6.6)	7 (5.5)	6 (4.7)
	PD	13 (6.1)	14 (7.7)	9 (7.1)	10 (7.9)
	PR	8 (3.8)	20 (10.9)	6 (4.7)	12 (9.4)

Abbreviations: CR, Complete response, CRu, Complete response unconfirmed; PR, Partial remission; SD, Stable disease; PD, Progressive disease; PFS, Progression-free survival; OS, Overall survival.

Table S2: Characteristics of patients that were excluded after primary matching.

		Un-matched	
	Levels	**R-CHOEP**	**R-CHOP**
Total number of Patients		86	56
Age at diagnosis	Median [range]	45 [18, 55]	58 [19, 60]
Sex n(%)	Female	31 (36.0)	23 (41.1)
	Male	55 (64.0)	33 (58.9)
Ann Arbor n(%)	I-II$^{\Omega}$	<5	<5
	III-IV	86 (100)	56 (100)
ECOG (%)	0	42 (48.8)	19 (33.9)
	1	26 (30.2)	21 (37.5)
	2	18 (20.9)	16 (28.6)
Tumor diameter n(%)	≤10cm and missing$^{\Omega}$	53 (61.6)	33 (58.9)
	>10cm	33 (38.4)	23 (41.1)
B-symptoms n(%)	No and missing$^{\Omega}$	29 (33.7)	21 (37.5)
	Yes	57 (66.3)	35 (62.5)
CCI n(%)	0	71 (82.6)	45 (80.4)
	1-2	15 (17.4)	11 (19.6)
	≥3	22 (25.6)	13 (23.2)
IPI (Grouped) (%)	2-3	48 (55.8)	20 (35.7)
	4	16 (18.6)	23 (41.1)

aaIPI n(%)	2	80 (93.0)	48 (85.7)
	3	6 (7.0)	8 (14.3)
Year of diagnosis n(%)	2006-2010	23 (26.7)	37 (66.1)
	2011-2015	30 (34.9)	10 (17.9)
	2016-2020	33 (38.4)	9 (16.1)
Extra nodal n(%)	0	32 (37.2)	13 (23.2)
	1-2	42 (48.8)	32 (57.1)
	≥3	12 (14.0)	11 (19.6)
LDH ratio	Median [range]	1.74 [1.02, 16.10]	1.53 [0.74, 13.81]
Lymphocytes (10^9/L)	Median [range]	1.24 [0.10, 32.70]	1.04 [0.00, 2.70]
Total WBC (10^9/L)	Median [range]	7.81 [0.55, 39.30]	7.45 [1.80, 16.50]
Hemoglobin (g/L)	Median [range]	124.80 [68.80, 169.60]	120.80 [76.80, 171.20]
Nodal (Grouped) (%)	0-1	9 (10.5)	10 (17.9)
	≥2	77 (89.5)	46 (82.1)
5-year OS	% [95% CI]	82% (74-91)	68% (56-82)
5-year PFS	% [95% CI]	75% (66-85)	61% (49-76)
Median follow-up	Years [IQR]		
Response n(%)	CR/CRu	75 (87.2)	38 (67.9)
	PR/PD/NC/missing[Ω]	11 (12.8)	18 (32.1)

Abbreviations: LDH, lactate dehydrogenase; WBC, White blood cell; ECOG, Eastern Cooperative Oncology Group; CCI, Charlson comorbidity index; IPI, International prognostic index; CR, Complete response, CRu, Complete response unconfirmed; PR, Partial remission; SD, Stable disease; PD, Progressive disease; PFS, Progression-free survival; OS, Overall survival.

[Ω]Missing values or small sample sizes were aggregated to other subgroups due to the limited number of patients in order to ensure patient anonymity when numbers were <5.

Table S3: Univariate and multivariate cox proportional hazard modelling overall survival and progression-free survival with different treatment definitions.

		Univariate		Multivariate[Ω]	
Treatment definition (Outcome)	**Treatment**	**HR (95%CI)**	**P-value**	**Adj. HR (95%CI)**	**P-value**
Majority (PFS)	R-CHOEP (Ref)	1		1	
	R-CHOP	1.26 (0.91-1.75)	0.165	0.98 (0.69-1.39)	0.893
Majority (OS)	R-CHOEP (Ref)	1		1	
	R-CHOP	1.31 (0.89-1.95)	0.174	1.07 (0.70-1.64)	0.752
Exclusive (PFS)	R-CHOEP (Ref)	1		1	
	R-CHOP	1.25 (0.88-1.77)	0.205	0.98 (0.69-1.39)	0.893
Exclusive (OS)	R-CHOEP (Ref)	1		1	
	R-CHOP	1.24 (0.81-1.88)	0.318	1.05 (0.66-1.65)	0.843
ASP (PFS)	R-CHOEP (Ref)	1		1	
	R-CHOP	1.27 (0.91-1.76)	0.155	0.98 (0.69-1.40)	0.915
ASP (OS)	R-CHOEP (Ref)	1		1	
	R-CHOP	1.28 (0.87-1.89)	0.217	1.04 (0.68-1.6)	0.845

Abbreviations: OS, Overall survival; PFS, Progression-free survival; Ref, Reference level; HR, Hazard ratio; Adj, adjusted ; CI, Confidence intervals; ASP, R-CHOEP at some point during treatment.
[Ω]Adjusted for age, sex, ECOG, year of diagnosis, CCI and LDH ratio.

Table S4: Cause-specific-hazard ratios modelling cardiovascular disease and second primary malignancies.

		Univariate		Multivariate[Ω]	
Disease	**Treatment**	**HR (95%CI)**	**P-value**	**Adj. HR (95%CI)**	**P-value**
Second primary malignancies	R-CHOEP (Ref)	1		1	
	R-CHOP	1.65 (0.74 - 3.72)	0.22	1.21 (0.50 - 2.93)	0.68

Cardiovascular disease	R-CHOEP (Ref)	1		1	
	R-CHOP	1.83 (0.72-4.69)	0.21	1.55 (0.55 - 4.34)	0.40

Abbreviations: Ref, reference; CI, Confidence intervals; HR, Hazard ratio; Adj; Adjusted.

[Ω]Adjusted for age, sex, Ann Arbor stage ECOG performance score, absolute LDH-ratio and number of extra nodal diseases.

Table S5: Response assessment of matched patients diagnosed after 2010

		Matched patients	
Covariates	**Levels**	**R-CHOEP**	**R-CHOP**
Total number of Patients		91	70
Year of diagnosis n(%)	2006-2010	0 (0.0)	0 (0.0)
	2011-2015	43 (47.3)	33 (47.1)
	2016-2020	48 (52.7)	37 (52.9)
Response n(%)	CR/CRu	78 (85.7)	59 (84.3)
	NC/Missing	4 (4.4)	3 (4.3)
	PD	6 (6.6)	4 (5.7)
	PR	3 (3.3)	4 (5.7)

Abbreviations: CR, Complete response, CRu, Complete response unconfirmed; PR, Partial remission; SD, Stable disease; PD, Progressive disease.